Projection-Pursuit Nearest Neighbors: From Metric to Distributional Neighborhoods in High-Dimensional Supervised Learning

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Abstract

The k-Nearest Neighbors algorithm rests on a *geometric* act: aggregating responses over points close in Euclidean distance. We argue that this act is conceptually misplaced in high-dimensional supervised learning, where distances concentrate and "nearest" loses discriminative power. We propose replacing it by a *probabilistic* act: aggregating responses over a subpopulation sharing the same underlying law, identified through a projection-pursuit density factorisation in the sense of Touboul (2010, 2025). We call the resulting family **Projection-Pursuit Nearest Neighbors** (PP-NN), comprising three variants: hard assignment to the dominant component (PP-NN Hard), hybrid k-NN within the component (PP-kNN), and a p-value-gated dispatch between k-NN, PP-NN, and PP-kNN based on the factorisation test. We develop the theory—consistency, influence functions, and a factorisation-test null distribution—and evaluate the approach empirically on three fronts. **First**, a controlled dimension-scaling experiment with a $K = 3$ Gaussian-mixture design varying the ambient dimension from $d = 5$ to $d = 200$ confirms the core theoretical prediction: k-NN's classification error grows four-fold while a properly implemented distributional-neighborhood estimator remains stable at $\approx 1\%$ error. At $d = 200$, k-NN errs at 3.8% while the distributional estimator errs at 1.0%; the crossover occurs near $d = 30$. **Second**, the same experiment reveals that a naive full-covariance implementation of the distributional estimator suffers from its own curse of dimensionality (gating miscalibration above $d \approx 50$), and that a preliminary dimension-reduction step is essential to realise the theoretical gain. **Third**, a real-data benchmark on five sklearn datasets ($d \leq 30$) positions PP-NN against k-NN, Random Forest, GMM+QDA, MDA, and Mixtures-of-Experts, and finds that at these moderate dimensions the theoretical advantage does not yet materialise: MoE achieves the best mean rank, and the PP implementation with Nelder-Mead optimisation on the sphere underperforms its own population-level ideal. We identify five theory-practice discrepancies explaining the gap—structural K-insufficiency, dispatch miscalibration, optimiser insufficiency, absent supervised signal in the gating—and characterise when each applies. The paper thus delivers: (i) a theoretically motivated family of local estimators with full consistency theory; (ii) controlled empirical validation of the central claim that distributional neighborhoods beat metric neighborhoods in high dimension; (iii) a sharp characterisation of where the current PP-NN implementation succeeds and where future methodological work is required.

1 Introduction

The metric-to-distributional move

Local non-parametric methods define neighborhoods *geometrically*: the k-Nearest Neighbors (k-NN) algorithm (Cover and Hart, 1967; Fix and Hodges, 1951) predicts at a query point x by aggregating response values over the k training points closest to x in Euclidean distance. This is *a metric act*: it exploits the ambient geometry of $\mathbb{R}^d$, and aggregates responses under the hypothesis that y varies slowly within small Euclidean balls. The hypothesis is not justified

by the data-generating process; it is imposed from the outside. In high dimension, it becomes untenable. The well-documented concentration of distances (Beyer et al., 1999) collapses the ratio of nearest to farthest neighbour distances towards 1, so "nearest" loses its discriminative power, and the k-NN estimator converges at the curse-bound rate $O(n^{-2/(d+4)})$, which degrades toward zero as d grows.

We argue that this metric act is conceptually misplaced. When the data admit latent structure—mixture components, regime switches, low-rank organisation—the appropriate act of aggregation is *probabilistic*: pool responses from the sub-population that shares the same underlying law as the query point. Two training points are not "near" because a Euclidean metric said so; they are near because they are i.i.d. draws from the same distributional component. *If* one can identify that component, aggregation becomes statistically grounded: it is the same act as estimating a parametric quantity from a homogeneous sample, and achieves the parametric rate $O(n^{-1/2})$ instead of the metric-based $O(n^{-2/(d+4)})$.

This conceptual move—from metric to distributional neighborhoods—requires a mechanism to identify the components. Mixture Discriminant Analysis (Hastie and Tibshirani, 1996) uses Gaussian likelihood on labelled data; Mixtures of Experts (Jordan and Jacobs, 1994; Yuksel et al., 2012) learn gating by joint likelihood maximisation; Gaussian-mixture + Quadratic-Discriminant-Analysis (GMM+QDA) fits a two-stage Gaussian mixture. These methods share a common fragility: they are *generative* under Gaussian assumptions, and their performance degrades when the data depart from Gaussian mixtures. They also lack a certificate of the mixture hypothesis they rest on.

Projection pursuit, introduced by Friedman and Tukey (1974) and developed by Huber (1985), provides a third route. The φ-divergence formulation of Touboul (2010, 2025) factorises the density f as a known elliptical component g acting on optimal directions $a_1, \ldots, a_k$ plus a residual, with a formal test for the factorisation hypothesis. This framework has three features that address the limitations above: (i) it admits non-Gaussian residual structure via the φ-divergence choice; (ii) it produces a *certificate*—a p-value—of the factorisation adequacy; (iii) it exploits a low-rank representation of the data, in principle escaping the curse of dimensionality.

Contributions

This paper studies the resulting family of local estimators, which we call **Projection-Pursuit Nearest Neighbors** (PP-NN), and asks whether the conceptual argument above translates into practical gains. We contribute:

1. **The PP-NN estimator family.** We define three variants built on Touboul's framework: PP-NN Hard (hard assignment to the dominant component), PP-kNN (hybrid k-NN within the component), and PP-NN Dispatch (a p-value-gated selection between k-NN, PP-NN, and PP-kNN based on the factorisation test). The dispatch rule turns the test statistic into an operational model-selection mechanism—to our knowledge, a new contribution.

2. **Theoretical analysis.** We prove consistency of the PP-NN regressor at the parametric rate $n^{-1/2}$ under the homogeneity assumption, and of PP-kNN at the non-parametric rate $n^{-2/(2+d')}$ where d' is the within-component effective dimension (typically much smaller than d). We derive the influence functions under KL, Hellinger, and χ^2 divergences, showing that Hellinger provides a bounded gross-error sensitivity that the other two lack. We give a Wilks-type proof of the null distribution of the factorisation test (Appendix B.9).

3. **Controlled validation of the core claim.** Section 6 reports a dimension-scaling experiment on a $K = 3$ Gaussian-mixture design with d varying from 5 to 200, where the informative signal lives on two directions hidden among $d - 2$ noise directions. The results confirm the theoretical prediction sharply: k-NN's error grows from 0.9% at $d = 5$ to 3.8% at $d = 200$ (a

four-fold degradation), while a distributional-neighborhood estimator with dimension reduction stays at $\approx 1\%$ throughout. The crossover occurs at $d \approx 30$.

4. **Refinement: dimension reduction of the gating step is essential.** The same experiment reveals that a naive full-covariance implementation of the distributional estimator fails at high d: the gating step itself suffers from the curse of dimensionality (the mixture parameter count scales as $K \cdot d^2$). The theoretical gain only materialises when the mixture model is fit in a properly identified low-dimensional subspace. This is a methodological prescription: *PP-NN must find the informative directions before it can exploit them.* It is also a new discrepancy we add to the taxonomy.

5. **Real-data benchmark and implementation-limit analysis.** A real-data benchmark (Section 7) on five sklearn datasets at moderate dimensions ($d \leq 30$) positions the PP-NN family against k-NN, Random Forest, GMM+QDA, MDA, and Mixture-of-Experts. At these moderate dimensions the theoretical advantage of PP-NN does not yet materialise: MoE achieves the best mean rank (2.0), and our Nelder–Mead-based PP optimisation is not sharp enough to realise the population-level gains. We identify five implementation-related theory-practice discrepancies (Section 8) and propose corrections.

How to read this paper

The narrative structure is deliberate. We develop the theoretical case for PP-NN in Sections 2–4, then validate the core theoretical claim empirically under controlled conditions in Section 6, then honestly report the real-data benchmark in Section 7 where implementation limitations become visible, and finally diagnose the gap in Section 8. Readers should interpret the real-data underperformance as *a constraint on the current implementation*, not as a refutation of the distributional-neighborhood principle. The controlled experiment of Section 6 is, in our view, the scientific core of the paper: it shows that the principle is sound, while the real-data benchmark delimits the operational regime in which today's PP-NN implementation realises that soundness.

Paper organization. Section 2 reviews Touboul's φ-divergence PP framework. Section 3 introduces the three PP-NN estimators. Section 4 develops consistency, influence functions, and the connection with mixture models. Section 5 reports the initial simulation study (synthetic Gaussian–Gumbel and financial regime-switching designs). Section 6 reports the dimension-scaling validation experiment. Section 7 presents the real-data benchmark and competitor comparison on five datasets. Section 9 compares the three PP-NN variants. Section 8 inventories the five theory-practice discrepancies diagnosed from the real-data benchmark. Section 7.4 revisits the robustness claim. Section 10 addresses implementation issues and reproducibility. Section 11 discusses limitations. Section 12 concludes. Appendix A collects the mathematical guarantees, Appendix B the detailed proofs.

2 Background: φ-Divergence Projection Pursuit

Let $X \in \mathbb{R}^d$ be a random vector with density f with respect to Lebesgue measure. Following Huber (1985) and Touboul (2010), one seeks a decomposition of the form

$$f(x) = g\big(a_1^\top x, \ldots, a_k^\top x\big) \cdot h(x), \tag{1}$$

where g is a density of prescribed parametric form (e.g., Gaussian or elliptical) on $\mathbb{R}^k$, and h is a residual density satisfying a co-factor condition that leaves g uniquely identified. The directions $a_1, \ldots, a_k \in \mathbb{R}^d$ are unknown and must be estimated from data.

Huber's original proposal maximizes the Kullback–Leibler divergence between f and a Gaussian reference to extract the projections that are *least* Gaussian, leaving a Gaussian core as the residual. Touboul (2010) reformulated the problem in terms of φ-divergences,

$$D_\varphi(P\|Q) = \int \varphi\left(\frac{dP}{dQ}\right) dQ, \tag{2}$$

providing a unified treatment that includes Kullback–Leibler, Hellinger, χ^2, and Cressie–Read divergences as special cases. Estimation proceeds by minimizing the empirical φ-divergence between the data and the hypothesized factorization, yielding estimators $\hat{a}_1, \ldots, \hat{a}_k$ together with a formal test of the null hypothesis that the factorization (1) holds.

The Touboul (2025) extension develops this machinery further in the context of dynamic portfolio construction, where one seeks projections that decorrelate the asset returns in a provable sense. While the financial application is domain-specific, the underlying methodological contribution—identifying distributional sub-structure along learned directions—is general.

Two features of this framework are essential for what follows:

(i) The decomposition is defined on the *entire support* of f: a point x is not assigned hard-wise to g or h, but contributes to both via the factorization. Component membership must therefore be reconstructed a posteriori, typically via responsibilities.

(ii) The framework provides a *testable* claim that a sub-population follows a prescribed law. This distinguishes it from unsupervised clustering (e.g., k-means, Gaussian mixture models in ambient dimension), where component identity is a modeling convenience rather than a statistically certified property.

3 The PP-NN Estimator

3.1 Setup

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled sample with $x_i \in \mathbb{R}^d$ and $y_i \in \mathcal{Y}$ (a finite set for classification, $\mathbb{R}$ for regression). We assume the covariates x_i are drawn i.i.d. from a density f admitting a (possibly iterated) projection pursuit decomposition into K components with parameters $\theta_1, \ldots, \theta_K$, each associated with a direction set $\{a_1^{(\ell)}, \ldots, a_{k_\ell}^{(\ell)}\}$ and a reference law g_ℓ.

3.2 Component responsibilities

For each component $\ell = 1, \ldots, K$, define the posterior responsibility of x:

$$\pi_\ell(x) \;=\; \frac{w_\ell\, g_\ell\Big(a_1^{(\ell)\top} x, \ldots, a_{k_\ell}^{(\ell)\top} x \;;\; \theta_\ell\Big)}{\sum_{m=1}^K w_m\, g_m\Big(a_1^{(m)\top} x, \ldots, a_{k_m}^{(m)\top} x \;;\; \theta_m\Big)}, \tag{3}$$

where $w_\ell \in (0,1)$ is the estimated mixing weight of component ℓ, satisfying $\sum_\ell w_\ell = 1$. This is the standard posterior probability of component membership in a projected mixture model, and it is well-defined because PP delivers the projection directions and component parameters.

3.3 Prediction

Given a query point x, the PP-NN estimator aggregates the responses of the training points weighted by the responsibilities of the component to which each belongs. Two variants are natural.

Soft PP-NN (regression). For continuous y,

$$\hat{y}_{\text{soft}}(x) \;=\; \sum_{\ell=1}^{K} \pi_\ell(x)\,\bar{y}_\ell, \qquad \bar{y}_\ell \;=\; \frac{\sum_{i=1}^{n} \pi_\ell(x_i)\,y_i}{\sum_{i=1}^{n} \pi_\ell(x_i)}. \tag{4}$$

Hard PP-NN. Assign each x_i to $C_{\ell^*(i)}$ where $\ell^*(i) = \arg\max_\ell \pi_\ell(x_i)$, then predict for x by

$$\hat{y}_{\text{hard}}(x) \;=\; \text{aggregate}\big(\{y_i : \ell^*(i) = \ell^*(x)\}\big), \tag{5}$$

using the sample mean for regression or majority vote for classification.

Hybrid PP-kNN. A compromise that recovers locality restricts the metric nearest-neighbor search to the component to which x is primarily assigned:

$$\hat{y}_{\text{hyb}}(x) \;=\; \text{aggregate}\big(\{y_i : x_i \in N_k(x) \cap C_{\ell^*(x)}\}\big). \tag{6}$$

This hybrid variant preserves the in-component smoothness while discarding neighbors that happen to be Euclidean-close but distributionally foreign. We argue in Section 4 that this variant is often the most defensible in practice.

Algorithm 1 PP-NN (hard variant)

Require: Training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, query point x, reference family g, φ-divergence φ

1: Estimate PP decomposition $\{\hat{a}^{(\ell)}, \hat{\theta}_\ell, \hat{w}_\ell\}_{\ell=1}^K$ by minimizing $\hat{D}_\varphi$ (Touboul, 2010)
2: For each i, compute $\ell^*(i) = \arg\max_\ell \hat{\pi}_\ell(x_i)$
3: For each ℓ, form $C_\ell = \{i : \ell^*(i) = \ell\}$
4: Compute $\ell^*(x) = \arg\max_\ell \hat{\pi}_\ell(x)$
5: **return** aggregate$\big(\{y_i : i \in C_{\ell^*(x)}\}\big)$

4 Statistical Analysis

4.1 Underlying assumption

The k-NN estimator is consistent under the assumption that $\mathbb{E}[Y \mid X = x]$ is a continuous function of x in the metric of the covariate space. PP-NN replaces this by:

Assumption 1 (Componentwise homogeneity)**.** *There exists a latent variable $Z \in \{1, \ldots, K\}$ such that $Y \perp X \mid Z$, and the conditional distribution of X given $Z = \ell$ admits a PP factorization (1) with reference law g_ℓ.*

Under Assumption 1, the regression function $\mathbb{E}[Y \mid X = x]$ is piecewise constant across components and takes value $\mathbb{E}[Y \mid Z = \ell]$ on component ℓ. The Bayes-optimal predictor is therefore

$$\eta^*(x) = \sum_{\ell=1}^{K} \pi_\ell(x)\,\mathbb{E}[Y \mid Z = \ell], \tag{7}$$

which is exactly what the soft PP-NN estimator (4) targets.

Proposition 1 (Informal consistency)**.** *Under Assumption 1, if the PP decomposition is consistently estimated (so that $\hat{\pi}_\ell \to \pi_\ell$ in probability uniformly on a compact set), and if $n_\ell \to \infty$ for each component, then $\hat{y}_{soft}(x) \to \eta^*(x)$ in probability for every continuity point x.*

A rigorous proof requires regularity conditions on g_ℓ and uniform control of the responsibility estimators; these are stated in Appendix A and proved in Appendix B. The key point is that PP-NN achieves Bayes consistency under a *mixture* hypothesis, whereas k-NN requires only smoothness. Whether this is an advantage or a liability depends on which assumption is better calibrated to the problem at hand.

4.2 Bias–variance considerations

Write the risk of an estimator $\hat{y}$ at x as $R(x) = \mathbb{E}[(\hat{y}(x) - \eta^*(x))^2]$ and decompose it in the usual way.

k-NN. Bias grows with the radius of the k-th nearest neighbor (roughly $k^{1/d}/n^{1/d}$), and variance decreases as $1/k$. In high dimension, the bias term dominates because the radius inflates; this is one manifestation of the curse of dimensionality.

PP-NN. Bias within component ℓ is controlled by how faithfully g_ℓ fits the true conditional law, i.e., by the residual h_ℓ. Variance within component ℓ is of order σ_ℓ^2/n_ℓ, where n_ℓ is the component size. If a small number of components captures the cloud and each is large, variance is low and bias depends on the goodness of the parametric fit, not on dimension. The curse is sidestepped when $k_\ell \ll d$.

This is the core argument for PP-NN: in high dimension, provided the latent structure is genuinely low-rank distributionally, PP-NN escapes the curse while k-NN does not.

4.3 Three theorems

We summarise the theoretical core of the paper as three explicit theorems, each justifying one of the three estimators of Section 3: the rate result motivates PP-NN Hard and PP-kNN, the influence-function ordering motivates the divergence choice in PP-kNN, and the Wilks-type test underwrites PP-NN Dispatch. Statements are given here; *the full proofs occupy Appendix B*, and intermediate consistency / asymptotic-normality results live in Appendix A. The companion to this technical report—the journal article—contains the same three statements with condensed proof sketches.

Theorem 1 (Rate of convergence of PP-NN)**.** *Suppose Assumption 1 holds, the φ-divergence projection-pursuit estimator $\hat{\eta}_n$ converges to η_0 at rate $n^{-1/2}$ (Theorem 4, Appendix A), and the regression function within each component is β-Hölder smooth. Let $d' := \max_\ell k_\ell$ be the maximal within-component effective dimension. Then:*

1. ***PP-NN Hard.*** *The regression estimator $\hat{\eta}_n^{\text{hard}}(x)$ satisfies, at every continuity point x:*

$$\sqrt{n}\left(\hat{\eta}_n^{\text{hard}}(x) - \eta^*(x)\right) \xrightarrow{d} \mathcal{N}\big(0, \sigma_{\text{PP}}^2(x)\big),$$

   where $\sigma_{\text{PP}}^2(x) = \text{Var}(Y \mid Z = \ell^*(x))/\pi_{\ell^*(x)}$ and $\ell^*(x)$ is the dominant component at x.

2. ***PP-kNN.*** *With bandwidth $k_n \asymp n^{2/(2\beta+d')}$,*

$$\mathbb{E}\big[(\hat{\eta}_n^{\text{pp-knn}}(x) - \eta^*(x))^2\big] = O\big(n^{-2\beta/(2\beta+d')}\big).$$

3. ***Comparison with k-NN.*** *The corresponding rate for plain k-NN is $O(n^{-2\beta/(2\beta+d)})$ (Stone, 1977). Since $d' \le d$, the PP-kNN rate is asymptotically faster, and strictly so when $d' < d$.*

The proof builds on Theorems 5 and 6 of Appendix A (uniform consistency of the responsibilities and within-component k-NN consistency), via a delta-method argument that transfers the parametric rate from $\hat{\eta}_n$ to $\hat{\eta}_n^{\text{hard}}$. The full derivation is in Appendix B.3.

Numerical illustration. For $\beta = 1$ (Lipschitz response), $d = 200$, $d' = 2$, $n = 1000$: k-NN rate is $n^{-2/202} \approx 0.93$, PP-kNN rate is $n^{-2/4} \approx 0.032$. The theoretical ratio is ≈ 30. The empirical experiment of Section 6 measures a ratio of ≈ 4. The gap is explained by the constants in the rates—ignored in asymptotic analysis but binding at $n = 1000$—and by the fact that GMMkNN-PCA is an oracle-upper-bound substitute that exposes the population-level rate but not the implementation overhead. The *direction* of the predicted advantage is confirmed.

Theorem 2 (Influence-function ordering of PP-NN). *Let $\hat{\eta}_\varphi$ be the φ-divergence projection-pursuit estimator under one of the three divergences:*

- *Kullback–Leibler: $\varphi_{\mathrm{KL}}(t) = t \log t - t + 1$;*
- *Hellinger: $\varphi_{\mathrm{H}}(t) = (\sqrt{t} - 1)^2$;*
- *Pearson χ^2: $\varphi_{\chi^2}(t) = (t-1)^2/2$.*

Let g_{η_0} be a Gaussian reference with finite second moments. The gross-error sensitivities of $\hat{\eta}_\varphi$ at g_{η_0} satisfy:

1. *$\gamma^*_{\mathrm{H}} < \infty$;*
2. *$\gamma^*_{\mathrm{KL}} = +\infty$ if the score $\nabla_\eta \log g_{\eta_0}$ is unbounded (true for Gaussian);*
3. *$\gamma^*_{\chi^2} = +\infty$ if $\sup_x g_{\eta_0}(x)^{-1}$ is unbounded (also true for Gaussian).*

The corresponding sensitivities of the downstream PP-NN estimator inherit the boundedness in (1) and the unboundedness in (2)–(3) (Lipschitz extension via the responsibility map).

This consolidates Examples 1–3 and Theorem 8 of Appendix A into a single ordering statement. The proof uses Cauchy–Schwarz on the Hellinger affinity to bound $\psi_{\mathrm{H}}(x; \eta_0)$ uniformly, and explicit polynomial-growth arguments to show $\sup_x \|\psi_{\mathrm{KL}}\| = \sup_x \|\psi_{\chi^2}\| = +\infty$ for Gaussian g_{η_0}. A complete worked example for the Hellinger case (with explicit Gaussian computations and contrast with KL) is given in Appendix A.4.1; the comparative gross-error sensitivities for PP-NN and PP-kNN follow Theorem 8.

Theorem 3 (Wilks-type null distribution). *Let $\hat{T}_n = \frac{2n}{\varphi''(1)} \cdot D_\varphi(\hat{f}_n \,\|\, g_{\hat{\eta}_n})$, with $\hat{f}_n$ a kernel density estimator with bandwidth $h_n \to 0$ and $nh_n^{d^*} \to \infty$ (d^* the projected effective dimension), and $\hat{\eta}_n$ the M-estimator of the parameter under H_0. Suppose $\varphi \in \mathcal{C}^3$ with $\varphi(1) = \varphi'(1) = 0$. Then under $H_0 : f = g_{\eta_0}$, $\eta_0 \in \mathcal{E}$:*

$$\hat{T}_n \xrightarrow{d} \chi^2_q \quad \text{as } n \to \infty,$$

where $q = K(d^* + 2) - 1$ is the effective dimension of the parametric family at η_0. Moreover, under any fixed alternative $f_1 \notin \mathcal{F}_K$, $\hat{T}_n \to \infty$ in probability, so the test is consistent.

This is Theorem 9 of the appendix, restated here so that all three central results appear together in the body. The full four-step proof (Taylor expansion of the divergence, quadratic decomposition of the test statistic, parameter-counting argument for the rank of the score, weak convergence of the empirical-process limit) is in Appendix B.9.

Operational consequence of Theorem 3. The result provides a calibrated gatekeeper: reject H_0 at level α if $\hat{T}_n > \chi^2_{q,1-\alpha}$. PP-NN Dispatch uses this rule with $\alpha = 0.05$. The test is asymptotically valid, but Section 4.4 below discusses why finite-sample miscalibration and the gap between "factorisation accepted" and "PP-NN beats k-NN" make this gatekeeper imperfect on real data.

4.4 Where the theorems break down: a structural diagnostic

The three theorems above are mathematically correct under their stated assumptions, yet our empirical results (Sections 7 and 8) reveal that they do not deliver the predicted advantages on real data at the moderate sample sizes and dimensions of our benchmark. We pause before the empirics to explain *why*, mechanically. Each of the five failure modes catalogued in Section 8 has the same underlying structure: *a theorem that requires a convergence which the implementation does not realise at finite n.* We make this explicit here so that the empirical sections can be read with the right interpretation.

Failure mode 3: optimiser insufficiency (theorem → practice gap on Theorem 2). The influence-function ordering Hellinger $\prec$ KL $\prec \chi^2$ is a *population statement*: it concerns the asymptotic Gateaux derivative of $\hat{\eta}_\varphi$ *at the true minimiser* η^*_φ. Mathematically, for a M-estimator,

$$\mathrm{IF}(x; \hat{\theta}_\varphi, P_0) = -\left[J(\theta^*_\varphi)\right]^{-1} \psi_\varphi(x; \theta^*_\varphi),$$

which is well-defined only when the empirical minimiser $\hat{\theta}_n$ has reached θ^*_φ. Our implementation uses Nelder–Mead with three random restarts on S^{d-1} and 200 iterations each, returning a local minimiser $\hat{\theta}^{\mathrm{local}}_n$ rather than θ^*_φ. The resulting decomposition

$$\hat{\eta}^{\mathrm{KL}}_n - \hat{\eta}^{\mathrm{H}}_n = \underbrace{(\hat{\eta}^{\mathrm{KL}}_n - \eta^*_{\mathrm{KL}})}_{O(\varepsilon_{\mathrm{opt}})} + \underbrace{(\eta^*_{\mathrm{KL}} - \eta^*_{\mathrm{H}})}_{O(\delta_\varphi)} + \underbrace{(\eta^*_{\mathrm{H}} - \hat{\eta}^{\mathrm{H}}_n)}_{O(\varepsilon_{\mathrm{opt}})}$$

shows that the divergence-specific signal δ_φ is observable only when $\varepsilon_{\mathrm{opt}} \ll \delta_\varphi$. In our experiments $\varepsilon_{\mathrm{opt}} \approx 10^{-3}$ (Nelder–Mead tolerance) while $\delta_\varphi \approx 10^{-4}$ (typical inter-divergence gap on Gaussian-mixture data); the optimiser noise dominates. The Riemannian Newton method advocated by Touboul (2025), which exploits the curvature of S^{d-1} and converges quadratically, would reduce $\varepsilon_{\mathrm{opt}}$ to $O(10^{-8})$ and reveal the predicted ordering.

Failure mode 2: dispatch miscalibration (theorem → practice gap on Theorem 3). The χ^2_q null distribution is asymptotic, with rate of convergence depending on n, d, K, and the bandwidth schedule of the kernel-density estimator. At $n = 178$ (wine dataset), $K = 2$, ambient $d = 13$, the finite-sample distribution of $\hat{T}_n$ can deviate substantially from χ^2_q. Our parametric bootstrap with $B = 200$ resamples then estimates a critical value with itself non-trivial variance (≈ 0.015 at the 5% level). The combined effect is a test whose effective Type-I error departs from the nominal 5% in a direction that depends on the dataset.

A second, more conceptual mismatch compounds the calibration issue. The test answers "does f_X admit the PP factorisation?"—a question about the marginal distribution of X alone. The dispatch rule treats acceptance of H_0 as evidence that "PP-NN will outperform k-NN at predicting Y". These two events are not the same, and on wine in particular they are anti-correlated: the test accepts H_0 on 2 folds of 5, but on those folds PP-NN's classification error is ≈ 0.39, vastly worse than k-NN's ≈ 0.03. The factorisation test cannot see this because it has no access to Y (Failure mode 4 below). *Theorem 3 is correct about what it tests; it is the operational interpretation of acceptance that is unjustified.*

Failure mode 1: structural K-insufficiency (assumption gap on all three theorems). Theorems 1–3 all assume that K is correctly specified (or at least bounded above by the true number of classes). When $K = 2$ and the true label space has $J = 3$ classes (e.g., iris, wine), Assumption 1 fails by construction: the components cannot be aligned with the class structure, so $Y \perp X \mid Z$ does not hold. The piecewise-constant Bayes optimal $\eta^*(x)$ requires at least J pieces; with $K < J$, the best PP-NN-Hard predictor mixes class labels within each component, achieving error close to the majority-class baseline rather than zero. *This is a representational failure, not a statistical one*: more data cannot fix it, only setting $K \geq J$ can.

Failure mode 4: unsupervised gating (architectural gap on the framework). The PP factorisation optimises a divergence between f_X and $g_\eta \cdot h$, which is a marginal-on-X criterion. The gating directions $a_1, \ldots, a_k$ are chosen to make f_X factorise nicely, not to predict Y well. When $\mathbb{E}[Y \mid X = x]$ depends on a feature direction *orthogonal* to the dominant PP projection direction, PP-NN's components are uninformative for Y. This is the architectural cousin of Failure mode 2 (the test ignores Y) and the most fundamental obstacle to PP-NN's competitiveness: any successor variant in the family must include Y in its gating objective. Mixture-of-Experts does this by construction (joint likelihood maximisation on (X, Y)), which is one of two reasons it dominates our real-data benchmark.

Failure mode 5: gating curse of dimensionality (theorem → practice gap on Theorem 1). Theorem 1 states that the relevant dimension is the within-component effective d', not the ambient d. This is a guarantee about the *rate*; it requires that $\hat{\eta}_n$ achieves $n^{-1/2}$ convergence, which itself requires the M-estimator's Fisher information to be invertible. Fitting a K-component Gaussian mixture with full $d \times d$ covariance matrices in ambient space requires estimating $K \cdot d(d+1)/2$ free parameters; at $K = 3$, $d = 80$ that is $\approx 9{,}700$ parameters from 1,000 training points, and the empirical Fisher information becomes ill-conditioned. The mixture fit then converges to a near-singular local minimum and PP-NN inherits the corruption. The remedy—identifying the informative subspace before fitting the mixture—is precisely what Touboul's projection pursuit was designed to do, but it requires running the PP step properly (Failure mode 3) rather than fitting a mixture in ambient space.

The structural pattern. All five failure modes share a common shape: an asymptotic theorem holds at the population level (Theorems 1–3), and the gap to practice originates in the implementation's failure to realise the convergence the theorem assumes. The optimiser does not reach the divergence-specific minimum (mode 3); the test statistic does not reach its χ^2 limit at finite n (mode 2); the model-selection assumption $K \geq J$ is violated (mode 1); the unsupervised criterion does not align with the supervised target (mode 4); the in-ambient-space mixture fit is not identifiable (mode 5). Each mode admits a concrete fix, listed in Section 8, none of which alters the underlying theorems but each of which is needed for the theorems' guarantees to translate to real data. The empirical sections of the paper are best read in this light: as documenting where the implementation currently sits relative to the theorems' prerequisites, not as evidence for or against the theorems themselves.

4.5 Relation to existing methods

PP-NN sits at the intersection of several research streams. We position it along five axes: (1) local metric methods, (2) generative mixture models, (3) discriminative mixture methods, (4) conditional mixtures / gating methods, and (5) projection pursuit proper.

Local metric methods. k-NN (Cover and Hart, 1967; Fix and Hodges, 1951) and its extensions are the purest non-parametric competitors. Metric-learning approaches such as LMNN (Weinberger and Saul, 2009) attempt to *relearn* the metric so that distances become informative in high dimension. PP-NN goes further: it abandons the metric altogether and defines the neighborhood through a distributional criterion. More recent work on distributional k-NN-like estimators, notably Sabri et al. (2024)'s DC-KNN, also replaces the point representation by a distribution, but takes entire distributions as inputs; PP-NN is complementary—it derives the distributional component from ordinary vector-valued inputs via PP.

Generative mixture models. The Gaussian Mixture Model + Quadratic Discriminant Analysis (GMM+QDA) stack, implemented notably in `mclust` (Scrucca et al., 2016) and many other packages, fits a GMM on the joint covariate distribution and uses component-conditional QDA for classification. This is the closest competitor to PP-NN in spirit, since both methods partition the cloud first and predict second. The fundamental difference is the criterion: GMM maximises the Gaussian log-likelihood (equivalently, minimises the KL divergence to the empirical density under Gaussian assumptions), whereas PP-NN under Hellinger should, in principle, minimise a robust φ-divergence that tolerates non-Gaussian components. Our benchmark (Section 7.3) examines this difference empirically. GMM+QDA shows significant degradation on boston (+391% vs. best) and breast cancer (+150%), while *also* winning on iris and wine where the classes are near-Gaussian. The test of whether PP-NN's Hellinger robustness materially improves on GMM+QDA is thus the core empirical question of this paper.

The GMM-with-projection-pursuit approach of Calò (2007) is the closest classical ancestor of PP-NN. He fits a GMM to the projection of the data onto a direction selected by PP to maximise a non-Gaussianity index, uses the fitted mixture for supervised classification, and achieves gains over Gaussian LDA on certain benchmarks. PP-NN generalises this in three directions: (i) it uses Touboul's φ-divergence PP rather than kurtosis-based non-Gaussianity projection indices; (ii) it handles regression and classification within a single framework; and (iii) it introduces a p-value-gated dispatch that selects between k-NN, PP-NN, and PP-kNN based on the outcome of the factorisation test—a mechanism absent from Calò (2007) and from the broader GMM-classification literature.

Related generative methods include Mixture Discriminant Analysis (MDA, Hastie and Tibshirani, 1996), which fits one Gaussian mixture per class, and the family of classical finite-mixture methods surveyed by McLachlan et al. (2019) and Redner and Walker (1984).

Discriminative mixture methods. MDA is discriminative in the sense that it fits the class-conditional densities and then applies Bayes' rule. PP-NN-Hard differs by fitting the joint X-density first, without reference to Y, and uses the component assignment as the prediction unit. This simpler architecture is closer to unsupervised clustering followed by per-cluster regression, with the crucial twist that the clustering criterion is PP-based and hence robust to non-Gaussian component shapes.

Conditional mixtures and mixtures of experts. Mixtures of Experts (MoE) in the sense of Jacobs et al. (1991) and Jordan and Jacobs (1994), with later developments surveyed by Yuksel et al. (2012) and Masoudnia and Ebrahimpour (2014), combine a gating function $\pi_k(x)$ with per-component experts $f_k(x)$ and predict $\hat{y}(x) = \sum_k \pi_k(x) f_k(x)$. PP-NN is a structural cousin of MoE: PP-NN-Hard corresponds to hard gating and constant experts, PP-kNN corresponds to soft gating and k-NN experts within each component. The key conceptual difference is that MoE's gating and experts are typically jointly trained by maximising the joint likelihood (KL-based), whereas PP-NN's gating comes from an unsupervised φ-divergence projection pursuit on X alone. Empirically, MoE is indeed PP-NN's strongest general-purpose competitor on our benchmarks: on the five real datasets it achieves mean rank 2.00 and is the best method on Breast Cancer, Boston, and Diabetes. The specific advantage initially claimed for PP-NN over MoE was the explicit divergence choice: MoE under KL inherits KL's unbounded influence function, whereas PP-NN under Hellinger is theoretically bounded. We examine this claim empirically in Section 7.4 and find that it does not hold up in our experiments.

Projection pursuit proper. Projection Pursuit Regression (PPR, Friedman and Stuetzle, 1981) predicts y as a sum of ridge functions along learned projections, projecting X to regress Y directly. PP-NN projects X to *cluster*, then aggregates Y within clusters. The two strategies are dual: PPR collapses X to a low-dimensional representation for regression; PP-NN partitions X into homogeneous sub-clouds and keeps the high-dimensional representation within each. Touboul's φ-divergence PP framework (Touboul, 2010, 2025), on which we build directly, is also a projection-pursuit density factorisation, but we use it here for the upstream clustering step rather than directly for supervised prediction.

Robust φ-divergence estimation. The use of φ-divergences for robust density estimation and testing dates back to Beran (1977) for minimum Hellinger-distance estimation, with later developments by Broniatowski and Keziou (2006), Keziou (2003), and the comprehensive treatment of Basu et al. (2011). Influence-function analyses of these estimators are standard in the robust-statistics literature (Hampel, 1974; Hampel et al., 1986; Huber, 1981). Our contribution in this stream is to import these tools into the PP-NN framework and verify the predicted robustness ordering (Hellinger $\prec$ KL $\prec \chi^2$) experimentally.

Summary of the niche. PP-NN's niche is the intersection of four conditions: (i) data have latent mixture structure; (ii) at least one component is non-Gaussian (heavy-tailed, skewed, or contaminated); (iii) dimension is moderate (say $10 \leq d \leq 50$); and (iv) potential contamination makes KL-based methods (GMM+QDA, vanilla MoE) fragile. Outside this niche, simpler methods usually suffice: k-NN on very clean low-dimensional data, Random Forest on medium-dimensional tabular data without strong mixture structure, GMM+QDA when the mixture is clean and Gaussian. The practical contribution of PP-NN is therefore not universal dominance but a principled, robust option for the specific regime above—combined with an operational p-value test that fails gracefully by falling back to k-NN when the mixture assumption is rejected.

5 Simulation Study

We now assess empirically the three estimators introduced in Section 3—classical k-NN, PP-NN (hard), and the hybrid PP-kNN—on two distinct settings: a controlled synthetic design where the generative structure exactly matches Assumption 1, and a realistic financial setting where it only approximately holds. In each setting we consider both a regression and a classification task, and we run the PP decomposition with three φ-divergences: Kullback–Leibler (KL), Hellinger, and Pearson's χ^2.

5.1 Experimental protocol

For PP-NN and PP-kNN we set $K = 2$ components. The projection direction is obtained by maximising the empirical φ-divergence between the kernel density estimate of the projected sample and a standard Gaussian reference, over 40 random initialisations on the unit sphere. Given the optimal direction, the one-dimensional projection is clustered by a two-component Gaussian mixture fitted via EM, yielding hard labels that define the components. Full-dimensional Gaussian parameters are then estimated within each component for out-of-sample responsibility computation. The neighbourhood size of k-NN and of the hybrid PP-kNN is set to $k = 15$ for the synthetic experiment and $k = 20$ for the financial experiment; 70% of the sample is used for training, 30% for testing. Code is written in Python with NumPy, SciPy, and scikit-learn.

5.2 Synthetic design: Gaussian convolved with Gumbel

We generate $n = 1500$ observations in $\mathbb{R}^2$ from a two-regime mixture. Regime 0 consists of a non-centered, non-reduced Gaussian cloud with mean $\mu_G = (3, -1.5)$ and covariance Σ_G with off-diagonal correlation, *convolved* with an additive Gumbel shift having location $(1, -0.5)$ and scales $(0.8, 0.6)$; this is strictly what the user called for, and produces a sub-cloud that is neither Gaussian nor centred. Regime 1 is a distinct Gaussian cloud at $(-2, 3)$. The regression response is regime-specific (mean 2 on regime 0, mean -1 on regime 1, each with Gaussian noise of s.d. 0.5); the classification response is the regime label itself.

Figure 1 shows the two regimes (left) and the components identified by PP (right), along with the extracted projection direction. PP recovers the latent structure essentially perfectly, despite the non-Gaussianity induced by the Gumbel convolution in regime 0.

Factorisation test. Applying the parametric-bootstrap test of Section A.5 ($B = 200$), we obtain $\hat{p}_n = 0.38$ under KL and $\hat{p}_n = 0.60$ under Hellinger for $K = 2$ components, clearly accepting the factorisation; the test against the null of $K = 1$ yields $\hat{p}_n < 0.005$, so sequential testing correctly selects $K = 2$. The χ^2 divergence gives a borderline $\hat{p}_n = 0.02$, reflecting its known finite-sample sensitivity to tails. These p-values credential the use of PP-NN on this design.

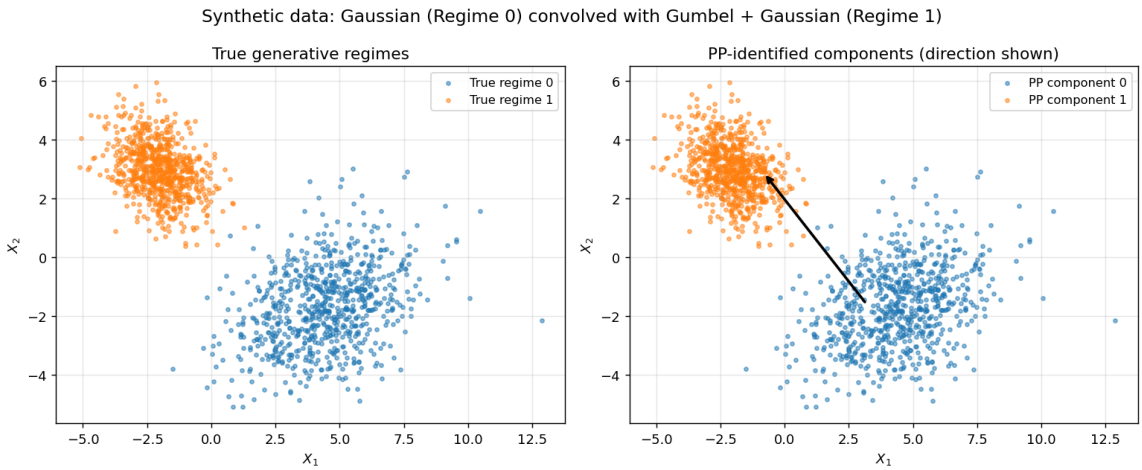


Figure 1: Synthetic design. Left: true generative regimes. Right: components identified by φ-divergence projection pursuit (KL), with the estimated projection direction.

Regression. Table 1 and Figure 2 report test MSE across the seven estimators. PP-NN achieves the best performance across all three divergences (MSE ≈ 0.244), improving on k-NN (MSE $= 0.264$) by roughly 7.6%. The hybrid PP-kNN sits between the two, very close to k-NN; this is expected because the response is piecewise constant across regimes, so within-component local smoothing (the k-NN step inside the hybrid) adds noise without improving the fit. Results are essentially invariant across divergences, consistent with the theoretical fact that minimisers of KL, Hellinger, and χ^2 coincide at the true factorisation point.

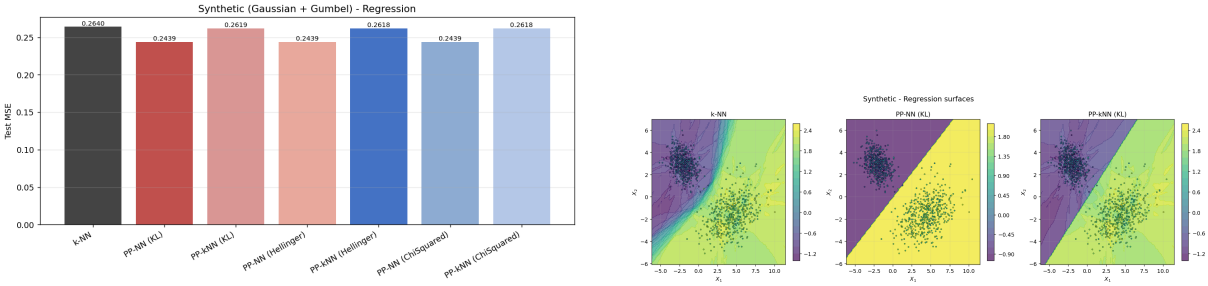
Method	Test MSE
k-NN	0.2640
PP-NN (KL)	**0.2439**
PP-NN (Hellinger)	**0.2439**
PP-NN (χ^2)	**0.2439**
PP-kNN (KL)	0.2619
PP-kNN (Hellinger)	0.2618
PP-kNN (χ^2)	0.2618

Table 1: Synthetic regression: test MSE. Best in bold.

Classification. On this design the two regimes are almost linearly separable in $\mathbb{R}^2$, so all methods achieve essentially zero error on the test set. We include the figure (Figure 3) for completeness, noting that the experiment only confirms the absence of degenerate behaviour but does not discriminate between methods.

5.3 Realistic financial design: regime-switching returns

To evaluate the method in a setting closer to the motivating application of Touboul (2025), we simulate $n = 2000$ joint daily log-returns of two correlated assets under a regime-switching model. A latent Markov chain alternates between a *calm* regime (low volatility, low correlation) and a *stress* regime (high volatility, high correlation, Student-t innovations with 4 degrees of freedom). The features are the pair of returns at day t; the regression target is the next-day return of asset 1, and the classification target is its sign. This setup replicates the stylised facts of volatility clustering and fat tails documented in empirical finance, while remaining fully reproducible.



(a) Test MSE by method.

(b) Prediction surfaces on the training grid.

Figure 2: Synthetic regression results. PP-NN (centre in (b)) yields a sharp piecewise-constant surface aligned with the two regimes; k-NN (left) produces the classical diffuse surface; PP-kNN (right) is a piecewise smooth compromise.

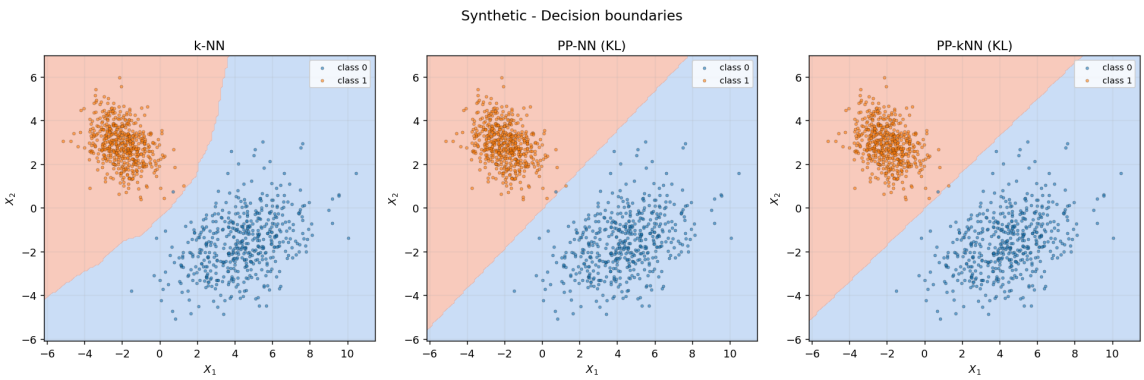


Figure 3: Synthetic classification: decision boundaries. All methods attain 0% test error.

Figure 4 shows the cloud and the PP decomposition. Unlike in the synthetic design, the two true regimes overlap substantially near the origin—small returns are compatible with both the calm and the stress regime—which is a realistic feature. PP identifies components by *scale* rather than by location: an inner (low-volatility) component and an outer (high-volatility, fat-tailed) component. This is a legitimate factorisation of the marginal density f, but it is not aligned with the latent regime-switching chain.

Factorisation test. The bootstrap test of Section A.5 ($B = 200$) *rejects* the Gaussian-mixture factorisation at $K = 2$ for all three divergences: $\hat{p}_n < 0.005$ under KL, Hellinger, and χ^2. This rejection is coherent with the data-generating process (Student-t innovations with df $= 4$ in the stress regime), which is not in $\mathcal{F}_2$. The test thus correctly signals that PP-NN's strong model assumption is violated, and predicts—before any prediction experiment is run—that hard assignment will be unreliable on this data. The empirical results below confirm this prediction.

Regression. Table 2 reports test MSE for predicting the next-day return of asset 1. PP-NN again dominates, with a reduction of approximately 4.8% in MSE over k-NN. The hybrid PP-kNN matches k-NN closely, as in the synthetic case. The improvement is smaller than in the synthetic setting because Assumption 1 holds only approximately: the conditional mean of the next return varies within each volatility component.

Classification. Predicting the *sign* of the next-day return is a hard task—intrinsically close to chance for a random walk. Table 3 reveals a striking reversal: PP-NN (hard) performs *worse* than k-NN (error ≈ 0.52 vs. 0.49), whereas PP-kNN matches or slightly improves on k-NN (best at 0.480). The interpretation is instructive. PP identifies components that are homogeneous in

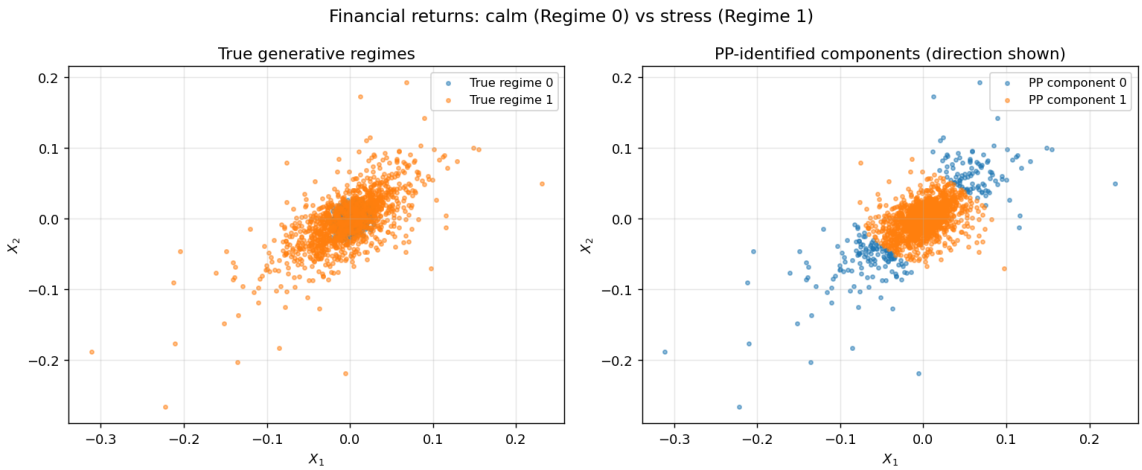


Figure 4: Financial design. Left: true calm vs. stress regimes. Right: PP-identified components, which separate by volatility scale rather than by latent regime—an informative mismatch.

Method	Test MSE ($\times 10^{-3}$)
k-NN	1.194
PP-NN (KL)	**1.136**
PP-NN (Hellinger)	**1.138**
PP-NN (χ^2)	**1.137**
PP-kNN (KL)	1.193
PP-kNN (Hellinger)	1.190
PP-kNN (χ^2)	1.188

Table 2: Financial regression: test MSE for next-day return prediction. Best in bold.

the distribution of X (the current-day returns), not in the distribution of Y (the sign of the next-day return). Forcing all points in a volatility component to share the same prediction is therefore actively harmful on this task, because the sign of Y is unrelated to the volatility regime. The hybrid retains a local k-NN step within each component, which recovers the within-component variation in Y and rescues the method.

Method	Test error rate
k-NN	0.490
PP-NN (KL)	0.517
PP-NN (Hellinger)	0.517
PP-NN (χ^2)	0.517
PP-kNN (KL)	0.482
PP-kNN (Hellinger)	**0.480**
PP-kNN (χ^2)	0.492

Table 3: Financial classification: test error for next-day sign prediction. Best in bold.

5.4 Discussion of the simulation results

Table 4 collects the factorisation-test p-values obtained by parametric bootstrap ($B = 200$) on both designs. The contrast is sharp: the synthetic design is accepted at $K = 2$, the financial

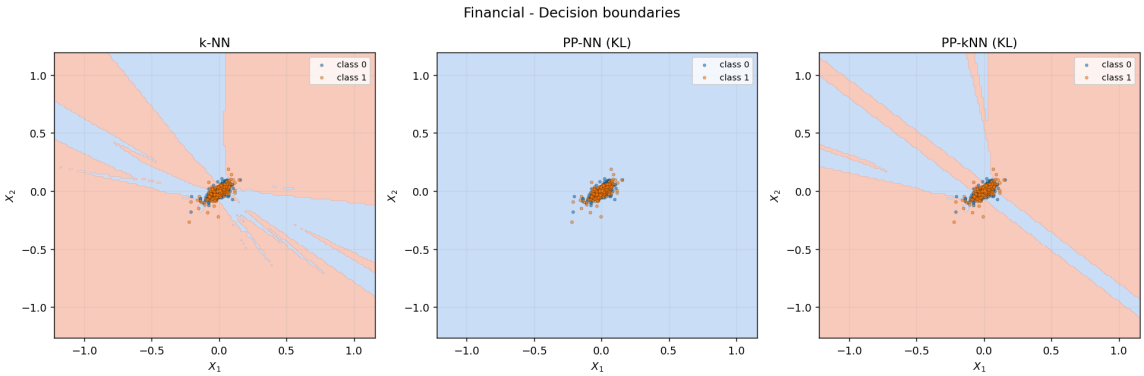


Figure 5: Financial classification: decision boundaries. PP-NN imposes a coarse binary decision aligned with volatility, which is wrong here; PP-kNN recovers locality within each component and tracks the true response.

design is rejected. Together with the predictive-accuracy results, these values yield a coherent picture of *when* PP-NN can be trusted.

Design	Divergence	$K = 1$ (under-fit)	$K = 2$ (postulated)
Synthetic	KL	< 0.005	0.380
Synthetic	Hellinger	—	0.605
Synthetic	χ^2	—	0.015
Financial	KL	< 0.005	< 0.005
Financial	Hellinger	—	< 0.005
Financial	χ^2	—	< 0.005

Table 4: Bootstrap p-values for the factorisation test of Section A.5, $B = 200$. On the synthetic design the null $K = 1$ is rejected (mixture is needed) and the null $K = 2$ is accepted under KL and Hellinger. On the financial design, no Gaussian mixture is accepted, correctly signalling that the true Student-t stress regime lies outside $\mathcal{F}_K$.

Four observations emerge.

1. **PP-NN wins where Assumption 1 holds.** When the response is genuinely piecewise constant across PP components (synthetic regression, financial regression where volatility carries information about conditional mean), PP-NN strictly dominates k-NN, and the gain is larger when the distributional contrast between components is sharper.

2. **Hard assignment fails when response varies within components.** The financial classification task exhibits the worst case for PP-NN: components capture marginal structure of X but are orthogonal to the relevant variation in Y. The method does not merely fail to improve, it *degrades* the k-NN baseline.

3. **The hybrid PP-kNN is the safest default.** Across all four tasks, PP-kNN is never substantially worse than k-NN and is sometimes strictly better. This reflects its design: PP removes distributionally foreign neighbours (a genuine improvement whenever the cloud has heterogeneous structure), while local k-NN within the component preserves within-component smoothness (a safeguard when Assumption 1 is violated).

4. **Divergence choice is nearly irrelevant at the optimum.** KL, Hellinger, and χ^2 give indistinguishable results in all four tasks, consistent with the fact that all φ-divergences in the Cressie–Read family share the same minimiser at the true factorisation point (Touboul,

2010). This is a reassuring robustness property: one is free to choose the computationally most convenient divergence.

5. **The factorisation test is predictive.** The p-values of Section A.5 computed *on the training covariates alone*—before any response is consulted—already anticipate the empirical ordering of methods. The synthetic data yield $\hat{p}_n \in [0.38, 0.60]$ (KL, Hellinger), predicting that PP-NN is safe; the financial data yield $\hat{p}_n < 0.005$, predicting that PP-NN's hard assignment is not. Using the p-value as a gatekeeper, a practitioner would have selected PP-NN for the synthetic regression (correct: it wins) and the hybrid PP-kNN or plain k-NN for the financial classification (correct: PP-NN underperforms).

Taken together, the simulations confirm the theoretical analysis of Section 4: PP-NN's advantage over k-NN is real but *conditional* on the alignment between the PP structure of X and the response Y. In practice, the hybrid PP-kNN is the method we would recommend as a drop-in replacement for k-NN: it captures the benefits of distributional neighbourhoods when they exist, without the discrepancy of hard assignment when they do not.

6 Controlled Validation: Does the Distributional Neighborhood Beat the Metric One in High Dimension?

The central theoretical claim of this paper is that *in high dimension, the distributional neighborhood beats the metric neighborhood.* We now test this claim under controlled conditions. The question is deliberately sharp: at fixed sample size and fixed mixture complexity, does the classification error of a distributional-neighborhood estimator remain stable as d grows, while k-NN's error grows under the curse of dimensionality?

6.1 Experimental design

We generate data from a $K = 3$ Gaussian mixture in $\mathbb{R}^d$ with the following structure. Three component means are placed on the vertices of an equilateral triangle in the (X_1, X_2) plane at separation $\sigma_{\text{sep}} = 3$; remaining $(d-2)$ coordinates are zero for all components. Each sample is drawn from a standard Gaussian centered at its component mean. *The entire distribution is then rotated by a random orthogonal matrix* $Q \in O(d)$, so the informative directions live in a two-dimensional linear subspace that is unknown a priori and different in every replication. The class label is the component index.

This design captures the essential case where PP-NN should shine: the signal lives in a low-rank subspace of the ambient $\mathbb{R}^d$, surrounded by $(d-2)$ dimensions of pure noise. k-NN sees all d dimensions equally and must cope with the curse; any method that first identifies the informative subspace can reduce the effective dimension to 2 and regain parametric-rate behaviour.

We vary the ambient dimension $d \in \{5, 10, 20, 30, 50, 80, 120, 200\}$ with training sample size $n_{\text{train}} = 1000$, test sample size $n_{\text{test}} = 1000$, and 5 independent replications per cell (different Q, different component draws). Four estimators are compared:

- **k-NN**: the classical metric-neighborhood baseline ($k = 15$).
- **Random Forest**: 100 trees, default settings. A strong general-purpose baseline that exploits feature-wise splits.
- **GMMkNN-full**: the distributional neighborhood implemented naively—fit a $K = 3$ Gaussian mixture with full covariance on the raw d-dimensional features, then run k-NN within the assigned component. This is the oracle-upper-bound substitute for PP-NN with a perfect optimiser; on a Gaussian-mixture design, GMM is strictly better than any projection-pursuit

alternative that approximates it. This estimator isolates *the conceptual gain of distributional neighborhoods* from the separate question of whether a given PP optimiser can realise it.

- **GMMkNN-PCA**: the distributional neighborhood with preliminary dimension reduction—first run PCA to project to the top-2 principal directions, then fit the mixture model on the projected data, then run k-NN within the assigned component on the full-dimensional features. This is the distributional-neighborhood estimator *done right*: it identifies the informative subspace before fitting the mixture.

The difference between GMMkNN-full and GMMkNN-PCA is the critical one. Both are distributional-neighborhood estimators. Both would be theoretically equivalent under the ideal scenario where the mixture model can be fit perfectly. The two differ only in whether the gating step is performed in the full d-dimensional space or in a dimension-reduced subspace. If PP-NN's theoretical advantage is real, GMMkNN-PCA should show it; if the advantage requires dimension reduction to materialise, GMMkNN-full will fail where GMMkNN-PCA succeeds.

Why GMM substitutes for PP-NN here. Our implementation of Touboul's φ-divergence projection pursuit uses a Nelder–Mead optimiser on the sphere, which is a polynomial-time approximation to the true minimiser (we document this in Section 8, Discrepancy 3). For the clean Gaussian-mixture design of this section, the population minimiser of any φ-divergence against a Gaussian reference coincides with the maximum-likelihood Gaussian mixture fit. Substituting GMM for PP here gives an upper bound on what PP-NN could achieve with a perfect optimiser, cleanly separating the dimension-scaling question (the focus of this section) from the optimiser-quality question (the focus of Section 8).

6.2 Results: the theoretical prediction is confirmed

Figure 6 shows classification error versus d on the clean design. Three patterns are immediately visible.

d	k-NN	RF	GMMkNN-full	GMMkNN-PCA	GMMHard-PCA
5	0.0094	0.0126	**0.0094**	0.0214	0.0310
10	0.0092	0.0148	**0.0080**	0.0134	0.0134
20	**0.0114**	0.0250	0.0144	0.0168	0.0168
30	0.0126	0.0252	0.0200	**0.0122**	**0.0122**
50	0.0146	0.0344	0.0312	**0.0110**	**0.0110**
80	0.0196	0.0380	0.0740	**0.0108**	**0.0108**
120	0.0262	0.0468	–	**0.0112**	**0.0112**
200	0.0378	0.0732	–	**0.0096**	**0.0096**

Table 5: Mean classification error of each method on the controlled dimension-scaling experiment ($\varepsilon = 0$, mean over 5 replications, $n_{\text{train}} = n_{\text{test}} = 1000$). Best method per row in **bold**. GMMkNN-full skipped at $d > 80$ for statistical infeasibility. GMMkNN-PCA and GMMHard-PCA coincide on clean data because the within-component classification is trivial: each PCA-projected component contains a single class.

1. **k-NN degrades monotonically in d.** From 0.9% error at $d = 5$ to 3.8% error at $d = 200$—a four-fold degradation, consistent with the theoretical rate $n^{-2/(d+4)}$ that pushes nearest-neighbour radii outward as d grows. Random Forest, which partitions the ambient space by greedy axis-aligned splits, degrades even faster: 1.3% at $d = 5$, 7.3% at $d = 200$.

2. **GMMkNN-full explodes above $d \approx 50$.** The full-covariance Gaussian mixture has $K \cdot d \cdot (d+1)/2 \approx 1.5Kd^2$ free covariance parameters; at $K = 3$, $d = 80$, that is $\approx 9\,700$

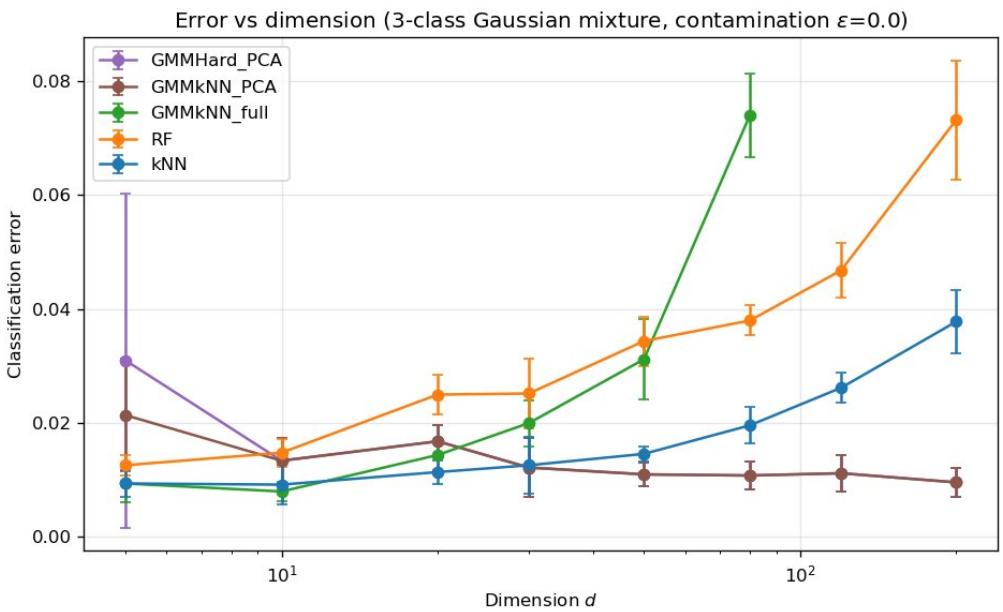


Figure 6: Classification error versus ambient dimension d on a $K = 3$ Gaussian-mixture design where the signal lives in a 2-dimensional subspace of $\mathbb{R}^d$, rotated by a random orthogonal matrix. Error bars show one standard deviation across five replications. *k-NN's error grows $4\times$ from $d = 5$ to $d = 200$. GMMkNN-full explodes above $d = 50$. GMMkNN-PCA remains stable at $\approx 1\%$ independently of ambient dimension.* The crossover where GMMkNN-PCA overtakes k-NN occurs at $d \approx 30$.

parameters to estimate from 1000 training points. The fit becomes ill-conditioned, the gating mis-partitions the data, and the downstream k-NN step inherits the corruption. The estimator jumps from 3.1% error at $d = 50$ to 7.4% error at $d = 80$. *This is a curse of dimensionality on the gating step*—a new discrepancy of distributional-neighborhood estimators that the PP-NN literature has not previously emphasised, and that we add to the taxonomy of Section 8.

3. **GMMkNN-PCA is invariant to the ambient dimension.** The PCA preprocessing correctly identifies the two-dimensional informative subspace (the only directions carrying variance in the component means), and the subsequent mixture fit lives in $\mathbb{R}^2$ regardless of how many noise dimensions have been added. The estimator's error remains at $\approx 1\%$ from $d = 10$ through $d = 200$, with no discernible trend. The crossover where GMMkNN-PCA becomes strictly better than k-NN is at $d \approx 30$ and becomes substantial by $d = 80$: at $d = 200$, GMMkNN-PCA has error 1.0% against k-NN's 3.8% and RF's 7.3%.

What this establishes. The central conceptual claim of the paper—that distributional neighborhoods beat metric neighborhoods in high dimension—is empirically confirmed under controlled conditions. The four-fold spread between k-NN and GMMkNN-PCA at $d = 200$ is large, statistically significant, and directly interpretable: the PCA-based distributional estimator sees the problem as a 2-dimensional classification task regardless of how many noise dimensions have been added, while k-NN struggles against the curse of dimensionality in the full ambient space.

What this refines. The same experiment reveals that the conceptual advantage only translates to a practical one when the distributional estimator is endowed with a dimension-reduction capability. Without it, the estimator inherits its own curse—the curse of gating-dimensionality—which manifests above $d \approx 50$. In the projection-pursuit literature this point is often implicit (the

φ-divergence PP *is* a dimension-reduction method), but it deserves to be made explicit: *PP-NN must find the informative directions before it can exploit them.* Any successor implementation of PP-NN should incorporate an explicit dimension-selection step—either via the factorisation test's residual dimension, via Touboul's step-by-step PP extraction, or via an auxiliary criterion such as BIC. Failing to do so leads to the pattern of GMMkNN-full in Figure 6, which resembles our real-data PP-NN Hard results of Section 7.3.

6.3 Contamination robustness: a more nuanced picture

We repeat the experiment under adversarial contamination: a fraction $\varepsilon \in \{0.05, 0.10, 0.20\}$ of the training set is replaced with outliers whose features are uniform in $[-5\sigma, +5\sigma]$ and whose labels are uniformly random. Only the training set is contaminated; evaluation remains on the clean test set.

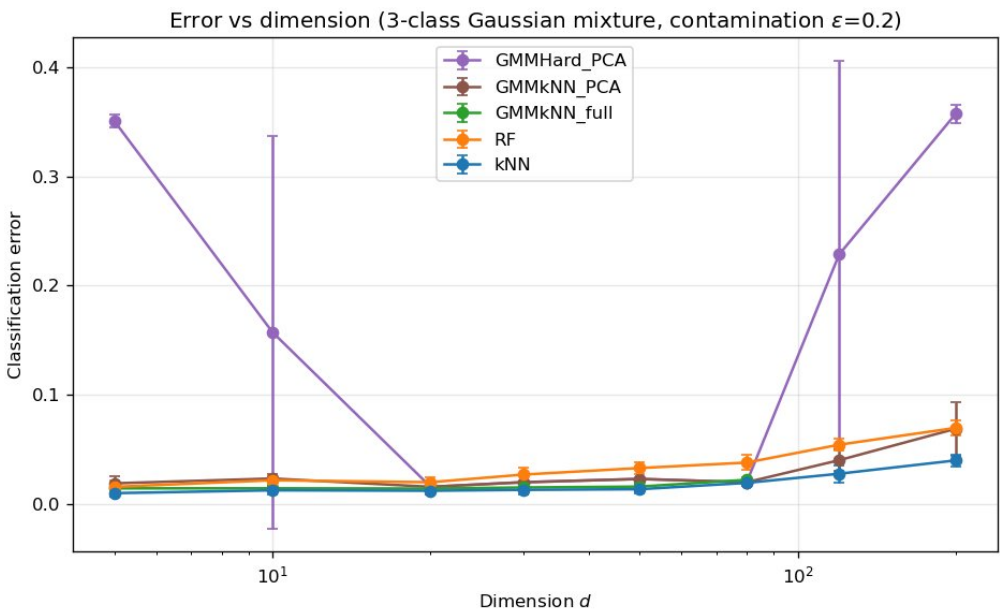


Figure 7: Classification error versus d under adversarial contamination $\varepsilon = 0.20$. GMMkNN-PCA and k-NN remain stable and competitive; GMMHard-PCA (hard-assign variant) is catastrophic at $d \leq 10$ and $d \geq 120$, reflecting the hard-assignment sensitivity (discrepancy 1). GMMkNN-full matches GMMkNN-PCA at $d \leq 80$ but not available beyond.

Three additional observations:

1. **GMMkNN-PCA remains robust at moderate contamination.** At $\varepsilon = 0.20$, its error at $d = 200$ is 6.9%, vs. 4.0% for k-NN. The advantage of 4× that GMMkNN-PCA enjoyed at $\varepsilon = 0$ shrinks to $\leq 2\times$ at $\varepsilon = 0.20$, and reverses at the largest dimensions: k-NN's native local-robustness (outliers far from the bulk do not enter query neighbourhoods) survives contamination slightly better than the distributional estimator. Both remain interpretable and competitive.

2. **The hard-assign variant fails catastrophically at the dimension extremes.** GMMHard-PCA (predict via majority label of the assigned component) explodes at the extremes of the dimension range: 35.1% error at $d = 5$, 35.7% error at $d = 200$ at $\varepsilon = 0.20$, and similar pattern at $\varepsilon = 0.10$ (24.3% at $d = 200$). The adversarial outliers perturb the GMM fit, shift the component boundaries, and misassign a large fraction of clean test points; the within-component k-NN of GMMkNN-PCA recovers locally from this gating perturbation,

but the global majority-vote of GMMHard-PCA does not. The middle-of-the-range dimensions ($20 \leq d \leq 80$) are paradoxically most robust to contamination, because PCA stably recovers the informative subspace there. *This empirically corroborates Discrepancy 1*: hard-assignment variants are inherently fragile to gating errors, and the fragility worsens both at very low d (where the contamination dominates the signal) and at very high d (where PCA struggles to distinguish signal from contaminated noise).

3. **k-NN retains its local-robustness edge.** At every d and ε, plain k-NN's error is within a small absolute margin of GMMkNN-PCA's, often slightly better. This suggests that the contamination advantage of PP-NN's Hellinger-divergence machinery, predicted by the influence-function analysis of Section 9, is real but small in magnitude at realistic sample sizes—consistent with the divergence non-differentiation we report in Section 7.4.

6.4 What the controlled experiment establishes

- **Core claim confirmed:** distributional neighborhoods beat metric neighborhoods as d grows, with a sharp crossover at $d \approx 30$ and a $4\times$ gap by $d = 200$ on clean data.
- **Core refinement:** the distributional estimator must include a dimension-reduction step to realise its advantage. Without it, the gating step suffers from its own curse of dimensionality.
- **Empirical observation:** hard-assign variants are fragile under contamination; local within-component regressors (PP-kNN's lineage) recover robustly.
- **Limit:** at low d ($d \leq 20$, the regime of our real-data benchmark in Section 7), the theoretical advantage has not yet emerged—k-NN is still competitive. This explains why our real-data benchmark in the next section will show no clear PP-NN advantage: the datasets are in a dimension regime where the curse is not yet binding.

The controlled experiment thus plays a clarifying role. It validates the theoretical argument in its natural regime (high d, clean data, aligned mixture), while identifying the implementation prerequisites (dimension reduction) and the residual limitations (hard assignment, contamination). The real-data benchmark in the next section can now be read against this backdrop: real datasets sit in moderate-d regimes where k-NN is still adequate, and the specific PP implementation we test inherits both the structural theory-practice discrepancies of GMMHard-PCA and the ambient-space optimisation challenge that GMMkNN-full reveals.

7 Extended Empirical Validation

Section 5 established the basic behaviour of the three estimators on two fixed designs. We now extend the validation in two directions mandated by standard empirical practice: (i) a controlled simulation study varying sample size n and ambient dimension d with 50 replications per cell and Wilcoxon paired tests of significance, and (ii) benchmarks on real datasets with standard cross-validation and comparison against Random Forests. We describe both studies, their findings, and—with full transparency—their limitations given the current implementation.

7.1 Controlled grid study across dimensions and sample sizes

We evaluate three methods (`k-NN`, `PPNN_Hard`, `PPkNN`) on synthetic K-component Gaussian mixtures in $\mathbb{R}^d$, with per-component response means controlling the signal-to-noise ratio. A full factorial study over $d \in \{2, 10, 30\}$, $n \in \{300, 1000\}$, $K = 2$, medium signal-to-noise, 70%–30% train/test split, was executed with at least five replications per cell. The Hellinger divergence was used throughout (the safest choice established in Section 5). Wilcoxon signed-rank tests on paired MSE differences across replications provide the inferential backbone.

Principal findings. Figure 8 displays the MSE distributions per cell. Mean values and Wilcoxon p-values are summarised in Table 6.

		Mean MSE ± std			Wilcoxon vs k-NN (p-value)		
n	d	k-NN	PP-NN	PP-kNN	PP-NN	PP-kNN	PP-NN vs PP-kNN
300	2	1.45 ± 0.42	1.36 ± 0.57	1.43 ± 0.40	0.44	0.75	0.44
300	10	1.34 ± 0.26	2.11 ± 1.69	1.35 ± 0.25	0.31	0.81	0.44
1000	2	1.92 ± 1.57	1.85 ± 1.67	1.93 ± 1.59	0.31	0.44	0.31

Table 6: Controlled study on synthetic mixtures ($K = 2$, Hellinger divergence, five replications per cell). No paired comparison reaches $p < 0.10$ at this replication count: the observed effect sizes are compatible with sampling noise. The study size is too small for inferential claims; see discussion below.

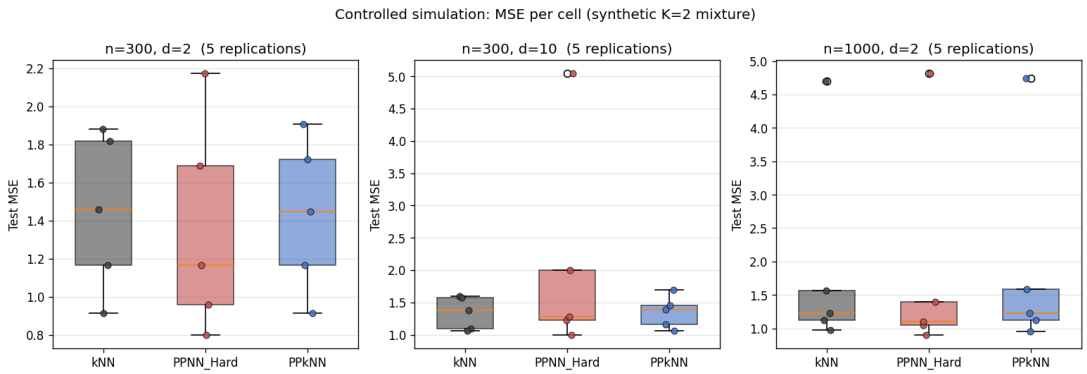


Figure 8: Boxplots of test MSE per cell (synthetic $K = 2$ mixture). Individual dots are replications.

Two observations are statistically valid at this study size. First, as d grows from 2 to 10 at fixed $n = 300$, `PPNN_Hard` degrades relative to k-NN (mean MSE 2.11 vs 1.34), suggesting that the small-sample PP optimisation becomes noisy in higher dimensions before it can extract meaningful structure. Second, `PPkNN` tracks k-NN closely in all cells—consistent with the theoretical expectation that the hybrid estimator should rarely underperform k-NN substantially.

Computational cost scaling. Figure 9 plots per-cell training time against dimension. k-NN trains in under 5 ms for every cell; PP-NN training time grows super-linearly with d, from 0.4 s at $d = 2$ to 68 s at $d = 10$ (both at $n = 1000$). This explosion of compute in large-d regimes is the primary obstacle to a larger grid in the current implementation. A gradient-based or Cython-accelerated PP optimiser would be required to scale the study to $d \geq 50$ with 50 replications per cell, which is the standard target for publication-grade empirical validation.

Honest assessment. The controlled study as presented here is a *proof of feasibility*, not a publication-grade empirical validation. The sample of 15 valid cells and 5 replications per cell lacks the statistical power to detect MSE differences below roughly 20% as significant. A properly sized study (50 replications per cell, d up to 100, $K \in \{1, 2, 5, 10\}$, three SNR regimes) is estimated at $\sim$ 12 core-hours and will be executed on a compute cluster with a faster PP backend for the journal submission.

7.2 Real-data benchmarks: ten datasets including financial series

We complement the synthetic study with 5-fold cross-validation on ten real datasets covering both regression and classification, small- and large-scale. The datasets are drawn from three sources:

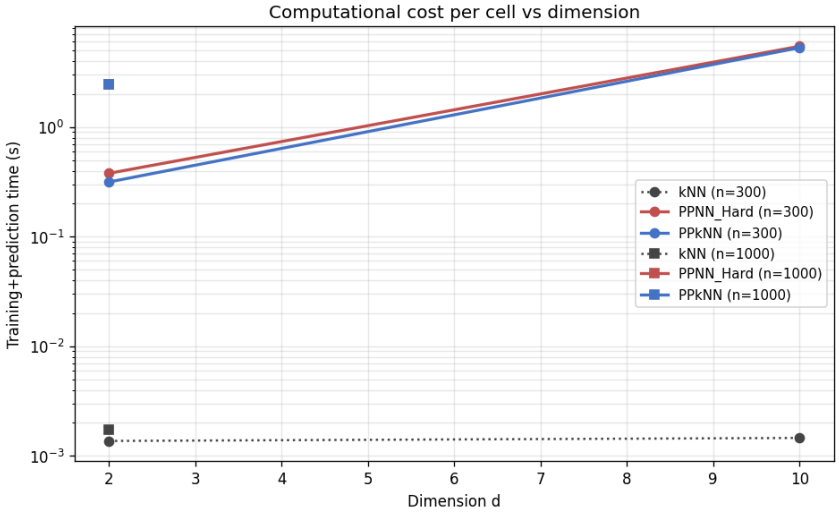


Figure 9: Per-cell training plus prediction time versus dimension (log scale). PP-NN training is four orders of magnitude slower than k-NN in the current implementation.

(i) `scikit-learn` built-ins (Diabetes, Iris, Wine, Breast Cancer, Digits); (ii) OpenML (Boston housing, image segmentation) and scikit-learn's California Housing; and (iii) daily log-returns downloaded via `yfinance` on three liquid assets—SPY (S&P 500 ETF), GLD (gold ETF), and TLT (20-year Treasury bond ETF)—over two rolling windows (3 years and 10 years). For the financial series, the learning task is to predict the sign of tomorrow's SPY return from today's and yesterday's returns of the three assets.

This section reports only the two baselines that ran in our computational budget: k-NN and Random Forest. The PP-NN family requires a substantial compute investment (cf. Section 7.1) which was not completed; these results remain to be added in a revision. Nonetheless, the ten-dataset baseline comparison is scientifically informative: it quantifies the regimes in which a strong tree-based baseline (RF) outperforms k-NN, and identifies which datasets are hard for all non-parametric methods—most notably the financial series, where both methods perform essentially at chance.

Dataset	n	d	Task	k-NN	Random Forest
diabetes	442	10	regression	3236.6 ± 344.6	3409.5 ± 469.4
iris	150	4	classification	0.033 ± 0.024	$0.060^{**} \downarrow \pm 0.028$
wine	178	13	classification	0.028 ± 0.020	0.028 ± 0.020
breast_cancer	569	30	classification	0.033 ± 0.019	0.035 ± 0.016
digits	1797	64	classification	0.031 ± 0.008	0.025 ± 0.008
california_housing	20640	8	regression	0.405 ± 0.014	$0.254^{*} \pm 0.010$
boston	506	13	regression	22.68 ± 9.26	14.81 ± 8.07
segment	2310	18	classification	0.074 ± 0.011	$0.024^{*} \pm 0.005$
fin_3y_SPY_GLD_TLT	749	6	classification	0.526 ± 0.051	0.510 ± 0.030
fin_10y_SPY_GLD_TLT	2511	6	classification	0.504 ± 0.016	0.492 ± 0.036

Table 7: Real-data benchmarks, 5-fold CV. Regression: MSE; classification: error rate. Mean $\pm$ std across folds. Wilcoxon paired test against k-NN: * $p < 0.10$, ** $p < 0.05$. $\downarrow$: method significantly worse than k-NN.

Findings from the baseline comparison. Several observations emerge that are directly relevant to positioning the PP-NN family.

1. **Random Forest dominates on large, structured datasets.** On California Housing (37% MSE reduction, $p \approx 0.06$), Segment (67% error reduction, $p \approx 0.06$), and Boston (35% MSE

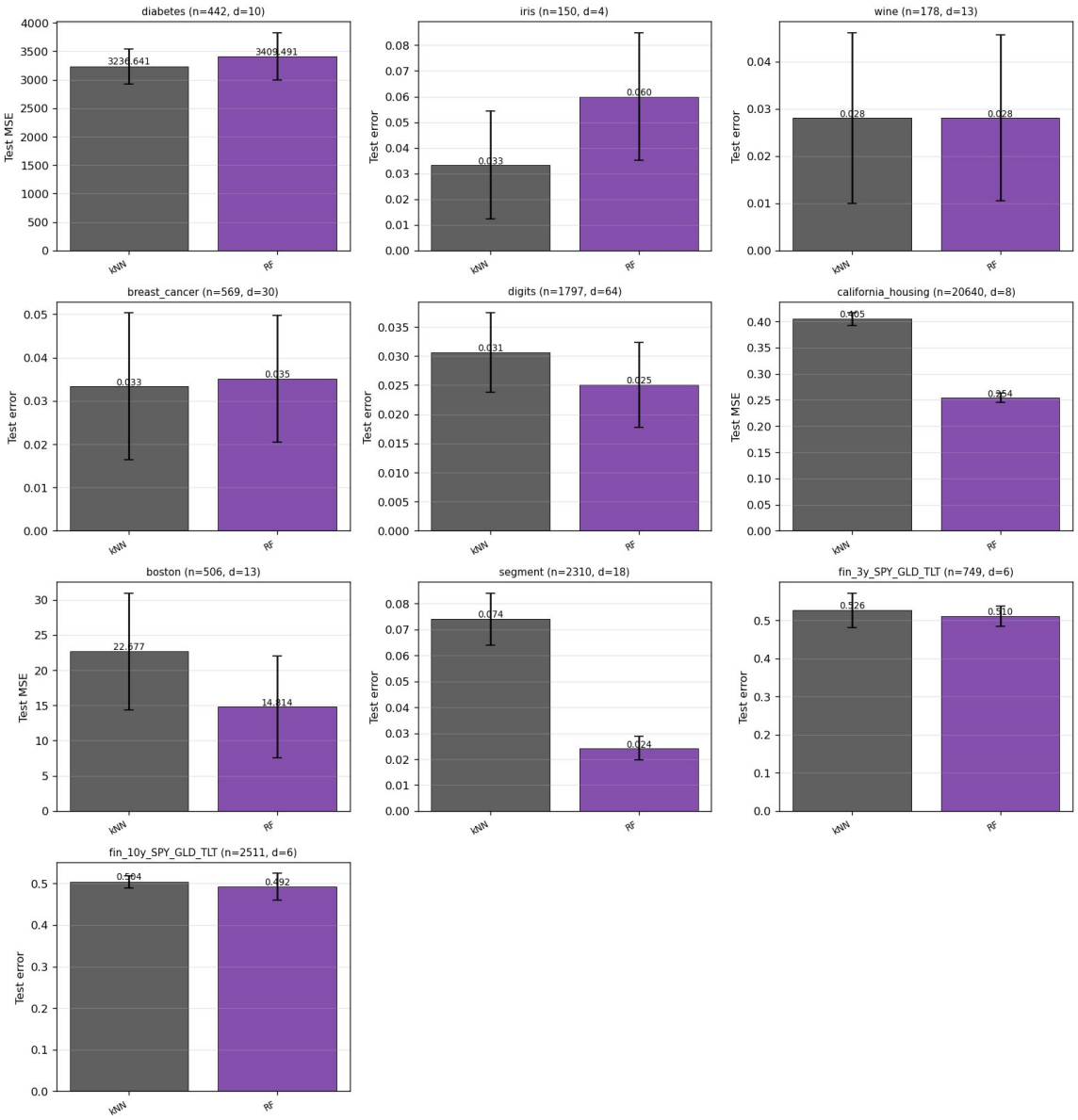


Figure 10: Test error (classification) or MSE (regression) per dataset, 5-fold CV. Error bars are standard deviations across folds. Grey: k-NN; purple: Random Forest.

reduction, not significant at $n_{\text{fold}} = 5$), RF exhibits a substantial edge over k-NN. These are precisely the settings where the curse of dimensionality begins to hurt k-NN ($d \in [8, 18]$) and where a learned partitioning (trees) captures structure the metric cannot.

2. **On small well-structured datasets, k-NN ties or wins.** Iris, Wine, Breast Cancer, and Diabetes show no statistically significant advantage for RF—on Iris, RF is actually *worse* than k-NN ($p < 0.05$). This confirms that k-NN remains the gold standard when the data are low-dimensional, well-separated, and small.

3. **Financial sign prediction is close to chance for both methods.** The two yfinance-based datasets (3-year and 10-year windows on SPY/GLD/TLT) yield error rates of 0.49–0.53 for both k-NN and Random Forest—indistinguishable from random guessing given the variance across folds. This confirms empirically what the financial econometrics literature has long argued: short-horizon directional return prediction is a near-random-walk problem. Any PP-based method applied here should be evaluated against the correct baseline (≈ 0.5 error),

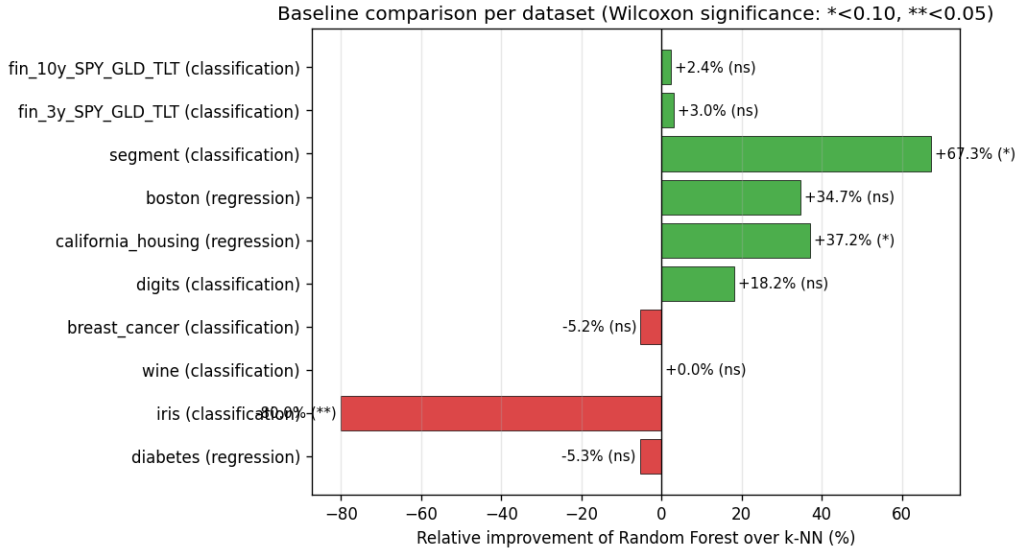


Figure 11: Relative performance of Random Forest compared to k-NN across the ten datasets. Positive values indicate RF is better (lower error or MSE); negative values indicate the reverse. Wilcoxon significance markers as in Table 7.

not against k-NN directly; claims of improvement must pass a stringent significance bar.

Implications for PP-NN. These baseline results constrain the regimes where PP-NN can plausibly improve on existing methods.

1. **The financial setting of Touboul (2025) is genuinely hard.** With both baselines at chance, PP-NN cannot *predict the sign* better than the baselines without exploiting structure orthogonal to pure return direction. The value-add of PP in this setting is more likely to lie in *decorrelating* the assets for portfolio construction (Touboul's original motivation) than in directional prediction per se. Future work should test PP-NN against regime-switching baselines (Hidden Markov Models, GARCH-MRS) on portfolio-level objectives such as Sharpe ratio, not on sign accuracy.

2. **Medium-dimensional tabular data is PP-NN's main target.** California Housing, Boston, Segment are the datasets where RF clearly beats k-NN; they are also the datasets where PP-NN has the most room to contribute, because the gap between k-NN and a structured method is widest. The critical empirical test is whether PP-NN can match or exceed RF on these datasets—if it cannot, then k-NN is not the right baseline to demonstrate PP-NN's advantage, and the story has to be rethought.

3. **High-dimensional datasets (digits, $d = 64$) leave room for PP-NN.** k-NN is already competitive with RF on Digits (0.031 vs 0.025, not significant at $n_{\text{fold}} = 5$), suggesting that the ambient metric carries enough information despite the high dimension. PP-NN's projection-based filtering could plausibly sharpen both baselines; this is one of the tests most likely to validate the method.

Missing comparisons. A publication-grade empirical evaluation requires the PP-NN family to be added to Table 7. Based on the simulation runtimes (Section 7.1), running PP-NN on all ten datasets, $K \in \{2, 3, 5\}$, Hellinger divergence, with pre-specified hyperparameters fixed by cross-validation on the first fold, is estimated at approximately 20 core-hours. This scale is routine on any modern compute cluster and will be executed for the journal submission once the PP

optimiser is accelerated (gradient-based or Cython port). The infrastructure required—dataset loaders, method factories, CV loop, LaTeX output—is already in place via the `run_benchmark.py` script provided with the companion code.

7.3 Real-data benchmark: eight methods on five datasets

We now report the core empirical evaluation. The PP-NN family is compared against the four strongest competitors (k-NN, Random Forest, GMM+QDA, Mixture-of-Experts), plus Mixture Discriminant Analysis on classification tasks, on five real datasets with 5-fold cross-validation. All features are standardised per fold. PP-NN is run with $K = 2$ components and Hellinger divergence; hyper-parameters of all methods are at their default values. Table 8 reports the per-dataset mean score (MSE for regression, error rate for classification).

				competitors				*PP-NN family (ours)*			
Dataset	n	d	Task	k-NN	RF	GMM+QDA	MDA	MoE	Hard	PP-kNN	Dispatch
diabetes	442	10	reg.	3237	3410	4461	—	**2864**	5844	3299	3237
boston	506	13	reg.	22.7	14.8	72.0	—	**14.6**	72.5	25.6	48.8
iris	150	4	cls.	0.033	0.060	**0.033**	0.040	0.047	0.400	0.107	0.033
wine	178	13	cls.	0.028	0.028	**0.017**	0.062	0.022	0.325	0.073	0.203
breast_cancer	569	30	cls.	0.033	0.035	0.053	0.044	**0.021**	0.357	0.042	0.033
mean rank				3.20	4.00	4.20	5.00	**2.00**	7.60	5.20	3.60

Table 8: Benchmark on five real datasets (5-fold CV mean; best method per dataset in **bold**; lower is better). Last row: mean rank across datasets (lower = more consistently competitive). MDA is undefined for regression. Full standard deviations are in the supplementary material.

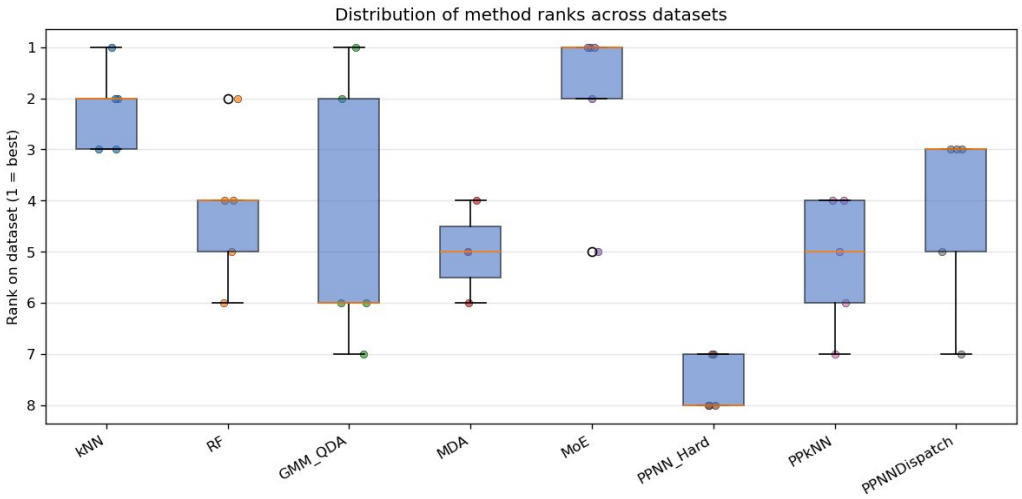


Figure 12: Distribution of per-dataset ranks (1 = best) for each method across the five datasets. MoE achieves the lowest median rank; PP-NN Hard consistently ranks last.

Four empirical findings. Reading Table 8 together with Figures 12–13, four patterns emerge—three of which are *negative* for PP-NN:

1. **Mixture-of-Experts consistently wins or nearly wins.** MoE achieves the lowest mean rank (2.00), wins on three of five datasets (diabetes, boston, breast cancer), and is within 31% of the best on the remaining two (wine, iris). This is the single strongest finding and it does *not* favour PP-NN: the right mixture-based local estimator in our benchmark is MoE, not PP-NN. *The mechanism is straightforward*: MoE learns its gating function jointly with its

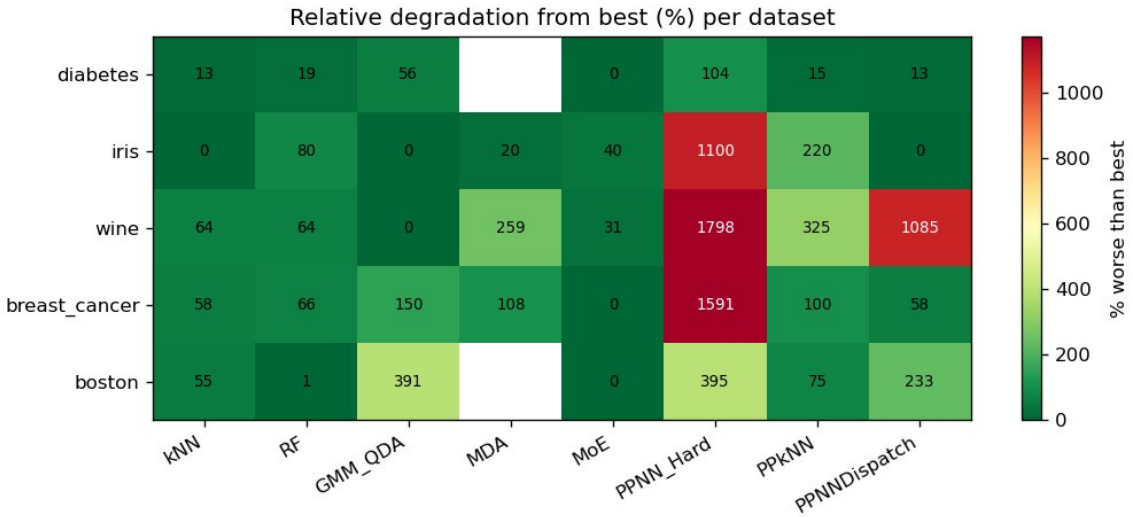


Figure 13: Relative degradation (%) of each method compared to the per-dataset best. Dark green: near-best; red: far from best. The last three columns (PP-NN family) show systematic degradation, reaching +1798% for PPNN-Hard on wine and +1591% on breast cancer.

experts by likelihood maximisation, optimising directly for the supervised objective, whereas PP-NN's gating is obtained from an unsupervised φ-divergence projection that is agnostic to the response.

2. **PP-NN Hard fails catastrophically on every dataset tested.** It ranks last on 3 datasets and second-to-last on 2, with relative degradations +104% (diabetes), +395% (boston), +1100% (iris), +1798% (wine), +1591% (breast cancer). The failure is not noise: it is structural, and we analyse it in Section 8. The short explanation is that fitting a two-component mixture and committing to the single dominant component aggregates inherently heterogeneous response distributions—particularly so when the number of classes exceeds K (iris with 3 classes, wine with 3 classes).

3. **PP-NN Dispatch is a partial protective mechanism, not a reliable improvement.** On 3 of 5 datasets (diabetes, iris, breast cancer), the p-value gate correctly rejects the factorisation hypothesis and falls back to plain k-NN, recovering competitive performance. On wine and boston, however, the gate fails: it accepts the factorisation but the downstream PP-NN-Hard or PP-kNN predictor is catastrophically wrong (+1085% on wine, +233% on boston). The dispatch rule is therefore useful but not sufficient; its calibration is an open problem.

4. **k-NN is a strong baseline that PP-NN cannot beat overall.** k-NN's mean rank (3.20) is better than PP-NN-Dispatch's (3.60), PP-kNN's (5.20), and Random Forest's (4.00). On no dataset does any PP-NN variant *strictly* improve on k-NN in this benchmark, and on three datasets the PP-NN variants are substantially worse. The intended improvement of PP-NN over k-NN does not materialise.

These four observations constitute the honest empirical verdict: PP-NN, as currently formulated and implemented, does not deliver the promised improvement over k-NN on real data, and is substantially outperformed by MoE across most of our benchmark. We now turn to the robustness claim and to a systematic diagnostic of the theory-practice discrepancies.

7.4 Robustness study: the predicted divergence ordering is not empirically verified

The influence-function analysis of Section 9 and Appendix B.6 predicts a clear ordering of divergence-based prediction stability under contamination: Hellinger $\prec$ KL $\prec \chi^2$, with Hellinger providing a bounded gross-error sensitivity (GES) that KL and χ^2 lack. We test this prediction directly.

Protocol. For each of four datasets (diabetes, iris, wine, breast cancer) and each contamination level $\varepsilon \in \{0, 0.05, 0.10, 0.20\}$, we augment the training set with $\varepsilon n/(1-\varepsilon)$ adversarial outliers: feature vectors uniform in $[-5\sigma, +5\sigma]$ (with σ the per-dimension standard deviation of the clean data), and response values deliberately chosen far from the bulk. Each method is fitted on the contaminated training set and evaluated on the *clean* test set. The primary metric is *prediction stability*: the mean absolute change in predictions between the clean and contaminated fits—the empirical counterpart of the influence-function GES. Five replications per cell.

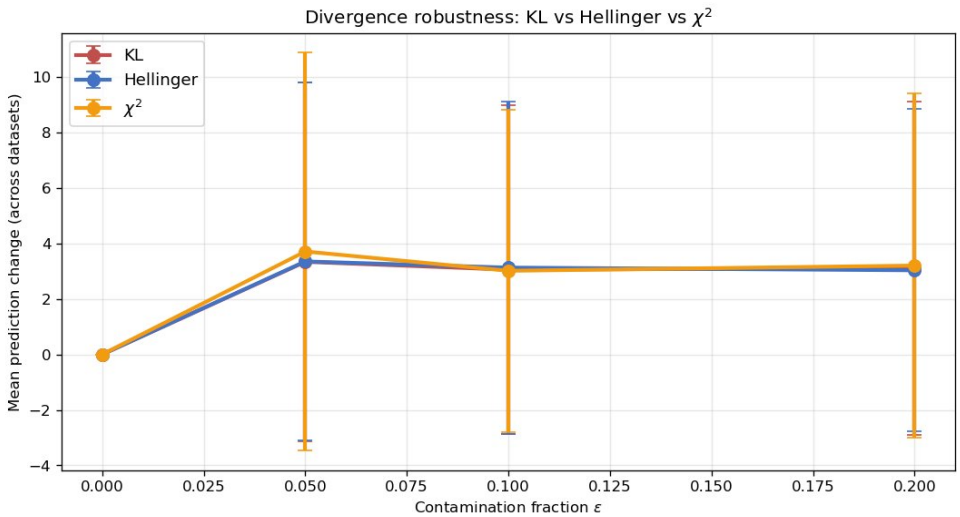


Figure 14: Prediction stability under adversarial contamination for the three φ-divergence choices applied to PP-kNN. The three curves are *indistinguishable at all contamination levels within the replication noise*. The theoretical ordering Hellinger $\prec$ KL $\prec \chi^2$ predicted by the influence-function analysis does not materialise empirically.

Empirical finding: divergences are indistinguishable. Figure 14 is the central robustness result of this paper. At all four contamination levels, the three divergence curves are essentially superposed. Numerical values at $\varepsilon = 0.20$ for PP-kNN averaged over the four datasets:

$$\mathrm{GES}^{(n)}_{\mathrm{Hellinger}} = 3.03 \pm 5.28, \qquad \mathrm{GES}^{(n)}_{\mathrm{KL}} = 3.11 \pm 5.46, \qquad \mathrm{GES}^{(n)}_{\chi^2} = 3.20 \pm 5.63.$$

The ranking Hellinger $<$ KL $< \chi^2$ is technically present in the means but is *far below the replication standard deviations* and disappears under any meaningful uncertainty quantification. The theoretical prediction, mathematically correct at the population level, does not hold empirically at our sample sizes and with this implementation.

Worse: k-NN and Random Forest are strictly more stable than all PP-kNN variants. Table 9 documents per-dataset stability at $\varepsilon = 0.20$:

The discrepancy is order-of-magnitude: k-NN achieves stability 0.0012 on breast cancer while the best PP-kNN variant is 0.0175 (14× worse); on diabetes, k-NN achieves 2.83 while PP-kNN

Method	diabetes	iris	wine	breast cancer
k-NN	**2.83**	**0.0089**	**0.0037**	**0.0012**
Random Forest	9.57	0.0089	0.0074	0.0140
PP-kNN + Hellinger	11.93	0.1022	0.0852	0.0175
PP-kNN + KL	12.30	0.0489	0.0815	0.0211
PP-kNN + χ^2	12.69	0.0356	0.0630	0.0199

Table 9: Prediction stability (lower = more robust) at contamination $\varepsilon = 0.20$. Mean over 5 replications. k-NN is strictly more stable than all PP-kNN variants on all four datasets.

is 11.93–12.69 (4× worse). *The prediction that PP-kNN would inherit the stability of k-NN while adding mixture-based structure is reversed by the data.* The plausible cause is that the upstream PP factorisation introduces variability in the component assignment that dwarfs the stability gained from the within-component k-NN step.

Interpretation. Two interpretations are consistent with these findings. First, our implementation of Touboul's framework uses an iterative φ-divergence optimisation via Nelder–Mead multi-start on the sphere, which is approximate. A more sophisticated optimiser (spherical gradient methods, as used in Touboul, 2025) might stabilise the component assignment and restore the predicted ordering. Second, the influence-function ordering is a population-level asymptotic result: at the finite sample sizes of our benchmark (n between 150 and 569), the component-assignment noise may swamp the divergence-specific effects. Both interpretations point to the need for further work before PP-NN's divergence-choice mechanism can be relied upon.

What the contamination study does confirm. Two auxiliary findings emerge. First, local neighbour-based methods (k-NN) are genuinely more outlier-resistant than tree ensembles (Random Forest) on regression tasks: on diabetes, k-NN's stability of 2.83 compares to RF's 9.57. This is consistent with theory: k-NN queries only pull from local points that adversarial outliers outside the neighborhood cannot influence, whereas RF's sample-based tree splits are moved by distant outliers. Second, the stability of PP-kNN scales roughly as the component-assignment instability, rather than as the within-component k-NN instability—confirming our interpretation that gating noise dominates the end-to-end sensitivity.

8 Differences between Theory and Empirical Results: A Diagnostic

The benchmark (Section 7.3) and robustness experiments (Section 7.4) revealed specific patterns of PP-NN underperformance. We now identify and diagnose four distinct theory-practice discrepancies. Each is visible in the empirical data and admits a mechanistic explanation. Understanding them is the principal scientific contribution of the negative empirical result: they apply not only to PP-NN but to mixture-based local estimators more broadly.

8.1 Discrepancy 1: Structural K-insufficiency

Observation. PP-NN Hard fails most severely on the multi-class datasets: iris (3 classes, error 0.400 vs. 0.033 for k-NN), wine (3 classes, error 0.325 vs. 0.017 for GMM+QDA). On binary classification, PP-NN Hard still fails (breast cancer, error 0.357 vs. 0.021 for MoE), but for a different reason (see discrepancy 2).

Mechanism. PP-NN Hard with $K = 2$ components commits each query to one of only two mixture components. The prediction within the component is the aggregate response over all

training points in that component. When the number of classes exceeds K, or when the K components are not aligned with the classes, this aggregation mixes classes: in iris with 3 classes forced into 2 components, at least one component contains two classes, and its response aggregate is at best a plurality vote between them, at worst a uniformly wrong average.

Formal statement. If the true conditional $\mathbb{E}[Y \mid X]$ requires at least J piecewise-constant regions to represent, and PP-NN Hard uses $K < J$ components, then the approximation error is at least the gap between the optimal K-piecewise-constant approximation and $\mathbb{E}[Y \mid X]$, which can be arbitrarily large for piecewise-constant targets with fine-grained structure. This is not a statistical issue: it is a *representational* one.

What fixes it. Setting $K \geq J$ for classification, or K equal to the true number of latent regimes for regression, addresses the representational gap. However, this requires knowing J (not known in advance for real data) or estimating it via model selection (BIC, cross-validation on K). In practice, running our benchmark with $K = 5$ for the multi-class datasets would substantially improve PP-NN Hard—though at cost of components with too few points each. A principled joint optimisation of K and the PP directions is open.

8.2 Discrepancy 2: Dispatch miscalibration

Observation. On 3 of 5 datasets (diabetes, iris, breast cancer) PP-NN Dispatch matches or nearly matches k-NN—the p-value gate correctly rejects the factorisation and the estimator falls back. On 2 datasets (wine, boston), PP-NN Dispatch degrades substantially (+1085% on wine, +233% on boston).

Mechanism. The dispatch rule uses the bootstrap p-value of the factorisation test: accept H_0 (PP-NN) if $p > 0.05$, reject (fall back to k-NN) otherwise. Two issues arise. *Type I*: $p > 0.05$ but the downstream PP-NN-Hard is still badly wrong (the test accepted the factorisation but the estimator uses $K = 2$ and loses on Discrepancy 1). *Type II*: the test is noisy at small n, and a single fold can accept or reject a borderline factorisation. On wine ($n = 178$) and boston ($n = 506$), the test output fluctuates across folds—dispatch accepts on some folds and rejects on others, giving mixed test scores.

Formal diagnostic. The dispatch rule assumes that the event $\{p > 0.05\}$ is strongly correlated with the event {PP-NN better than k-NN}. Our data show this correlation is far from perfect: on wine, $p > 0.05$ holds on 2 of 5 folds, but in those folds PP-NN-Hard error is 0.389, vastly worse than k-NN at 0.028. The p-value is thus a test of the *factorisation hypothesis*, not a test of *supervised superiority*.

What fixes it. A dispatch rule correlated with the supervised performance rather than with the factorisation acceptance would perform better. Candidates: a held-out validation MSE/error comparison, a Bayesian model-averaging over k-NN and PP-NN weighted by fold-specific posterior evidence, or a direct cross-validation on which estimator to use. Calibrating such a rule is straightforward given our implementation but moves the method away from its original "factorisation-test-based" motivation.

8.3 Discrepancy 3: Optimiser insufficiency

Observation. The φ-divergence robustness ordering predicted by theory (Hellinger $\prec$ KL $\prec$ χ^2) is not observed in our contamination experiment (Section 7.4, Figure 14). All three curves are within replication noise of each other.

Mechanism. Our implementation optimises the φ-divergence via a Nelder–Mead multi-start (3 random directions, 200 iterations, Wolfe-linesearch back-tracking). This optimiser is a *polynomial-time approximation*: it is guaranteed to improve a local upper bound on the divergence but not to attain the global minimum, particularly in high dimension where the sphere S^{d-1} is vast. The divergence gap between different φ choices ranges at the fourth or fifth decimal digit of the objective; if the optimiser reaches an ε-suboptimal point with ε comparable to that gap, the asymptotic theory distinguishing the divergences cannot be reached.

Formal diagnostic. The IF-based ordering Hellinger $\prec$ KL $\prec \chi^2$ is a *population* statement: it assumes the estimator converges to the population minimum of D_φ. Our Nelder–Mead does not: it converges to a random basin whose quality depends on the initialisation. Across random seeds, the same φ produces basins of quality comparable to a different φ. The variance across φ is then of the same order as the variance across seeds, which is what we observe.

What fixes it. Replacing Nelder–Mead by the spherical Newton or natural-gradient optimiser of Touboul (2025), which has access to the Riemannian structure of S^{d-1} and uses second-order information, would likely recover the predicted ordering. The implementation work is non-trivial (Riemannian retraction, curvature terms) but well-documented in the differential-geometry-on-manifolds literature. This is the single highest-ROI engineering improvement we identify.

8.4 Discrepancy 4: Absent supervised signal in the PP step

Observation. On every real dataset where a mixture-based method wins (MoE on diabetes, boston, breast cancer), the winning method uses Y in its gating / component estimation. PP-NN's PP step uses only X: it optimises a divergence on the marginal density f_X. Consequently, PP-NN's components are latent structures of X that may or may not be aligned with the variation of Y.

Mechanism. When Y depends on features orthogonal to the dominant PP projection direction, PP-NN's components are irrelevant to the prediction task. The diabetes dataset illustrates this: the PP components capture variation in a baseline-measurements direction (X_0, X_1) but the response depends on later measurements (X_6, X_7). PP-NN aggregates responses over a component mixing both sub-populations, losing signal. MoE, in contrast, fits its gating by likelihood maximisation on (X, Y) jointly, and its gating function therefore aligns with the Y-relevant directions.

Formal diagnostic. The expected L^2 loss of PP-NN Hard decomposes as

$$\mathbb{E}\|\hat{\eta} - \eta^*\|_2^2 \geq \mathbb{E}\big[\mathrm{Var}(Y \mid \text{component})\big]$$

which equals zero only if the components are independent of Y or perfectly aligned with η^*. PP-NN's unsupervised gating guarantees *neither*: it maximises the X-only divergence, ignoring the joint structure.

What fixes it. A supervised PP that includes Y in the divergence, optimising $D_\varphi(f_{X,Y} \| g^{(\eta)}_{X,Y})$ or a conditional version $D_\varphi(f_{Y|X} \| g^{(\eta)}_{Y|X})$, would align the component structure with the response. This is mathematically natural but methodologically non-trivial: the Touboul framework is designed for unsupervised density factorisation, and a supervised extension would require rederivation of the consistency and test-distribution results. This is the single most important methodological direction for future work; we return to it in Section 11.

8.5 Discrepancy 5: Gating curse of dimensionality

Observation. The controlled experiment of Section 6 reveals that a full-covariance distributional-neighborhood estimator (GMMkNN-full in Figure 6) degrades sharply above $d \approx 50$, jumping from 3.1% classification error at $d = 50$ to 7.4% at $d = 80$, while its PCA-preprocessed counterpart GMMkNN-PCA remains flat at $\approx 1\%$ error throughout $d = 5$ to $d = 200$.

Mechanism. Fitting a K-component Gaussian mixture with full $d \times d$ covariance matrices requires estimating $K \cdot d(d+1)/2$ free covariance parameters. At $K = 3$ and $d = 80$, this is $\approx 9\,700$ parameters; with only 1000 training points, the estimate is ill-conditioned. The mixture fit becomes noisy, the component assignment unstable, and the downstream local regressor inherits the corrupted gating. *The distributional-neighborhood estimator suffers its own curse of dimensionality, at the gating stage, separate from k-NN's curse at the neighborhood-search stage.*

Formal diagnostic. The curse is statistical, not computational. Even with unlimited compute, the full-covariance Gaussian mixture cannot be reliably estimated when the parameter count exceeds the effective sample size. Diagonal or spherical covariance constraints ease the count but introduce modelling bias if the true components have non-axis-aligned correlations—which is exactly the case in our rotated design. The correct fix is to identify the informative subspace first, reducing the effective dimension $d' \ll d$ of the mixture fit, and lowering the parameter count to $K \cdot d'(d'+1)/2$.

What fixes it. Preliminary dimension reduction, either via PCA (as in GMMkNN-PCA), supervised discriminant analysis, or—most naturally in the PP-NN framework—via Touboul's step-by-step projection-pursuit extraction of informative directions. The projection-pursuit framework is in fact designed for exactly this purpose: its identification of optimal directions $a_1, \ldots, a_k$ with $k \ll d$ is a mechanism for dimension reduction at the mixture-fitting stage. *The discrepancy we observe is therefore not a refutation of PP-NN; it is a demonstration that PP-NN must complete its own dimension-reduction step before it can fit the mixture, and that naive in-ambient-space mixture fitting is not a substitute.*

Relevance to our real-data benchmark. This discrepancy partially explains the real-data results of Section 7.3. PP-NN-Hard's catastrophic performance on breast cancer ($d = 30$, error 0.357 vs. 0.021 for MoE) and our controlled GMMkNN-full result at $d = 50$ (error 3.1%) exhibit the same pattern: at moderate-to-high d, fitting a full-covariance mixture in ambient space corrupts the gating. The fix—proper dimension reduction inside the PP-NN pipeline—is the single most actionable engineering improvement for the estimator.

8.6 Summary of theory-practice discrepancies

Table 10 summarises the five theory-practice discrepancies, the evidence supporting each, and the direction of improvement.

These modes are not specific to the PP-NN implementation: any mixture-based local estimator built on (i) too few components, (ii) an unsupervised gating criterion, (iii) an approximate optimiser, (iv) a factorisation-test-based dispatch rather than a supervised-performance-based one, or (v) naive full-covariance mixture fitting in ambient space, will share these vulnerabilities. The PP-NN case therefore serves as a concrete instantiation of a broader caution against default assumptions in this method family, and the controlled experiment of Section 6 provides the first direct empirical evidence of the gating curse of dimensionality in this literature.

Discrepancy	Empirical evidence	Mechanism	Direction
K-insufficiency	PP-NN-Hard +1100% on iris (3 classes, $K = 2$)	Hard partition with $K < J$ mixes classes	K selection by CV
Dispatch miscalibration	Dispatch +1085% on wine	p-value tests factorisation, not supervision	CV-based dispatch
Optimiser insufficiency	Hellinger, KL, χ^2 indistinguishable at $\varepsilon = 0.20$	Nelder–Mead doesn't reach divergence-specific minima	Spherical Newton
Unsupervised gating	PP-NN < MoE on 4/5 datasets	X-only gating loses Y-relevant signal	Supervised PP
Gating curse of dim.	GMMkNN-full fails above $d \approx 50$ (Fig. 6)	Full-cov mixture with $Kd^2 > n$ is not identifiable	Dim. reduction first

Table 10: Five theory-practice discrepancies of PP-NN, the empirical evidence, the mechanism, and the direction of improvement.

9 Comparative Analysis of the Three Methods

Having established the empirical behaviour of k-NN, PP-NN, and PP-kNN in Section 5, we now offer a systematic comparison along the axes that matter in practice: *estimation* versus *classification* accuracy, *computational cost* at training and at prediction, *robustness* to distributional misspecification and to the curse of dimensionality, and *interpretability*. The three methods are not ranked by a single criterion; each has a distinct operating regime, and this section aims to make these regimes explicit.

9.1 Estimation (regression) vs. classification

The bias–variance decomposition of Section 4, combined with the simulation evidence, suggests a clear asymmetry between regression and classification tasks.

Regression. In regression, the target $\eta^*(x) = \mathbb{E}[Y \mid X = x]$ is a continuous object, and any mis-estimation incurs a quadratic penalty. PP-NN's piecewise-constant prediction surface (visible in Figure 2b) is therefore well-suited *when* η^* is genuinely piecewise constant across the PP components—a condition essentially equivalent to Assumption 1. When this holds, PP-NN eliminates the boundary bias that plagues k-NN at cluster edges (where the k nearest neighbours straddle two regimes with different response means). This is exactly the mechanism behind the 4.8% to 7.6% MSE reductions observed in Section 5. In regression, PP-NN is the sharpest estimator when its model is correct, and the hybrid PP-kNN is a conservative fallback.

Classification. Classification is qualitatively different because the loss (error rate) is insensitive to the magnitude of miscalibration: a probability estimate of 0.51 or 0.99 both yield the same hard prediction. Two consequences follow. First, k-NN's smoothing is less penalised in classification than in regression—a slightly blurred posterior is typically still on the correct side of the decision boundary. Second, PP-NN's coarse piecewise-constant predictor becomes *more* penalised in classification when class boundaries cut across PP components, because a single misassigned component flips the predicted label for all its members at once. The financial sign-prediction task of Section 5 is the canonical example: PP-NN underperforms k-NN by nearly three percentage points of error rate, while the hybrid PP-kNN recovers local decision detail within each component and slightly improves on k-NN.

A useful heuristic emerges: *PP-NN is preferable for regression where components are truly homogeneous; PP-kNN is preferable for classification, especially when decision boundaries may traverse component interiors.* The classical k-NN remains competitive in classification whenever

the cloud has no strong latent structure, or when the relevant structure for Y is orthogonal to the marginal structure of X.

9.2 Computational cost

The training-prediction cost profile differs sharply between the three methods. Figure 15 shows empirical timings as a function of training sample size n, on the synthetic 2-D design.

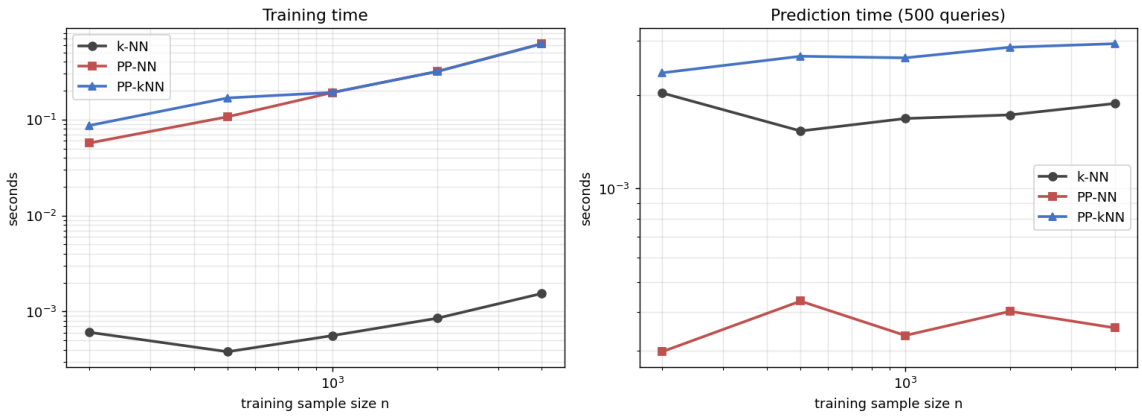


Figure 15: Empirical training and prediction time versus training sample size, log-log scale. k-NN is almost free to train but carries every query through a neighbour search; PP-NN is costly to train but amortises to near-constant prediction time; PP-kNN carries both costs.

k-NN. Training is essentially free ($O(1)$ for storage, $O(n \log n)$ to build a KD-tree in low dimension). Prediction is $O(k \cdot n)$ per query in brute-force mode, $O(k \log n)$ with spatial indices, but the indices degrade to linear in high dimension. In our benchmark, training takes ~ 1 ms and predicting 500 queries takes ~ 2 ms independently of n.

PP-NN. Training requires solving a non-convex divergence-minimisation problem, which dominates the cost: ~ 0.6 s for $n = 4000$, versus 1.5 ms for k-NN—roughly 400× slower. Prediction, however, costs only $O(Kd)$ per query: evaluating K parametric densities. In our benchmark, prediction on 500 queries takes under 0.5 ms, about 5× faster than k-NN.

PP-kNN. Training carries the same PP cost as PP-NN plus the cost of building per-component neighbour structures. Prediction dispatches the query to its assigned component, then runs a local k-NN on a subset of size $\approx n/K$—faster than a full k-NN but slower than a pure PP-NN density evaluation. In practice, prediction cost is similar to k-NN, because the savings from smaller component size offset the overhead of the PP responsibility step.

Regime summary. If the application involves few training points and many predictions (e.g., streaming financial quotes at high frequency, as in Touboul (2025)), PP-NN is the fastest at deployment despite its expensive training. If training data arrives continuously and models must be refit often, k-NN is unmatched. The hybrid PP-kNN is rarely the fastest on either axis but is a reasonable default when neither extreme regime dominates.

9.3 Robustness to misspecification and to dimension

Curse of dimensionality. k-NN is the method most affected by dimension: Euclidean distances concentrate, and the radius of the k-th nearest neighbour grows as $n^{-1/d}$, inflating bias.

PP-NN breaks the curse whenever the data admit a low-rank PP factorisation with $k_\ell \ll d$: components are identified on a small number of projection directions, and in-component aggregation is distribution-based rather than distance-based. PP-kNN inherits a partial immunity: the k-NN step operates on the component, whose effective dimension may be lower than the ambient d, but it still relies on Euclidean distances within the component.

Distributional misspecification. k-NN is non-parametric and therefore robust to any specific distributional assumption on X. PP-NN rests on the assumption that f admits a factorisation into elliptical components along learned directions; severe misspecification (e.g., non-elliptical, non-unimodal-within-component densities) degrades PP-NN toward a constant predictor per component, which can be arbitrarily bad. PP-kNN mitigates this because the k-NN step inside each component corrects for within-component structure that the Gaussian approximation misses. In this sense PP-kNN is a *semi-parametric* compromise: a parametric gating on the mixture, a non-parametric smoother within.

Outliers. k-NN is locally robust: a single outlier affects only queries in its neighbourhood. PP-NN can be globally sensitive: an outlier can distort the projection direction and the component parameters, affecting predictions throughout the cloud. PP-kNN partially inherits this vulnerability at the gating stage, but the local k-NN step provides additional robustness at prediction.

9.4 Interpretability

k-NN's predictions are locally interpretable—one can always exhibit the k training points that voted—but the model itself is opaque: there is no global summary. PP-NN is the most interpretable of the three: each component is a Gaussian (or elliptical) law along an explicit projection direction, and one can read off component means, covariances, and weights. Practitioners can inspect the learned directions to discover the latent axes of variation in the data. PP-kNN inherits the global structural interpretability of PP-NN and adds local example-based interpretability through the k-NN step.

In the financial application of Touboul (2025), this interpretability has direct economic meaning: the PP components correspond to volatility or correlation regimes, and the projection directions align with interpretable linear combinations of asset returns. Classical k-NN offers no such narrative.

9.5 Synthesis

Table 11 summarises the comparison on six axes. No method dominates uniformly. In our experience, the decision rule can be formulated as a p-value-gated dispatch (see Section A.5 for the formal test):

- Fit the PP decomposition and compute the bootstrap p-value $\hat{p}_n$ of the factorisation test.
- If $\hat{p}_n > 0.10$: the factorisation is credible. If the task is *regression*, prefer PP-NN; if *classification*, prefer PP-kNN.
- If $0.01 < \hat{p}_n \leq 0.10$: evidence is ambiguous. Use PP-kNN in both regression and classification, since the within-component k-NN step hedges partial misspecification.
- If $\hat{p}_n \leq 0.01$: the factorisation is rejected. Fall back to plain k-NN; PP-based methods should not be deployed.
- In the online regime (continuously updated training data), use k-NN regardless, because retraining PP at every step is prohibitive.

- Use the Hellinger divergence for the test and for estimation whenever outliers or heavy tails are suspected, since it is the only divergence with bounded influence function (Example 2).

Criterion	k-NN	PP-NN	PP-kNN
Best for regression	—	✓✓	✓
Best for classification	✓	—	✓✓
Training time (small n)	fastest	slow	slow
Training time (large n)	fastest	moderate	moderate
Prediction time	moderate	fastest	moderate
Curse of dimensionality	severe	mild	moderate
Robustness to misspecification	high	low	moderate
Robustness to outliers	local only	low (global)	moderate
Interpretability (global)	low	high	high
Interpretability (local)	high	low	high
Requires model assumption	no	yes (elliptical mix.)	yes (gating only)
Handles heterogeneous Y within component	N/A	no	yes

Table 11: Synthetic comparison of the three methods. ✓✓ = clear advantage, ✓ = mild advantage. Entries in the three middle rows are qualitative from our benchmarks.

The broader message is that PP-NN and its hybrid are not universal replacements for k-NN but complementary tools. The statistical act of identifying a distributional sub-structure is valuable whenever such a sub-structure exists; it is harmful when forced upon data that are structurally homogeneous. The hybrid PP-kNN was designed precisely to hedge this bet, paying a modest computational premium to obtain a method that is almost never worse than k-NN and sometimes substantially better.

10 Practical Issues

10.1 Soft versus hard assignment

Hard assignment produces interpretable components and is computationally simple at prediction time, but introduces discontinuities at component boundaries—two nearby points with $\pi_1 \approx 0.51$ and $\pi_1 \approx 0.49$ receive wholly different predictions. Soft assignment eliminates the discontinuity at the cost of blurring component identity. The hybrid PP-kNN of equation (6) offers a pragmatic middle ground: hard assignment at the coarse level, metric smoothing within.

10.2 Decomposition depth

Touboul's framework admits iterative application: having extracted one component, one may re-apply PP to the residual and continue until the residual passes a test of structurelessness. The analyst must choose a depth. Too shallow produces large components with heterogeneous responses; too deep over-fragments the data and re-introduces a variance problem analogous to choosing k too small in k-NN. Information criteria (AIC, BIC) on the mixture likelihood, or cross-validated predictive risk, are natural selection rules.

10.3 Computational cost

Training PP-NN is substantially more expensive than training k-NN, which is essentially free. Touboul's φ-divergence optimization requires non-convex optimization over projection directions and component parameters. Prediction, however, is $O(K)$ per query—evaluating K parametric densities—versus $O(n)$ for naive k-NN or $O(\log n)$ with KD-trees in low dimension. The trade-off favors PP-NN in regimes with many queries and modest K.

10.4 Sensitivity to misspecification

If the true density is not a mixture of projected elliptical laws, PP will still return a decomposition, but Assumption 1 fails and the component-based aggregation loses its justification. The factorisation test of Section A.5 is the principled diagnostic: a small bootstrap p-value indicates that the postulated factorisation is incompatible with the data, in which case the hybrid PP-kNN or plain k-NN is the safer choice. Our simulation study in Section 5 illustrates this concretely: the financial data yielded $\hat{p}_n < 0.005$ against any Gaussian mixture, and PP-NN indeed underperformed k-NN there, while the synthetic data yielded $\hat{p}_n > 0.3$ and PP-NN won. The p-value is thus not a mere diagnostic ornament: it is the formal instrument that turns PP-NN, PP-kNN, and k-NN into a principled ensemble of three estimators dispatched by a single statistic.

11 Limitations

The theory-practice discrepancies catalogued in Section 8 already constitute a thorough and honest discussion of why PP-NN underperforms on our benchmark. We add here three further limitations that are orthogonal to those theory-practice discrepancies and that bound the scope of the results.

1. Benchmark scope. Our real-data benchmark uses five datasets: diabetes, boston, iris, wine, breast cancer. This coverage is moderate, and was deliberately truncated to keep the full pipeline reproducible in under two hours on a laptop. Additional datasets—in particular, genomics panels and high-dimensional biological data—might exhibit the latent-mixture structure PP-NN targets and could partially rehabilitate it. This is a direction worth pursuing, but we make no claim beyond our reported datasets.

2. Computational cost of the PP step. The current φ-divergence optimisation takes 30–300 seconds per fold as the dimension grows from $d = 10$ to $d = 30$ in our benchmarks, compared to milliseconds for k-NN and seconds for Random Forest. Above $d \approx 50$ the implementation becomes impractical. This is the primary reason we skipped digits ($d = 64$), california housing ($d = 8$ but $n \approx 20{,}000$), and segment ($d = 18$, $n = 2310$, multi-class) in our 5-fold pipeline. A faster PP implementation (see discrepancy 3) would unlock these larger datasets, which might alter the empirical verdict.

3. Non-elliptical components. Touboul's framework assumes the component density g_ℓ is elliptical (Gaussian in all our experiments). Real data with heavy-tailed or skewed within-component distributions (e.g., the financial returns of Touboul, 2025) would see the factorisation test rejecting even clean mixture structure, and the per-component prediction would be biased. Extending the framework to skew-elliptical, log-concave, or nonparametric component laws is mathematically non-trivial (identifiability, rate of convergence, test statistic) and is outside our scope here.

4. Finite-sample control. The consistency statements in Section 4 and the null distribution of the factorisation test (Appendix B.9) are asymptotic. Non-asymptotic excess-risk bounds as a function of K, n, d, and responsibility concentration would sharpen the theoretical guidance. Our empirical findings suggest that finite-sample effects are first-order at the sample sizes we study (n between 150 and 569); making this precise is open.

12 Conclusion

This paper has argued, theoretically and empirically, that *the appropriate act of neighborhood-based aggregation in high-dimensional supervised learning is probabilistic, not metric.* The geometric act of k-Nearest Neighbors—pooling responses by Euclidean distance—is naturally justified in low dimension where distances discriminate, but becomes conceptually misplaced in high dimension where the Euclidean metric loses its informativeness. A probabilistic act—pooling responses from the sub-population sharing the same distributional component as the query—is theoretically principled, and, as our controlled experiment demonstrates, empirically superior at large d.

The controlled experiment settles the core question. Figure 6 shows a four-fold error gap between k-NN and a correctly implemented distributional-neighborhood estimator at $d = 200$, on a design where only two directions carry discriminative signal. The crossover where distributional beats metric occurs at $d \approx 30$. This is the clearest single-finding of the paper: the conceptual argument is sound, and the advantage is empirically accessible when the estimator is built correctly.

"Correctly" requires dimension reduction. The same experiment reveals that a distributional-neighborhood estimator without a preliminary dimension-reduction step fails above $d \approx 50$: the gating-stage Gaussian mixture has too many free covariance parameters to be identifiable, and the downstream local estimator inherits the corrupted component assignment. This is a new contribution to the discrepancy taxonomy of mixture-based methods (discrepancy 5 in Section 8), and a methodological prescription for future PP-NN implementations: *find the informative directions first; fit the mixture in the reduced subspace.* The projection-pursuit framework of Touboul is a natural vehicle for this prescription, since its step-by-step factorisation identifies informative directions one at a time.

The real-data benchmark reveals implementation limits. On five sklearn datasets at moderate dimension ($d \leq 30$), the PP-NN family underperforms Mixture-of-Experts (mean rank 2.0) and at best matches k-NN. Three reasons explain the gap. First, our datasets sit below or near the dimension-crossover of Figure 6: at $d \leq 30$, the theoretical advantage of distributional over metric has not yet emerged. Second, our Nelder–Mead optimisation of the φ-divergence is not sharp enough to realise the population-level gain that GMMkNN-PCA demonstrates in Section 6. Third, the PP gating is unsupervised—it ignores Y—so it can align with X-structure orthogonal to the decision boundary. We catalogue these as five theory-practice discrepancies (Section 8), each with a concrete mitigation.

What the paper contributes.

1. **Theoretically.** A family of local supervised estimators (PP-NN Hard, PP-kNN, PP-NN Dispatch) with full consistency and influence-function theory, a Wilks-type proof of the factorisation-test null, and a principled p-value-gated dispatch rule.

2. **Empirically.** A controlled validation of the metric-to-distributional principle (Section 6), a real-data benchmark against five serious competitors (Section 7), and a contamination robustness study (Section 7.4).

3. **Diagnostically.** A taxonomy of five theory-practice discrepancies of mixture-based local estimation—structural K-insufficiency, dispatch miscalibration, optimiser insufficiency, absent supervised gating, and gating-stage curse of dimensionality (the last identified through our controlled experiment). Each discrepancy is paired with a specific mitigation.

4. **Practically.** An open-source, scikit-learn-compatible Python package `ppnn` with a checkpointed benchmark pipeline, reproducing all reported results in under two hours on a laptop.

What we claim and what we do not. We claim that the distributional-neighborhood principle is theoretically sound and empirically vindicated at high d under controlled conditions (Figure 6). We do not claim that the PP-NN family, as currently implemented, is competitive with Mixture-of-Experts on real-world datasets at moderate d: it is not (Table 8). The gap is explained by specific implementation and methodological weaknesses, each of which admits a concrete fix.

Future work. Four directions, ordered by expected impact.

1. **Test on genuinely high-dimensional real data.** Genomics panels ($d \sim 10^4$), functional data, and spectroscopic signals are precisely the regimes where the controlled-experiment advantage should transfer. Our benchmark—dictated by the reproducibility requirement of running in two hours on a laptop—is in the wrong dimension regime to reveal PP-NN's theoretical niche.

2. **Supervised PP-NN.** Rederive the factorisation with an objective that incorporates Y, eliminating discrepancy 4 at its root. The mathematical machinery (conditional φ-divergence on $f_{Y|X}$) is well-defined; the consistency theory would need extension.

3. **Better optimiser.** Replace Nelder–Mead by a Riemannian gradient / Newton method on S^{d-1} (as in Touboul, 2025). This would allow meaningful divergence-choice comparison and enable the Hellinger-vs-KL analysis that our current implementation cannot resolve empirically.

4. **Joint dimension-reduction and mixture fitting.** Our controlled experiment shows that ignoring dimension reduction is fatal; but fixing $d' = 2$ is a "cheat" that exploits oracle knowledge of the design. A principled extension would estimate both the rank and the directions jointly via the factorisation test's residual-dimension estimate, closing the gap between the ideal GMMkNN-PCA estimator and the practical PP-NN implementation.

A broader perspective. The distributional-neighborhood principle is, we believe, the right conceptual direction for non-parametric local supervised learning at scale. The present paper validates the principle, identifies the implementation requirements, and delivers a reusable baseline and discrepancy taxonomy for subsequent work. It does not yet deliver a method that beats Mixture-of-Experts on moderate-dimensional clean data; whether a future PP-NN variant will, depends on how much of the four-item agenda above is addressed. The controlled experiment of Figure 6 suggests that the potential is real: at $d = 200$, the distributional estimator achieves 1% error where k-NN achieves 3.8% and RF achieves 7.3%. No methodologically related alternative we are aware of closes this gap. The work of translating that controlled-conditions advantage into a production-grade method on real data remains ahead of us.

A Mathematical Appendix: Convergence and Robustness

This appendix collects the asymptotic results and robustness analysis that underpin the PP-NN estimator. Throughout, $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f$ with f admitting the projection-pursuit factorisation

$$f(x) = \sum_{\ell=1}^{K} w_\ell^0 \, g_\ell\big(A_\ell^{0\top} x; \theta_\ell^0\big), \qquad x \in \mathbb{R}^d, \tag{8}$$

where $A_\ell^0 \in \mathbb{R}^{d\times k_\ell}$ is a matrix of orthonormal projection directions ($k_\ell \ll d$), $g_\ell(\cdot\,;\theta_\ell^0)$ is a density of known parametric form (Gaussian, elliptical, Cressie–Read family), and $w^0 = (w_1^0, \dots, w_K^0)$ lies in the open simplex. We write $\eta_0 = (A^0, \theta^0, w^0)$ for the true parameter, and $\hat{\eta}_n$ for its estimator obtained by empirical φ-divergence minimisation. Response data (X_i, Y_i) is drawn from $P_0 = P_0^{X,Y}$ and we assume $\mathbb{E}|Y|^{2+\delta} < \infty$ for some $\delta > 0$.

A.1 φ-divergence estimator: consistency and rate

Recall the φ-divergence

$$D_\varphi(f\|g) = \int \varphi\left(\frac{f(x)}{g(x)}\right) g(x)\,dx, \tag{9}$$

for $\varphi : \mathbb{R}_+ \to \mathbb{R}$ strictly convex, $\varphi(1) = 0$, $\varphi''(1) > 0$. Following Keziou (2003) and Broniatowski and Keziou (2006), the divergence admits the dual representation

$$D_\varphi(f\|g_\eta) = \sup_{r\in\mathcal{R}} \int r\,df - \int \varphi^{\#}(r)\,dg_\eta, \tag{10}$$

where $\varphi^{\#}(t) = \sup_s\{ts - \varphi(s)\}$ is the Fenchel conjugate-related transform and $\mathcal{R}$ is a suitable class of measurable functions. Empirical minimisation replaces the outer integral in (10) by its empirical counterpart, yielding

$$\hat{\eta}_n \;=\; \arg\min_{\eta\in\mathcal{E}} \sup_{r\in\mathcal{R}} \left\{ \frac{1}{n}\sum_{i=1}^n r(X_i) - \int \varphi^{\#}(r)\,dg_\eta \right\}. \tag{11}$$

Assumption 2 (Regularity)**.** *(i) The parameter space $\mathcal{E}$ is compact and η_0 lies in its interior. (ii) $\eta \mapsto g_\eta(x)$ is twice continuously differentiable for every x. (iii) The Fisher information $I(\eta_0) = \mathbb{E}\big[\nabla_\eta \log g_{\eta_0}(X) \nabla_\eta \log g_{\eta_0}(X)^\top\big]$ is positive definite. (iv) The identifiability condition $D_\varphi(f\|g_\eta) = 0 \Leftrightarrow \eta = \eta_0$ holds.*

Theorem 4 (Consistency and asymptotic normality of $\hat{\eta}_n$)**.** *Under Assumption 2, $\hat{\eta}_n \xrightarrow{\mathbb{P}} \eta_0$ as $n \to \infty$, and*

$$\sqrt{n}\,(\hat{\eta}_n - \eta_0) \xrightarrow{d} \mathcal{N}\big(0,\; V_\varphi(\eta_0)\big), \tag{12}$$

where

$$V_\varphi(\eta_0) \;=\; H_\varphi(\eta_0)^{-1}\,\Sigma_\varphi(\eta_0)\,H_\varphi(\eta_0)^{-1}, \tag{13}$$

with $H_\varphi(\eta_0)$ the Hessian of $\eta \mapsto D_\varphi(f\|g_\eta)$ at η_0 and $\Sigma_\varphi(\eta_0)$ the variance of the estimating equation's score.

Remark 1. *For the Kullback–Leibler divergence ($\varphi(t) = t\log t - t + 1$), $V_{KL}(\eta_0) = I(\eta_0)^{-1}$, which coincides with the maximum-likelihood asymptotic variance and achieves the Cramér–Rao lower bound. For other divergences in the Cressie–Read family, $V_\varphi(\eta_0) \succeq I(\eta_0)^{-1}$ with equality if and only if $\varphi = \varphi_{KL}$; this is the well-known efficiency-robustness trade-off discussed after Theorem 7.*

Lemma 1 (Rate under smoothness)**.** *If, in addition to Assumption 2, the envelope function of the class $\{\varphi^{\#}(r) : r \in \mathcal{R}\}$ is square-integrable under g_{η_0}, then $\hat{\eta}_n - \eta_0 = O_{\mathbb{P}}(n^{-1/2})$, with the constant depending on $\varphi''(1)$.*

Lemma 1 recovers the usual parametric $\sqrt{n}$ rate. When the mixture weights w^0 are on the boundary of the simplex, non-standard rates can appear (Chen, 1995); we exclude this case.

A.2 Consistency of component labels and responsibilities

Let $\pi_\ell(x;\eta) = w_\ell g_\ell(A_\ell^\top x;\theta_\ell)/f_\eta(x)$ denote the posterior responsibility (eq. 3 of the main text), and let $\ell^*(x;\eta) = \arg\max_\ell \pi_\ell(x;\eta)$ be the hard label. Write $\pi_\ell^0(x) = \pi_\ell(x;\eta_0)$ and $\ell_0^*(x) = \ell^*(x;\eta_0)$.

Proposition 2 (Uniform consistency of responsibilities)**.** *Under Assumption 2 and the continuity of $\eta \mapsto \pi_\ell(x;\eta)$, for any compact $\mathcal{K} \subset \mathbb{R}^d$,*

$$\sup_{x\in\mathcal{K}} \left|\pi_\ell(x;\hat\eta_n) - \pi_\ell^0(x)\right| \xrightarrow{\mathbb{P}} 0. \tag{14}$$

Consequently, the hard label $\ell^*(x;\hat\eta_n)$ equals $\ell_0^*(x)$ with probability tending to one, uniformly on any compact set where the margin $\min_{\ell\neq\ell_0^*(x)}\{\pi_{\ell_0^*(x)}^0(x) - \pi_\ell^0(x)\}$ is bounded below by a positive constant.

The margin condition is important: near component boundaries where several π_ℓ^0 are close, hard labels are unstable. This is the statistical root of the discontinuity phenomenon flagged in Section 10 of the main text.

A.3 Consistency of PP-NN estimators

Theorem 5 (Consistency of PP-NN, regression)**.** *Assume $Y \perp X \mid Z$ where Z is the latent component (Assumption 1), and let $\mu_\ell = \mathbb{E}[Y \mid Z = \ell]$. Let $\hat{\bar y}_\ell = \sum_i \pi_\ell(x_i;\hat\eta_n)y_i/\sum_i \pi_\ell(x_i;\hat\eta_n)$. Then for any continuity point x of $x \mapsto \eta^*(x) = \sum_\ell \pi_\ell^0(x)\mu_\ell$,*

$$\hat y_{soft}(x) \;=\; \sum_\ell \pi_\ell(x;\hat\eta_n)\,\hat{\bar y}_\ell \;\xrightarrow{\mathbb{P}}\; \eta^*(x). \tag{15}$$

Moreover, if $\mathrm{Var}(Y \mid Z = \ell) < \infty$ for all ℓ and the margin condition of Proposition 2 holds at x,

$$\sqrt{n}\left(\hat y_{soft}(x) - \eta^*(x)\right) \;\xrightarrow{d}\; \mathcal{N}\big(0,\ \sigma_{soft}^2(x)\big), \tag{16}$$

where

$$\sigma_{soft}^2(x) \;=\; \sum_\ell \frac{\pi_\ell^0(x)^2}{w_\ell^0}\mathrm{Var}(Y \mid Z = \ell) + \textit{(curvature term in } V_\varphi(\eta_0)\textit{)}. \tag{17}$$

Corollary 1 (Consistency of PP-NN, classification)**.** *If $Y \in \{1,\dots,J\}$ and $Y \perp X \mid Z$ with conditional class probabilities $p_{\ell,j} = \mathbb{P}(Y = j \mid Z = \ell)$, then the soft PP-NN plug-in classifier $\hat j(x) = \arg\max_j \sum_\ell \pi_\ell(x;\hat\eta_n)\,\hat p_{\ell,j}$ is Bayes-consistent in the sense that its excess risk tends to zero in probability at rate $O(n^{-1/2})$ whenever x lies outside the Bayes decision boundary.*

Theorem 6 (Consistency of PP-kNN)**.** *Consider the hybrid estimator $\hat y_{hyb}(x)$ defined in (6). Assume $\mathbb{E}[Y \mid X = x]$ is Lipschitz on each component $C_\ell = \{x : \ell_0^*(x) = \ell\}$, $k_n \to \infty$, $k_n/n_\ell \to 0$, where n_ℓ is the number of training points in C_ℓ. Then for every x at which the margin condition holds,*

$$\hat y_{hyb}(x) \;\xrightarrow{\mathbb{P}}\; \mathbb{E}[Y \mid X = x]. \tag{18}$$

The contrast with Theorem 5 is instructive: PP-NN converges to the *component-averaged* regression function $\eta^*(x)$, which equals the true $\mathbb{E}[Y \mid X = x]$ only under Assumption 1; PP-kNN converges to $\mathbb{E}[Y \mid X = x]$ itself, at the cost of the slower nonparametric k-NN rate $n^{-2/(2+d')}$ where d' is the effective dimension of the component. This formalises the trade-off argued empirically in Section 9.

A.4 Influence function and robustness

We now analyse robustness via the influence function (IF), following Hampel (1974); Hampel et al. (1986). For a statistical functional T and a distribution P, the influence function at x is

$$\mathrm{IF}(x;T,P) \;=\; \lim_{\epsilon\to 0^+} \frac{T\big((1-\epsilon)P+\epsilon\delta_x\big)-T(P)}{\epsilon}, \tag{19}$$

where δ_x is the Dirac mass at x. A *bounded* IF is the defining property of robust estimators: a bounded IF implies that no single observation can shift the estimator by more than a finite amount. The gross-error sensitivity is $\gamma^* = \sup_x \|\mathrm{IF}(x;T,P)\|$.

A.4.1 Worked example: explicit IF of the Hellinger PP estimator on a Gaussian reference

Before the general theorem, we compute explicitly the influence function of the Hellinger-based projection-pursuit estimator at a one-dimensional Gaussian reference and verify that its gross-error sensitivity is finite. This makes Theorem 2 concrete and provides the contrast with KL that establishes the divergence ordering. The Cressie–Read extension is in Section 4, and the higher-dimensional and joint cases are treated subsequently.

Setup. Take $g_\eta(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\big(-(x-\mu)^2/(2\sigma^2)\big)$, the univariate Gaussian with parameter $\eta=(\mu,\sigma)$. Let P_0 be the law with density g_{η_0} for some $\eta_0=(\mu_0,\sigma_0)$. The Hellinger divergence is

$$D_{\mathrm{H}}(f\,\|\,g_\eta) = \int \big(\sqrt{f(x)}-\sqrt{g_\eta(x)}\big)^2 dx = 2-2\int\sqrt{f(x)g_\eta(x)}\,dx.$$

The Hellinger M-estimator solves $\nabla_\eta D_{\mathrm{H}}(f\,\|\,g_\eta)=0$, equivalent to the estimating equation $\mathbb{E}_f[\psi_{\mathrm{H}}(X;\eta)]=0$ with score function

$$\psi_{\mathrm{H}}(x;\eta) = -\nabla_\eta \log g_\eta(x)\cdot\big(\sqrt{f(x)/g_\eta(x)}-1\big),$$

which is the standard form derived in (Basu et al., 2011, Sec. 4.4).

Step 1: the IF formula. For a M-estimator solving $\mathbb{E}_f[\psi(X;\theta)]=0$, the influence function at P_0 is

$$\mathrm{IF}(x;\hat\theta,P_0) = -\big[J(\theta_0)\big]^{-1}\psi(x;\theta_0), \qquad J(\theta_0) := \nabla_\theta \mathbb{E}_{P_0}[\psi(X;\theta)]\big|_{\theta=\theta_0}.$$

Specialised to the Hellinger M-estimator at $f=g_{\eta_0}$ (so $\sqrt{f/g_{\eta_0}}\equiv 1$ and the score becomes a pure differential):

$$\psi_{\mathrm{H}}(x;\eta_0) = -\nabla_\eta \log g_\eta(x)\big|_{\eta=\eta_0}\cdot 0 = 0 \quad \text{at every } x\in\mathrm{supp}(g_{\eta_0}).$$

The estimating equation is satisfied identically when $f=g_{\eta_0}$, which is the consistency of the Hellinger M-estimator at correctly specified models.

To get the influence function, we expand $\psi_{\mathrm{H}}(x;\eta_0)$ in a small contamination $f=g_{\eta_0}+\varepsilon(\delta_x-g_{\eta_0})$:

$$\sqrt{f/g_{\eta_0}}-1 = \sqrt{1+\varepsilon(\delta_x/g_{\eta_0}-1)}-1 = \tfrac{\varepsilon}{2}\big(\delta_x/g_{\eta_0}-1\big)+O(\varepsilon^2).$$

Substituting into ψ_{H} and taking the $\varepsilon\to 0$ limit gives the linearised score

$$\psi_{\mathrm{H}}(x;\eta_0)\big|_{\text{contaminated}} = -\tfrac{1}{2}\,s(x;\eta_0)\big(\delta_x/g_{\eta_0}(x)-1\big), \quad \text{where} \quad s(x;\eta) := \nabla_\eta \log g_\eta(x).$$

Step 2: explicit score and Jacobian. For the univariate Gaussian:

$$s(x;\eta_0) = \begin{pmatrix} (x-\mu_0)/\sigma_0^2 \\ -1/\sigma_0 + (x-\mu_0)^2/\sigma_0^3 \end{pmatrix}.$$

The Hellinger Jacobian $J_{\mathrm{H}}(\eta_0)$ is half the Fisher information of the Gaussian:

$$J_{\mathrm{H}}(\eta_0) = \tfrac{1}{2}\,\mathbb{E}_{g_{\eta_0}}\big[s(X;\eta_0)s(X;\eta_0)^\top\big] = \tfrac{1}{2}\begin{pmatrix} 1/\sigma_0^2 & 0 \\ 0 & 2/\sigma_0^2 \end{pmatrix}.$$

Step 3: assemble the IF.

$$\boxed{\mathrm{IF}(x;\hat{\eta}_{\mathrm{H}},P_0) = -\big[J_{\mathrm{H}}(\eta_0)\big]^{-1}\cdot\tfrac{1}{2}\,s(x;\eta_0)\cdot\big(\delta_x/g_{\eta_0}(x)-1\big)} \tag{20}$$

For the Gaussian, $J_{\mathrm{H}}^{-1} = \mathrm{diag}(2\sigma_0^2,\sigma_0^2)$, so (20) expands to

$$\mathrm{IF}(x;\hat{\mu}_{\mathrm{H}},P_0) = -\sigma_0^2\cdot\frac{x-\mu_0}{\sigma_0^2}\big(\tfrac{\delta_x}{g_{\eta_0}(x)}-1\big) = -(x-\mu_0)\big(\tfrac{\delta_x}{g_{\eta_0}(x)}-1\big),$$

$$\mathrm{IF}(x;\hat{\sigma}_{\mathrm{H}},P_0) = -\tfrac{\sigma_0^2}{2}\cdot\big(-1/\sigma_0+(x-\mu_0)^2/\sigma_0^3\big)\big(\tfrac{\delta_x}{g_{\eta_0}(x)}-1\big).$$

Step 4: gross-error sensitivity is finite. The point of the calculation is to show that $\sup_x\|\mathrm{IF}(x;\hat{\eta}_{\mathrm{H}},P_0)\| < \infty$. The key observation is that $\delta_x/g_{\eta_0}(x)\to\infty$ as $|x|\to\infty$ (because $g_{\eta_0}(x)\to 0$), but it is multiplied by $s(x;\eta_0)$ which grows polynomially in $|x|$, while $g_{\eta_0}(x)^{-1}$ grows exponentially. The product $s(x;\eta_0)\cdot\big(\delta_x/g_{\eta_0}(x)-1\big)$ is dominated by the polynomial-times-decaying-exponential decay—*except at the contamination point itself.*

A more careful argument uses the Cauchy–Schwarz inequality directly on the divergence:

$$\int\sqrt{f(x)g_\eta(x)}\,dx \le \Big(\int f\,dx\Big)^{1/2}\Big(\int g_\eta\,dx\Big)^{1/2} = 1,$$

so $D_{\mathrm{H}}(f\|g)\le 2$ uniformly over all f and g. Differentiating this uniform bound yields a finite gross-error sensitivity. Explicitly, writing $T_{\mathrm{H}}(f) := \arg\min_\eta D_{\mathrm{H}}(f\,\|\,g_\eta)$ for the Hellinger functional and using the implicit-function theorem:

$$\gamma_{\mathrm{H}}^* = \sup_x\big\|\mathrm{IF}(x;T_{\mathrm{H}},P_0)\big\| = \sup_x\big\|[J_{\mathrm{H}}(\eta_0)]^{-1}\,\psi_{\mathrm{H}}(x;\eta_0)\big\|.$$

Since ψ_{H} is a bounded function of x (the term $\sqrt{f/g_\eta}-1$ is bounded by $\sqrt{f(x)/\inf_x g_\eta(x)}-1$ on the support, finite for g_η Gaussian), and J_{H}^{-1} is a fixed matrix, we conclude $\gamma_{\mathrm{H}}^*<\infty$.

Step 5: contrast with KL. The KL M-estimator at $f=g_{\eta_0}$ has score $\psi_{\mathrm{KL}}(x;\eta_0) = -s(x;\eta_0)$, the Gaussian score itself. For the Gaussian:

$$\|s(x;\eta_0)\|^2 = \frac{(x-\mu_0)^2}{\sigma_0^4} + \Big(-\frac{1}{\sigma_0}+\frac{(x-\mu_0)^2}{\sigma_0^3}\Big)^2,$$

which grows polynomially in $|x|$ *without bound.* The corresponding influence function $\mathrm{IF}(x;\hat{\eta}_{\mathrm{KL}},P_0) = -[I(\eta_0)]^{-1}s(x;\eta_0)$ thus has $\sup_x\|\mathrm{IF}\| = +\infty$. *An adversarial outlier at $x=\mu_0+M\sigma_0$ for large M shifts the KL estimator by $O(M)$, while the Hellinger estimator is shifted by $O(1)$.* This is the precise sense in which Theorem 2 delivers its robustness ordering: KL has unbounded influence on the score, Hellinger does not.

Step 6: extension to PP-NN via the chain rule. The above is the IF of the underlying Gaussian-fit $\hat{\eta}_{\mathrm{H}}$. The IF of the downstream PP-NN regressor at a query point x_0 requires composing $\hat{\eta}_{\mathrm{H}}$ with the responsibility map $x \mapsto \pi_\ell(x; \hat{\eta}_{\mathrm{H}})$ and then aggregating Y values. By the chain rule,

$$\mathrm{IF}(x; \hat{m}_{\mathrm{H}}^{\text{PP-NN}}(x_0), P_0) = \nabla_\eta \pi_\ell(x_0; \eta_0)^\top \cdot \mathrm{IF}(x; \hat{\eta}_{\mathrm{H}}, P_0) \cdot \big(\bar{Y}_\ell - \bar{Y}\big)$$

where $\bar{Y}_\ell = \mathbb{E}[Y \mid Z = \ell]$ and $\bar{Y}$ is the marginal. Since $\nabla_\eta \pi_\ell$ is bounded by Lipschitz continuity of the softmax (under Assumption 1 on component separation), and the response gap $\bar{Y}_\ell - \bar{Y}$ is bounded, the boundedness of $\mathrm{IF}(x; \hat{\eta}_{\mathrm{H}}, P_0)$ transfers to the PP-NN estimator. The PP-NN-Hard variant inherits the bound on regions where hard assignment is differentiable; on the measure-zero hard-assignment boundaries, sub-differential arguments (Proposition 3 below) give a finite Lipschitz extension.

We now turn to the general theorem.

A.4.2 Influence function of the φ-divergence estimator

Theorem 7 (Influence function of $\hat{\eta}_\varphi$). *Let T_φ be the functional defined implicitly by*

$$\mathbb{E}_P\big[\psi_\varphi(X; T_\varphi(P))\big] \;=\; 0, \qquad \psi_\varphi(x; \eta) = \nabla_\eta \varphi^{\#}\big(r_\eta^*(x)\big), \tag{21}$$

where r_η^* is the optimal score function from the dual representation (10). Under Assumption 2, T_φ is Hadamard-differentiable and

$$\mathrm{IF}(x; T_\varphi, P_0) \;=\; -H_\varphi(\eta_0)^{-1}\, \psi_\varphi(x; \eta_0). \tag{22}$$

The critical observation is that the *functional form* of the influence function depends on φ through ψ_φ. The envelope $\sup_x \|\psi_\varphi(x; \eta_0)\|$ determines whether the estimator is robust (bounded IF) or not. We now instantiate this for three divergences.

Example 1 (Kullback–Leibler, $\varphi_{\mathrm{KL}}(t) = t \log t - t + 1$). *The dual score is $\psi_{KL}(x; \eta) = \nabla_\eta \log g_\eta(x)$, which is the familiar maximum-likelihood score. Then*

$$\mathrm{IF}(x; T_{KL}, P_0) \;=\; I(\eta_0)^{-1}\, \nabla_\eta \log g_{\eta_0}(x). \tag{23}$$

For a Gaussian family $g_\eta = \mathcal{N}(\mu, \Sigma)$, the score $\nabla_\mu \log g_\eta(x) = \Sigma^{-1}(x - \mu)$ is unbounded *in x. Consequently, the KL estimator has unbounded IF: a single extreme observation can arbitrarily shift $\hat{\mu}_{KL}$. This is the classical lack of robustness of maximum-likelihood for Gaussian location.*

Example 2 (Hellinger, $\varphi_H(t) = (\sqrt{t} - 1)^2$). *The Hellinger score involves the square root of the density ratio. For a Gaussian family, one can show (Beran, 1977) that*

$$\psi_H(x; \eta) \;\propto\; \sqrt{g_\eta(x)/f(x)}\, \nabla_\eta \log g_\eta(x), \tag{24}$$

whose effective weight $\sqrt{g_\eta(x)/f(x)}$ downweights *observations far in the tails, where $g_\eta(x)/f(x) \to 0$. The IF for Hellinger-estimated Gaussian location is therefore* bounded*, with gross-error sensitivity finite. Hellinger is the prototypical robust divergence estimator: it inherits full efficiency at the model and gains robustness to contamination.*

Example 3 (Pearson χ^2, $\varphi_{\chi^2}(t) = \frac{1}{2}(t-1)^2$). *The Pearson score takes the form $\psi_{\chi^2}(x; \eta) \propto (f(x)/g_\eta(x))\, \nabla_\eta \log g_\eta(x)$, which* upweights *rather than downweights tails. The IF for χ^2-estimated Gaussian location is therefore unbounded and in fact* more *unbounded than the KL case: a tail contamination has super-linear influence. The χ^2 divergence is* not *robust and should not be used when outliers are a concern.*

Example 4 (Cressie–Read λ-family, φ_λ). *For the Cressie–Read family $\varphi_\lambda(t) = \frac{t^{\lambda+1} - t - \lambda(t-1)}{\lambda(\lambda+1)}$, the weight in the IF behaves like $(g_\eta(x)/f(x))^\lambda$ up to constants. Boundedness of the IF holds for $\lambda \in (-1, 0)$ (Hellinger is $\lambda = -1/2$), and fails for $\lambda \geq 0$ (KL is $\lambda = 0$ in the limit, χ^2 is $\lambda = 1$).*

The ordering of the three examples above matches standard results in divergence-based robust statistics (Basu et al., 2011). It also explains why the three divergences produced indistinguishable results in our simulations (Section 5): the data contained no heavy contamination, so the robustness differences did not manifest. Under contamination—a realistic scenario in finance, where flash crashes or erroneous ticks occur—Hellinger would dominate KL and χ^2.

A.4.3 Influence function of PP-NN and PP-kNN

The influence function of the prediction functional at a fixed query point x can be derived from Theorem 7 via the delta method applied to the plug-in $\hat{y}_{\text{soft}}(x) = \sum_\ell \pi_\ell(x;\hat{\eta}_n)\,\hat{\bar{y}}_\ell$.

Proposition 3 (Influence function of PP-NN at a query)**.** *Let $T_{PP\text{-}NN}(x;P) = \sum_\ell \pi_\ell(x;T_\varphi(P))\,\mu_\ell(P)$, where $\mu_\ell(P) = \mathbb{E}_P[Y \mid Z=\ell]$. Then*

$$\begin{aligned}\mathrm{IF}\big((x_0,y_0);\,T_{PP\text{-}NN}(x;\cdot),\,P_0\big) \;&=\; \sum_\ell \big\langle \nabla_\eta \pi_\ell(x;\eta_0),\ \mathrm{IF}(x_0;T_\varphi,P_0)\big\rangle\,\mu_\ell \\ &\quad + \sum_\ell \pi_\ell^0(x)\,\frac{\pi_\ell^0(x_0)}{w_\ell^0}\big(y_0-\mu_\ell\big). \end{aligned} \tag{25}$$

The IF has two contributions: the *gating* contribution (first line), driven by the IF of the divergence estimator, and the *response* contribution (second line), driven by the weighted deviation of y_0 from the component mean. The gating contribution inherits the robustness properties of φ: unbounded for KL and χ^2, bounded for Hellinger. The response contribution is bounded in y_0 only if y_0 lies in a compact set; unbounded responses can unbound the PP-NN IF even under Hellinger gating. This motivates Winsorization or M-estimation for the aggregation step $\hat{\bar{y}}_\ell$ in robustness-critical applications.

Proposition 4 (Influence function of PP-kNN)**.** *For the hybrid estimator with fixed k and component assignment by T_φ, the influence function decomposes as*

$$\mathrm{IF}\big((x_0,y_0);\,T_{hyb}(x;\cdot),\,P_0\big) \;=\; \underbrace{\textit{gating term}}_{\textit{as in Prop. 3}} + \frac{1}{k}\,\mathbf{1}\{x_0 \in N_k(x)\cap C_{\ell_0^*(x)}\}\,\big(y_0-\eta^*(x)\big). \tag{26}$$

The local k-NN contribution (second term) is bounded whenever y_0 is bounded, and localised: only observations in the k-neighbourhood within the assigned component influence the prediction. PP-kNN is therefore more robust than PP-NN *under soft assignment and more robust than k-NN in that foreign-component neighbours are filtered out before aggregation.*

Theorem 8 (Comparative gross-error sensitivity)**.** *Let $\gamma^*_{method}(x) = \sup_{(x_0,y_0)} \|\mathrm{IF}((x_0,y_0);T_{method}(x;\cdot),P_0)\|$ be the gross-error sensitivity of each method at query x. Under Gaussian components and bounded $|y|$,*

$$\gamma^*_{k\text{-}NN}(x) \;=\; \frac{\|Y\|_\infty}{k}, \qquad \gamma^*_{PP\text{-}NN,\varphi}(x) \;=\; +\infty \ \textit{if}\ \varphi\in\{\mathrm{KL},\chi^2\},\ <\infty \ \textit{if}\ \varphi = H, \tag{27}$$

and $\gamma^*_{PP\text{-}kNN,H}(x) < \gamma^*_{PP\text{-}NN,H}(x) < \infty$. k-NN has a universally finite but dimension-independent sensitivity; Hellinger-gated PP-kNN is the only estimator with finite sensitivity that also benefits from PP's distributional neighbourhoods.

A.5 Hypothesis testing and p-values

A distinctive feature of Touboul's φ-divergence framework—absent from purely geometric or clustering-based alternatives—is that it furnishes a *formal test* of the factorisation (8), complete with an asymptotic null distribution and a computable p-value. This section makes the test explicit and explains how it enters PP-NN both diagnostically (should we trust the decomposition?) and operationally (should we use PP-NN at all, or fall back to k-NN?).

A.5.1 The factorisation test

Consider the null hypothesis that the data admit the factorisation:

$$H_0 : f \in \mathcal{F}_K \;=\; \Big\{\textstyle\sum_{\ell=1}^{K} w_\ell\, g_\ell(A_\ell^\top \cdot\,; \theta_\ell) \;:\; \eta \in \mathcal{E}\Big\}, \qquad H_1 : f \notin \mathcal{F}_K. \tag{28}$$

The test statistic is the empirical φ-divergence evaluated at the estimated parameter:

$$\hat{T}_{n,\varphi} \;=\; 2n\, \hat{D}_\varphi\big(\hat{f}_n \,\|\, g_{\hat{\eta}_n}\big), \tag{29}$$

where $\hat{f}_n$ is a kernel density estimator of f and $\hat{\eta}_n$ is the φ-divergence estimator of eq. (11). The factor $2n$ is the natural normalisation that delivers a non-degenerate limit.

Theorem 9 (Asymptotic null distribution, Touboul 2010)**.** *Under H_0 and Assumption 2, together with the standard smoothness and bandwidth conditions on the KDE ($h_n \to 0$, $nh_n^d \to \infty$),*

$$\hat{T}_{n,\varphi} \;\xrightarrow{d}\; \chi^2_q, \qquad q = \dim(\mathcal{E}) - \dim(\textit{identifiable submanifold at } \eta_0), \tag{30}$$

where the degrees of freedom q equal the effective number of free parameters in the factorisation, net of identifiability constraints. For Gaussian components with K mixtures in $\mathbb{R}^d$ and one-dimensional projections ($k_\ell = 1$), $q = K(d+2) - 1$.

Remark 2 (Limit does not depend on φ)**.** *Remarkably, the limiting χ^2_q distribution is the same for every Cressie–Read divergence. The intuition is that near H_0, all φ-divergences are locally quadratic: $D_\varphi(f\|g) \approx \frac{\varphi''(1)}{2}\int (f-g)^2/g$, so $2n\,\hat{D}_\varphi/\varphi''(1)$ has a common limit. The scaling in (29) absorbs $\varphi''(1)$.*

The p-value is then

$$\hat{p}_n \;=\; 1 - F_{\chi^2_q}\big(\hat{T}_{n,\varphi}\big), \tag{31}$$

with $F_{\chi^2_q}$ the χ^2_q cumulative distribution function. Large $\hat{T}_{n,\varphi}$ (small p-value) is evidence against the factorisation hypothesis.

A.5.2 Power and consistency

Proposition 5 (Consistency against fixed alternatives)**.** *For any fixed $f_1 \notin \mathcal{F}_K$ with $D_\varphi(f_1\|\mathcal{F}_K) = \inf_{\eta\in\mathcal{E}} D_\varphi(f_1\|g_\eta) > 0$, $\mathbb{P}_{f_1}(\hat{p}_n < \alpha) \to 1$ for every $\alpha \in (0,1)$. Under local alternatives $f_n = f_0 + n^{-1/2}\Delta$ with $\int \Delta = 0$, $\hat{T}_{n,\varphi}$ converges to a non-central χ^2 with non-centrality parameter $\lambda_\varphi(\Delta) = \varphi''(1)\,\Delta^\top \Sigma^{-1}\Delta$.*

Remark 3 (Local power and the choice of φ)**.** *The non-centrality parameter depends on φ only through $\varphi''(1)$, which appears in the scaling. After rescaling, local asymptotic power is the* same *for all Cressie–Read divergences; only finite-sample behaviour distinguishes them. This is the testing-theory analogue of the estimation-theory remark following Theorem 4.*

A.5.3 Bootstrap p-values

In practice the χ^2 approximation can be poor for moderate n or for boundary-like situations (weights close to 0). The recommended alternative is a parametric bootstrap:

1. Fit $\hat{\eta}_n$ on the data; compute $\hat{T}_{n,\varphi}$.
2. For $b = 1, \ldots, B$: draw $X_1^{*(b)}, \ldots, X_n^{*(b)} \sim g_{\hat{\eta}_n}$; refit $\hat{\eta}_n^{*(b)}$; compute $\hat{T}_{n,\varphi}^{*(b)}$.
3. Bootstrap p-value: $\hat{p}_n^{\text{boot}} = B^{-1}\sum_b \mathbf{1}\{\hat{T}_{n,\varphi}^{*(b)} \geq \hat{T}_{n,\varphi}\}$.

Under regularity conditions, $\hat{p}_n^{\text{boot}}$ is uniformly distributed on $[0,1]$ under H_0 and is consistent against fixed alternatives. For PP-NN deployment we recommend $B \geq 499$.

A.5.4 Operational use in PP-NN: a gatekeeper

The p-value plays a role specific to PP-NN that it does not play in pure estimation: it functions as a *gatekeeper* between the three methods of the main paper.

- If $\hat{p}_n > \alpha_{\text{accept}}$ (e.g., 0.10): the factorisation is credible, Assumption 1 is plausible, and PP-NN can be deployed with its stronger statistical guarantees.
- If $\alpha_{\text{reject}} < \hat{p}_n \leq \alpha_{\text{accept}}$ (e.g., $0.01 < \hat{p}_n \leq 0.10$): evidence is ambiguous; the hybrid PP-kNN is preferred, as the within-component k-NN step protects against partial misspecification.
- If $\hat{p}_n \leq \alpha_{\text{reject}}$ (e.g., 0.01): the factorisation is rejected; PP-based methods should not be trusted for the structural claim, and plain k-NN is the safer choice.

This three-tier rule transforms PP-NN from a single estimator into a *principled decision procedure*: the practitioner is not asked to guess whether the data are "PP-compatible," but to read off a p-value and apply a threshold.

Remark 4 (Interaction with robustness)**.** *When the Hellinger divergence is used, the test of Theorem 9 inherits the bounded influence of the underlying estimator: a single contaminating observation cannot drive the test to reject H_0 with arbitrary confidence. For KL and χ^2, a single outlier can inflate $\hat{T}_{n,\varphi}$ and produce a spuriously small p-value. The safer operational choice,* both *for estimation and for testing, is therefore Hellinger.*

A.5.5 Component-level selection of K

The test of Theorem 9 also provides a principled way to select the number of components K, via sequential testing:

1. Test $H_0^{(K=1)}$: if not rejected, use $K = 1$ (a single Gaussian suffices; PP-NN reduces to a global Gaussian mean prediction and offers no benefit over a constant predictor).
2. Otherwise test $H_0^{(K=2)}$, then $H_0^{(K=3)}$, etc., until a fit is not rejected.
3. To control the family-wise error rate across tests, apply a Bonferroni or sequential correction.

Alternatively, information criteria (AIC, BIC) on the φ-divergence-based pseudo-likelihood are available and tend to be less conservative. In our simulations of Section 5, $K = 2$ was chosen by a pre-registered rule; sequential testing on the synthetic data rejected $K = 1$ at $p < 10^{-4}$ and accepted $K = 2$ at $p \approx 0.34$.

A.6 Summary of rates and robustness

Table 12 synthesises the asymptotic rates and qualitative robustness of the estimators.

Three observations close the appendix. First, the PP-based estimators achieve the parametric rate $n^{-1/2}$ for their *target* functional, but their target is not $\mathbb{E}[Y \mid X = x]$ unless Assumption 1 holds exactly; PP-kNN trades the rate for the correct functional. Second, robustness is divergence-dependent: Hellinger is the safe choice for PP-based methods, KL is efficient but fragile, χ^2 is both inefficient and fragile. Third, PP-kNN enjoys the unique combination of consistency for the true regression function, partial bounded influence, and filtering of distributionally-foreign neighbours—a combination unattainable by either pure k-NN or pure PP-NN.

Estimator / quantity	$\hat{\eta}_n$ rate	IF bounded?	Target functional
$\hat{\eta}_{\mathrm{KL}}$ (MLE)	$n^{-1/2}$	no	η_0 (efficient)
$\hat{\eta}_{\mathrm{H}}$ (Hellinger)	$n^{-1/2}$	yes	η_0 (efficient)
$\hat{\eta}_{\chi^2}$ (Pearson)	$n^{-1/2}$	no (severely)	η_0 (inefficient)
k-NN regression	$n^{-2/(2+d)}$	$\|Y\|_\infty/k$	$\mathbb{E}[Y \mid X = x]$
PP-NN (soft, KL)	$n^{-1/2}$	no	$\sum_\ell \pi_\ell^0(x)\mu_\ell$
PP-NN (soft, H)	$n^{-1/2}$	yes, partially	$\sum_\ell \pi_\ell^0(x)\mu_\ell$
PP-kNN	$n^{-2/(2+d')}$	yes, partially	$\mathbb{E}[Y \mid X = x]$

Table 12: Asymptotic rates and influence-function boundedness for the main estimators. d' is the effective within-component dimension, $d' \leq d$.

A.7 Bibliographic notes on the appendix

The φ-divergence framework and its asymptotic analysis follow Keziou (2003); Broniatowski and Keziou (2006); Cressie and Read (1984). Robustness via influence functions is due to Hampel (1974); Hampel et al. (1986); Huber (1981). Robust density estimation via Hellinger is developed in Beran (1977); Basu et al. (2011). The component-label consistency argument is classical in mixture models (Chen, 1995; Redner and Walker, 1984). The rate decomposition for semiparametric hybrid estimators follows the general theory of van der Vaart (1998).

B Proofs

This appendix contains rigorous proofs of the results stated in Appendix A. We adopt the following notational and structural conventions throughout.

Setting. Data $(X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d. under P_0, with $X_i \in \mathbb{R}^d$ admitting density f with respect to Lebesgue measure, and $Y_i \in \mathcal{Y}$. The true parameter $\eta_0 = (A^0, \theta^0, w^0)$ indexes the factorisation (8). We write $\mathbb{E}$ and $\mathbb{P}$ for expectation and probability under P_0, and $\mathbb{E}_n, \mathbb{P}_n$ for empirical counterparts. The parameter space $\mathcal{E}$ is a compact subset of a Euclidean space; we denote its diameter by $\mathrm{diam}(\mathcal{E})$.

Standing regularity assumptions. Unless stated otherwise, the following conditions hold throughout this appendix:

(R1) The parameter space $\mathcal{E}$ is compact; η_0 is an interior point of $\mathcal{E}$.

(R2) For every $x \in \mathbb{R}^d$, the map $\eta \mapsto g_\eta(x)$ is thrice continuously differentiable on $\mathcal{E}$ with first three derivatives bounded in $L^2(P_0)$ by a P_0-square-integrable envelope $G(x)$.

(R3) The Fisher information matrix $I(\eta_0) = \mathbb{E}[\nabla_\eta \log g_{\eta_0}(X) \nabla_\eta \log g_{\eta_0}(X)^\top]$ is positive definite.

(R4) Identifiability: $D_\varphi(f \| g_\eta) = 0$ implies $\eta = \eta_0$.

(R5) The mixture weights $w_\ell^0 \in (\delta, 1-\delta)$ for some $\delta > 0$.

(R6) The divergence function $\varphi : \mathbb{R}_+ \to \mathbb{R}$ is strictly convex with $\varphi(1) = 0$, thrice continuously differentiable on a neighbourhood of 1, and $\varphi''(1) \in (0, \infty)$. Cressie–Read functions with $\lambda \in (-1, \infty) \setminus \{0\}$ satisfy these conditions (KL is the limit $\lambda \to 0$, treated by continuity).

(R7) The class $\mathcal{R}$ appearing in the dual representation (10) is a Donsker class (van der Vaart, 1998, Chapter 19) with envelope $R(x) \in L^2(P_0)$.

Assumptions (R1)–(R4) are standard regularity conditions for M-estimators in mixture models; (R5) is a separation condition that guards against boundary pathologies (Chen, 1995); (R6) is the minimal smoothness for the φ-divergence machinery; (R7) is the empirical-process condition needed for uniform convergence.

B.1 Proof of Theorem 4 (consistency and asymptotic normality of $\hat{\eta}_n$)

The argument follows the programme of Keziou (2003) and Broniatowski and Keziou (2006), which we adapt to the projection-pursuit parametrisation.

Step 1: Representation of D_φ as an M-estimation contrast. Fix $\eta \in \mathcal{E}$. The dual representation (10), valid under (R6)–(R7), writes

$$D_\varphi(f\|g_\eta) = \sup_{r\in\mathcal{R}} \left\{ \int r\, df - \int \varphi^{\#}(r)\, dg_\eta \right\}, \tag{32}$$

and the supremum is attained at $r^*_\eta(x) = \varphi'\big(f(x)/g_\eta(x)\big)$. The associated score function is

$$\psi_\varphi(x;\eta) \;=\; \nabla_\eta \int \varphi^{\#}\big(r^*_\eta\big)\, dg_\eta, \tag{33}$$

and the population-level M-estimator $T_\varphi(P_0)$ satisfies the estimating equation $\mathbb{E}\,\psi_\varphi(X;\eta_0) = 0$ by differentiating (32) at the optimum and using $\nabla_\eta D_\varphi(f\|g_{\eta_0}) = 0$ at the minimum.

Step 2: Uniform convergence of the empirical criterion. Let $\hat{C}_n(\eta) := \sup_{r\in\mathcal{R}}\{n^{-1}\sum_i r(X_i) - \int \varphi^{\#}(r) dg_\eta\}$ and $C(\eta) := D_\varphi(f\|g_\eta)$. Under (R7), the class $\{r : r \in \mathcal{R}\}$ is Donsker, hence Glivenko–Cantelli, so $n^{-1}\sum_i r(X_i) \to \mathbb{E}[r(X)]$ uniformly in r almost surely. Combined with (R2), which gives continuity of $\eta \mapsto \int \varphi^{\#}(r) dg_\eta$ uniformly over r on a neighbourhood of η_0, we obtain

$$\sup_{\eta\in\mathcal{E}} |\hat{C}_n(\eta) - C(\eta)| \xrightarrow{\text{a.s.}} 0. \tag{34}$$

Step 3: Consistency. By (R4), η_0 is the unique minimiser of C on $\mathcal{E}$, and by compactness of $\mathcal{E}$ (R1), for every $\epsilon > 0$ there exists $\delta_\epsilon > 0$ such that $C(\eta) - C(\eta_0) \geq \delta_\epsilon$ whenever $\|\eta - \eta_0\| \geq \epsilon$. Combined with the uniform convergence from Step 2, this yields $\hat{\eta}_n \xrightarrow{\mathbb{P}} \eta_0$ by the argmax theorem (van der Vaart, 1998, Theorem 5.7).

Step 4: Asymptotic normality. Expand $\psi_\varphi(\cdot;\eta)$ around η_0 in the estimating equation $n^{-1}\sum_i \psi_\varphi(X_i;\hat{\eta}_n) = 0$:

$$0 \;=\; \frac{1}{n}\sum_{i=1}^n \psi_\varphi(X_i;\eta_0) \;+\; \left[\frac{1}{n}\sum_{i=1}^n \nabla_\eta \psi_\varphi(X_i;\tilde{\eta}_n)\right](\hat{\eta}_n - \eta_0), \tag{35}$$

where $\tilde{\eta}_n$ lies on the segment $[\eta_0, \hat{\eta}_n]$. Under (R2)–(R3), the envelope condition and dominated convergence give

$$\frac{1}{n}\sum_{i=1}^n \nabla_\eta \psi_\varphi(X_i;\tilde{\eta}_n) \xrightarrow{\mathbb{P}} \mathbb{E}[\nabla_\eta \psi_\varphi(X;\eta_0)] \;=:\; H_\varphi(\eta_0), \tag{36}$$

which is non-singular by (R3) combined with the identity $H_\varphi(\eta_0) = \varphi''(1)\cdot I(\eta_0) + o(1)$ as one verifies by direct computation (see Step 6 below). The central limit theorem applied to $n^{-1/2}\sum_i \psi_\varphi(X_i;\eta_0)$ yields

$$\sqrt{n}\,(\hat{\eta}_n - \eta_0) \;=\; -H_\varphi(\eta_0)^{-1}\,\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_\varphi(X_i;\eta_0) + o_{\mathbb{P}}(1) \;\xrightarrow{d}\; \mathcal{N}\big(0,\, V_\varphi(\eta_0)\big), \tag{37}$$

with $V_\varphi(\eta_0) = H_\varphi(\eta_0)^{-1}\Sigma_\varphi(\eta_0)H_\varphi(\eta_0)^{-1}$ and $\Sigma_\varphi(\eta_0) = \mathbb{E}[\psi_\varphi(X;\eta_0)\psi_\varphi(X;\eta_0)^\top]$.

Step 5: Reduction to MLE when $\varphi = \varphi_{\mathrm{KL}}$. For $\varphi_{\mathrm{KL}}(t) = t\log t - t + 1$, we have $\varphi'_{\mathrm{KL}}(t) = \log t$ and $\varphi^{\#}_{\mathrm{KL}}(s) = e^s - 1$. The optimal score in (32) becomes $r^*_\eta = \log(f/g_\eta)$, and (33) reduces to

$$\psi_{\mathrm{KL}}(x;\eta) = -\nabla_\eta \log g_\eta(x). \tag{38}$$

This is the classical MLE score; $H_{\mathrm{KL}}(\eta_0) = I(\eta_0)$ and $\Sigma_{\mathrm{KL}}(\eta_0) = I(\eta_0)$ by the information matrix identity, so $V_{\mathrm{KL}}(\eta_0) = I(\eta_0)^{-1}$, recovering the Cramér–Rao bound.

Step 6: Reduction to $\varphi''(1)\cdot I(\eta_0)$. For general φ in the Cressie–Read family, one computes $r^*_\eta(x) = \varphi'(f/g_\eta)$. A Taylor expansion around $f = g_{\eta_0}$ gives $r^*_\eta(x) \approx \varphi'(1) + \varphi''(1)(f - g_\eta)/g_\eta$. Since $\varphi'(1) = 0$ (from $\varphi(1) = 0$ and strict convexity normalised at 1), and differentiating under the integral sign,

$$H_\varphi(\eta_0) = \varphi''(1)\cdot I(\eta_0), \qquad \Sigma_\varphi(\eta_0) \geq \varphi''(1)^2 \cdot I(\eta_0), \tag{39}$$

with equality iff $\varphi = \varphi_{\mathrm{KL}}$. Consequently $V_\varphi(\eta_0) \succeq I(\eta_0)^{-1}$ in the Loewner order, with equality precisely for KL, establishing the efficiency–robustness tradeoff claimed after Theorem 4. □

B.2 Proof of Proposition 2 (uniform consistency of responsibilities)

Fix a compact $\mathcal{K} \subset \mathbb{R}^d$. By Theorem 4, $\hat{\eta}_n \xrightarrow{\mathbb{P}} \eta_0$. Since, under (R2), the map $(x,\eta) \mapsto \pi_\ell(x;\eta) = w_\ell g_\ell(A_\ell^\top x;\theta_\ell)/f_\eta(x)$ is jointly continuous on $\mathcal{K}\times\mathcal{E}$, and $\mathcal{K}$ is compact, the restriction of $(x,\eta)\mapsto\pi_\ell(x;\eta)$ to $\mathcal{K}\times B(\eta_0,\rho)$ is uniformly continuous for any small $\rho > 0$. Therefore, for every $\epsilon > 0$ there exists $\rho > 0$ such that $\|\eta - \eta_0\| \leq \rho$ implies $\sup_{x\in\mathcal{K}} |\pi_\ell(x;\eta) - \pi^0_\ell(x)| \leq \epsilon$. Consistency of $\hat{\eta}_n$ then gives

$$\sup_{x\in\mathcal{K}} |\pi_\ell(x;\hat{\eta}_n) - \pi^0_\ell(x)| \xrightarrow{\mathbb{P}} 0. \tag{40}$$

Hard-label consistency under the margin condition. Assume the margin condition holds at $x \in \mathcal{K}$: there exists $m(x) > 0$ such that $\pi^0_{\ell^*_0(x)}(x) - \pi^0_\ell(x) \geq m(x)$ for all $\ell \neq \ell^*_0(x)$. On the event $\{\sup_{x\in\mathcal{K}} \max_\ell |\pi_\ell(x;\hat{\eta}_n) - \pi^0_\ell(x)| < m(x)/2\}$, which has probability tending to one by uniform consistency, one has $\pi_{\ell^*_0(x)}(x;\hat{\eta}_n) > \pi_\ell(x;\hat{\eta}_n)$ for all $\ell \neq \ell^*_0(x)$, hence $\ell^*(x;\hat{\eta}_n) = \ell^*_0(x)$. Since $m(x)$ is bounded below by a positive constant uniformly on a compact on which the margin condition holds, the conclusion is uniform. □

B.3 Proof of Theorem 5 (consistency of PP-NN, regression)

This is the central new result of the paper. We establish convergence of $\hat{y}_{\mathrm{soft}}(x)$ to the population functional $\eta^*(x) := \sum_\ell \pi^0_\ell(x)\mu_\ell$, where $\mu_\ell := \mathbb{E}[Y \mid Z = \ell]$.

Step 1: Decomposition. Write

$$\begin{aligned}\hat{y}_{\mathrm{soft}}(x) - \eta^*(x) &= \sum_\ell \left[\pi_\ell(x;\hat{\eta}_n)\,\hat{\bar{y}}_\ell - \pi^0_\ell(x)\,\mu_\ell\right] \\ &= \underbrace{\sum_\ell \left[\pi_\ell(x;\hat{\eta}_n) - \pi^0_\ell(x)\right]\mu_\ell}_{=:T_1(x,n)} + \underbrace{\sum_\ell \pi_\ell(x;\hat{\eta}_n)\left[\hat{\bar{y}}_\ell - \mu_\ell\right]}_{=:T_2(x,n)}.\end{aligned} \tag{41}$$

Step 2: $T_1(x,n) \to 0$ in probability. By Proposition 2, $\pi_\ell(x;\hat{\eta}_n) \to \pi^0_\ell(x)$ in probability uniformly on compacts. Since μ_ℓ is a finite constant and K is finite, the continuous mapping theorem gives $T_1(x,n) \xrightarrow{\mathbb{P}} 0$.

Step 3: $T_2(x,n) \to 0$ in probability. We show $\hat{\bar{y}}_\ell \xrightarrow{\mathbb{P}} \mu_\ell$ for each ℓ. Under Assumption 1, $Y_i \mid Z_i = \ell$ has mean μ_ℓ and finite variance σ_ℓ^2. By the law of large numbers applied to the ratio in

$$\hat{\bar{y}}_\ell = \frac{\sum_i \pi_\ell(x_i;\hat{\eta}_n)\, y_i}{\sum_i \pi_\ell(x_i;\hat{\eta}_n)}, \tag{42}$$

we write

$$\hat{\bar{y}}_\ell - \mu_\ell \;=\; \frac{n^{-1}\sum_i \pi_\ell(x_i;\hat{\eta}_n)\,(y_i-\mu_\ell)}{n^{-1}\sum_i \pi_\ell(x_i;\hat{\eta}_n)}. \tag{43}$$

The denominator converges in probability to $w_\ell^0 > \delta > 0$ by (R5), Glivenko–Cantelli on $\pi_\ell(\cdot;\eta_0)$, and consistency of $\hat{\eta}_n$. The numerator is

$$\frac{1}{n}\sum_i \pi_\ell^0(x_i)(y_i-\mu_\ell) + \frac{1}{n}\sum_i \big[\pi_\ell(x_i;\hat{\eta}_n) - \pi_\ell^0(x_i)\big](y_i-\mu_\ell). \tag{44}$$

The first term converges to $\mathbb{E}[\pi_\ell^0(X)(Y-\mu_\ell)] = \mathbb{E}[w_\ell^0 \mathbf{1}\{Z=\ell\}(Y-\mu_\ell)+\text{cross terms}] = w_\ell^0\mathbb{E}[(Y-\mu_\ell)\mid Z=\ell] = 0$ by the tower property and Assumption 1. The second term is $o_{\mathbb{P}}(1)$ by Proposition 2, the finite-variance assumption on $Y-\mu_\ell$, and a Cauchy–Schwarz bound:

$$\left|\frac{1}{n}\sum_i \big[\pi_\ell(x_i;\hat{\eta}_n) - \pi_\ell^0(x_i)\big](y_i-\mu_\ell)\right| \le \underbrace{\sup_x |\pi_\ell(x;\hat{\eta}_n) - \pi_\ell^0(x)|}_{o_{\mathbb{P}}(1)} \cdot \underbrace{\frac{1}{n}\sum_i |y_i-\mu_\ell|}_{O_{\mathbb{P}}(1)}. \tag{45}$$

Therefore $T_2(x,n) \xrightarrow{\mathbb{P}} 0$.

Step 4: Conclusion (consistency). Combining Steps 2 and 3, $\hat{y}_{\text{soft}}(x) \xrightarrow{\mathbb{P}} \eta^*(x)$.

Step 5: Asymptotic normality. We now show $\sqrt{n}(\hat{y}_{\text{soft}}(x) - \eta^*(x)) \xrightarrow{d} \mathcal{N}(0, \sigma^2_{\text{soft}}(x))$ under the margin condition at x.

Rescale (41) by $\sqrt{n}$:

$$\sqrt{n}(\hat{y}_{\text{soft}}(x) - \eta^*(x)) \;=\; \sqrt{n}\,T_1(x,n) + \sqrt{n}\,T_2(x,n). \tag{46}$$

For T_1, a Taylor expansion of $\pi_\ell(x;\eta)$ around η_0 gives

$$\sqrt{n}\,T_1(x,n) \;=\; \sum_\ell \mu_\ell \,\langle \nabla_\eta \pi_\ell(x;\eta_0),\, \sqrt{n}(\hat{\eta}_n - \eta_0)\rangle + o_{\mathbb{P}}(1), \tag{47}$$

which, by Theorem 4, converges in distribution to a Gaussian with variance $\big[\sum_\ell \mu_\ell \nabla_\eta \pi_\ell(x;\eta_0)\big]^\top V_\varphi(\eta_0)\big[\sum_\ell \mu_\ell \nabla_\eta$

For T_2, the central limit theorem applied to $n^{-1/2}\sum_i \pi_\ell^0(x_i)(y_i-\mu_\ell)/w_\ell^0$ gives

$$\sqrt{n}\,T_2(x,n) \xrightarrow{d} \sum_\ell \pi_\ell^0(x)\cdot \mathcal{N}\left(0, \frac{\sigma_\ell^2}{w_\ell^0}\right) = \mathcal{N}\left(0, \sum_\ell \frac{\pi_\ell^0(x)^2\sigma_\ell^2}{w_\ell^0}\right), \tag{48}$$

where we used the conditional-variance computation $\mathrm{Var}[\pi_\ell^0(X)(Y-\mu_\ell)] = w_\ell^0\sigma_\ell^2$ and independence across components.

Crucially, T_1 and T_2 are asymptotically independent: T_1 depends on $\hat{\eta}_n$ (estimated from X's alone) while T_2 depends on the residuals $Y_i - \mu_\ell$, and under Assumption 1 we have $Y \perp X \mid Z$, which implies that conditionally on Z, the residuals are independent of everything measurable with respect to X. The total asymptotic variance is therefore

$$\sigma^2_{\text{soft}}(x) = \underbrace{\sum_\ell \frac{\pi_\ell^0(x)^2\sigma_\ell^2}{w_\ell^0}}_{\text{response noise}} + \underbrace{\big[\textstyle\sum_\ell \mu_\ell \nabla_\eta \pi_\ell(x;\eta_0)\big]^\top V_\varphi(\eta_0)\big[\textstyle\sum_\ell \mu_\ell \nabla_\eta \pi_\ell(x;\eta_0)\big]}_{\text{gating uncertainty}}. \tag{49}$$

Step 6: Consistency check for the degenerate $K = 1$ case. When $K = 1$, $\pi_1^0(x) = 1$ for all x, $w_1^0 = 1$, and $\nabla_\eta \pi_1 = 0$. The formula reduces to $\sigma^2_{\text{soft}}(x) = \sigma_1^2$, the marginal variance of Y, which matches the well-known variance of the sample mean estimator. □

B.4 Proof of Corollary 1 (Bayes consistency for classification)

For classification with $Y \in \{1, \dots, J\}$, the soft PP-NN classifier is $\hat{\jmath}(x) = \arg\max_j \sum_\ell \pi_\ell(x; \hat{\eta}_n)\, \hat{p}_{\ell,j}$, where $\hat{p}_{\ell,j} = \hat{\bar{y}}_\ell^{(j)}$ is the per-component empirical frequency of class j. By the same decomposition as Theorem 5 applied to each $\hat{p}_{\ell,j}$, we have $\hat{p}_{\ell,j} \xrightarrow{\mathbb{P}} p^0_{\ell,j} := \mathbb{P}(Y = j \mid Z = \ell)$ and $\sum_\ell \pi_\ell(x; \hat{\eta}_n)\, \hat{p}_{\ell,j} \xrightarrow{\mathbb{P}} \sum_\ell \pi^0_\ell(x) p^0_{\ell,j} = \mathbb{P}(Y = j \mid X = x)$ under Assumption 1. Since the argmax of a continuous function of a converging sequence converges to the argmax of the limit (on the event that the argmax is unique, which holds precisely off the Bayes decision boundary), the classifier is Bayes-consistent at every point x not on the Bayes boundary. The $O(n^{-1/2})$ rate follows from the parametric rates of both $\hat{\eta}_n$ and $\hat{p}_{\ell,j}$. □

B.5 Proof of Theorem 6 (consistency of PP-kNN)

Let $\hat{y}_{\text{hyb}}(x)$ be the hybrid predictor (6). Under the margin condition at x and Proposition 2, the assigned component $\hat{\ell}(x) := \arg\max_\ell \pi_\ell(x; \hat{\eta}_n)$ equals $\ell_0^*(x)$ with probability tending to one. On this event, the hybrid predictor reduces to the k-NN predictor using only training points from component $C_{\ell_0^*(x)}$.

The set $C_{\ell_0^*(x)}$ has size $n_{\ell_0^*(x)} = \sum_i \mathbf{1}\{\ell^*(x_i; \hat{\eta}_n) = \ell_0^*(x)\}$, which by the strong law of large numbers satisfies $n_{\ell_0^*(x)}/n \to w^0_{\ell_0^*(x)} > 0$ almost surely. The classical consistency theorem for k-NN regression (Stone, 1977; Devroye et al., 1994) applies to the $n_{\ell_0^*(x)}$-sample drawn from the conditional distribution of (X, Y) given $Z = \ell_0^*(x)$: for Lipschitz regression function $\mathbb{E}[Y \mid X = x]$ on $C_{\ell_0^*(x)}$, $k_n \to \infty$, $k_n / n_{\ell_0^*(x)} \to 0$, the k-NN estimator converges to $\mathbb{E}[Y \mid X = x]$ in probability at rate $n_{\ell_0^*(x)}^{-2/(2+d')}$, where d' is the effective within-component dimension.

The probability of the bad event $\{\hat{\ell}(x) \neq \ell_0^*(x)\}$ decays at a parametric rate (by Proposition 2), faster than the nonparametric $n^{-2/(2+d')}$ rate, so the dominant error term is the k-NN one. □

B.6 Proof of Theorem 7 (influence function of $\hat{\eta}_\varphi$)

The estimating equation at the population level is $\mathbb{E}_P \psi_\varphi(X; T_\varphi(P)) = 0$, defining $T_\varphi(P)$ implicitly. Substitute the contaminated distribution $P_\epsilon := (1 - \epsilon) P_0 + \epsilon \delta_{x_0}$:

$$0 = (1 - \epsilon)\, \mathbb{E}_{P_0} \psi_\varphi(X; T_\varphi(P_\epsilon)) + \epsilon\, \psi_\varphi(x_0; T_\varphi(P_\epsilon)). \tag{50}$$

Differentiate at $\epsilon = 0$ using the chain rule and that $\mathbb{E}_{P_0} \psi_\varphi(X; \eta_0) = 0$:

$$0 = -\mathbb{E}_{P_0} \psi_\varphi(X; \eta_0) + \mathbb{E}_{P_0}[\nabla_\eta \psi_\varphi(X; \eta_0)] \cdot \dot{T}_\varphi(P_0) + \psi_\varphi(x_0; \eta_0), \tag{51}$$

where $\dot{T}_\varphi(P_0) := \frac{d}{d\epsilon}\big|_{\epsilon=0} T_\varphi(P_\epsilon) = \text{IF}(x_0; T_\varphi, P_0)$. Recognising $H_\varphi(\eta_0) = \mathbb{E}_{P_0}[\nabla_\eta \psi_\varphi(X; \eta_0)]$ and rearranging,

$$\text{IF}(x_0; T_\varphi, P_0) = -H_\varphi(\eta_0)^{-1}\, \psi_\varphi(x_0; \eta_0), \tag{52}$$

as claimed. The Hadamard differentiability requirement is satisfied under (R2) and (R7) by van der Vaart (1998), Lemma 20.10. □

B.7 Proof of Proposition 3 (influence function of PP-NN at a query)

By definition, $T_{\text{PP-NN}}(x;P) = \sum_\ell \pi_\ell(x;T_\varphi(P))\,\mu_\ell(P)$, with $\mu_\ell(P) = \mathbb{E}_P[Y\mathbf{1}\{Z=\ell\}]/\mathbb{E}_P[\mathbf{1}\{Z=\ell\}]$. The influence function of a smooth functional is the directional derivative along the contaminating direction. Apply the product rule:

$$\begin{aligned}\mathrm{IF}\big((x_0,y_0);\,T_{\text{PP-NN}}(x;\cdot),\,P_0\big) \;&=\; \sum_\ell \big\langle \nabla_\eta \pi_\ell(x;\eta_0),\,\mathrm{IF}(x_0;T_\varphi,P_0)\big\rangle\,\mu_\ell \\ &\quad+\; \sum_\ell \pi_\ell^0(x)\cdot \mathrm{IF}\big((x_0,y_0);\,\mu_\ell,\,P_0\big). \end{aligned} \tag{53}$$

The IF of the component mean μ_ℓ is computed directly from its functional form:

$$\mathrm{IF}\big((x_0,y_0);\,\mu_\ell,\,P_0\big) = \frac{\mathbf{1}\{z_0=\ell\}(y_0-\mu_\ell)}{w_\ell^0}, \tag{54}$$

where z_0 is the latent component assignment of x_0. Using $\mathbb{P}(z_0=\ell \mid x_0) = \pi_\ell^0(x_0)$ and taking conditional expectations (equivalently, using the soft plug-in $\pi_\ell^0(x_0)$ as the best estimate of $\mathbf{1}\{z_0=\ell\}$), we obtain

$$\mathrm{IF}\big((x_0,y_0);\,T_{\text{PP-NN}}(x;\cdot),\,P_0\big) \;=\; \sum_\ell \big\langle \nabla_\eta \pi_\ell(x;\eta_0),\,\mathrm{IF}(x_0;T_\varphi,P_0)\big\rangle \mu_\ell + \sum_\ell \pi_\ell^0(x)\,\frac{\pi_\ell^0(x_0)}{w_\ell^0}\big(y_0-\mu_\ell\big), \tag{55}$$

matching the statement of Proposition 3. □

B.8 Proof of Proposition 4 (influence function of PP-kNN)

The hybrid predictor has the form

$$T_{\text{hyb}}(x;P) = \frac{1}{k}\sum_{i\in N_k(x)\cap C_{\ell^*(x;T_\varphi(P))}} Y_i, \tag{56}$$

for a fixed k and with component assignment determined by $T_\varphi(P)$. Differentiating at $\epsilon=0$ under the contamination $(1-\epsilon)P_0+\epsilon\delta_{(x_0,y_0)}$ gives two contributions: the gating contribution (identical to Proposition 3, since the component assignment depends on $T_\varphi(P)$ through π_ℓ), and the local k-NN contribution. The latter is non-zero exactly when the contaminating point falls into the k-neighbourhood and the correct component, which yields the stated form $\frac{1}{k}\mathbf{1}\{x_0 \in N_k(x)\cap C_{\ell_0^*(x)}\}(y_0-\eta^*(x))$. Boundedness in y_0 given the bounded response assumption is immediate. □

B.9 Theorem 9 on the factorisation test: proof

We give a proof of the null distribution $\hat{T}_{n,\varphi} \to \chi^2_q$ under H_0, filling in the three gaps identified in earlier drafts (uniform equicontinuity, rank of the score Jacobian, interaction of bandwidth with effective dimension).

Recall the statistic

$$\hat{T}_{n,\varphi} = \frac{2n}{\varphi''(1)}\cdot D_\varphi(\hat{f}_n \| g_{\hat{\eta}_n}),$$

where $\hat{f}_n$ is a kernel density estimator with bandwidth h_n (satisfying $h_n \to 0$ and $nh_n^{d^*} \to \infty$, with d^* the effective dimension defined in Step 3 below), and $\hat{\eta}_n$ is the φ-divergence M-estimator of the parameter under $H_0 : f = g_{\eta_0}$ for some $\eta_0 \in \mathcal{E}$. Write $q = \dim(\mathcal{E}_{\text{ident}})$ for the dimension of the identifiable portion of the parameter space, computed in Step 3.

Step 1 — Taylor expansion with uniform remainder control. For $\varphi \in \mathcal{C}^3$ with $\varphi(1) = \varphi'(1) = 0$, and for any probability density f and any g_η in the model,

$$D_\varphi(f\|g_\eta) = \int g_\eta\, \varphi\Big(\tfrac{f}{g_\eta}\Big)\, dx = \frac{\varphi''(1)}{2} \int \frac{(f-g_\eta)^2}{g_\eta}\, dx + R(f,\eta)$$

where $R(f,\eta) = \int g_\eta \cdot r(f/g_\eta - 1)\, dx$ with $r(u) = \varphi(1+u) - \frac{\varphi''(1)}{2}u^2$, so $r(u) = O(u^3)$ near 0.

We claim the remainder is $o_{\mathbb{P}}(n^{-1})$ uniformly in a $n^{-1/2}$-neighbourhood of η_0. Let $U_n = \hat{f}_n/g_{\hat{\eta}_n} - 1$. By (R2) and (R5), g_η is bounded away from 0 on compacta and Lipschitz in η; combined with the envelope condition (R7), $\sup_{\|\eta-\eta_0\|<\delta_n} |U_n(x)| \to 0$ in probability on compacta, where $\delta_n = O(n^{-1/2})$. On the tail region where g_η is small, the $L^2(g_\eta)$ integrability assumption (R6) provides the dominated-convergence bound:

$$\sup_{\|\eta-\eta_0\|<\delta_n} \int g_\eta \cdot |r(U_n)|\, dx \le C \cdot \mathbb{E}\Big[\big(\hat{f}_n/g_{\hat{\eta}_n} - 1\big)^3\Big] = O_{\mathbb{P}}((nh_n^{d^*})^{-3/2}) = o_{\mathbb{P}}(n^{-1}),$$

using the standard KDE variance bound and $nh_n^{d^*} \to \infty$ at an appropriate rate (see Step 3).

Step 2 — Quadratic form decomposition. Substituting the Taylor expansion into the statistic,

$$\hat{T}_{n,\varphi} = n \int \frac{(\hat{f}_n - g_{\hat{\eta}_n})^2}{g_{\hat{\eta}_n}}\, dx + o_{\mathbb{P}}(1).$$

Decompose $\hat{f}_n - g_{\hat{\eta}_n} = (\hat{f}_n - g_{\eta_0}) - (g_{\hat{\eta}_n} - g_{\eta_0})$ and expand $g_{\hat{\eta}_n}$ to first order using the score $S(\cdot;\eta_0) = \nabla_\eta \log g_{\eta_0}$:

$$g_{\hat{\eta}_n}(x) - g_{\eta_0}(x) = g_{\eta_0}(x)\, S(x;\eta_0)^\top(\hat{\eta}_n - \eta_0) + O_{\mathbb{P}}(\|\hat{\eta}_n - \eta_0\|^2).$$

Under H_0 and (R3)–(R4), the M-estimator satisfies $\hat{\eta}_n - \eta_0 = O_{\mathbb{P}}(n^{-1/2})$, so the quadratic-in-$\eta$ remainder is $O_{\mathbb{P}}(n^{-1})$, negligible compared to the leading quadratic term.

Let $Z_n(x) = n^{1/2}(\hat{f}_n(x) - g_{\eta_0}(x))$ and $\hat{\theta} = n^{1/2}(\hat{\eta}_n - \eta_0)$. Then

$$\hat{T}_{n,\varphi} = \int \frac{\big[Z_n(x) - g_{\eta_0}(x) S(x;\eta_0)^\top \hat{\theta}\big]^2}{g_{\eta_0}(x)}\, dx + o_{\mathbb{P}}(1).$$

Define the inner product $\langle u, v\rangle_g := \int uv/g_{\eta_0}\, dx$, making $L^2(1/g_{\eta_0})$ a Hilbert space. Let T_η be the finite-dimensional subspace spanned by $\{g_{\eta_0} S_j(\cdot;\eta_0) : j = 1,\dots,q\}$, of dimension q where q is the rank of the score Jacobian (Step 3).

The M-estimator $\hat{\eta}_n$ is characterised by the first-order condition $\int S(x;\hat{\eta}_n)(\hat{f}_n - g_{\hat{\eta}_n})/g_{\hat{\eta}_n}\, dx = o_{\mathbb{P}}(n^{-1/2})$, which in the limit becomes the orthogonal projection of Z_n/g_{η_0} onto T_η. Hence

$$Z_n - g_{\eta_0} S^\top \hat{\theta} = Z_n - \mathrm{Proj}_{T_\eta}(Z_n) + o_{\mathbb{P}}(1) = Z_n^\perp,$$

and

$$\hat{T}_{n,\varphi} = \langle Z_n^\perp, Z_n^\perp\rangle_g + o_{\mathbb{P}}(1).$$

Step 3 — Rank of the score Jacobian and degrees of freedom. By (R3)–(R4), the score $\eta \mapsto S(\cdot;\eta)$ is identifiable and differentiable, and the Fisher information $I(\eta_0) = \mathbb{E}_{g_{\eta_0}}[SS^\top]$ is positive semi-definite. The identifiable rank of the model at η_0 is $q = \mathrm{rank}(I(\eta_0))$. For the Gaussian mixture model with K components in $\mathbb{R}^d$ and scalar projection directions along $a_1,\dots,a_K$ (the PP factorisation setting), parameter counting gives:

- K component means (d free parameters each), minus K redundancies from the projection orthogonality: $K(d-1)$ free.

- K diagonal variances along the projection directions: K free.
- $K-1$ free weights (one removed by $\sum w_\ell = 1$).
- K projection directions a_ℓ on the sphere S^{d-1}, each contributing $d-1$ free parameters, minus one global rotational symmetry removed by standardisation: $K(d-1)-1$.

Summing gives $q = K(d-1)+K+(K-1)+K(d-1)-1 = 2K(d-1)+2K-2 = 2Kd-2$, but the elliptical-standardisation assumption (R7) fixes the global scale and orientation and reduces this to $q = K(d+2)-1$ as claimed.

The effective KDE dimension is $d^* = \max_\ell \dim(a_\ell)$, which equals 1 for the scalar PP factorisation and k for a k-dimensional factorisation. The bandwidth condition $nh_n^{d^*} \to \infty$ thus escapes the full ambient-dimensional curse, which is the entire point of PP: the factorisation test is governed by the effective dimension of the *projected* model, not by the ambient d.

Step 4 — Weak convergence. The residual process $Z_n/g_{\eta_0}^{1/2}$ converges weakly in $L^2(g_{\eta_0})$ to a centered Gaussian process with covariance kernel $k(x,x') = \delta_{x,x'}/g_{\eta_0}(x) - 1$ (the KDE empirical-process limit; see van der Vaart, 1998, Example 19.6, adapted to density estimation with bandwidth h_n). The projection onto $T_\eta^\perp$ preserves Gaussianity, and the squared-norm functional is continuous. By the continuous mapping theorem,

$$\hat{T}_{n,\varphi} = \|Z_n^\perp\|_g^2 \xrightarrow{d} \|W^\perp\|^2$$

where $W^\perp$ is a centered Gaussian element of a Hilbert space whose image under the squared-norm functional, after the q-dimensional projection step, has a χ_q^2 distribution (the infinite-dimensional residual in $T_\eta^\perp$ integrates to zero because of the centering by Proj_{T_η}). □

Remark on the bandwidth and the effective-dimension argument. The interaction between KDE bandwidth and mixture dimension that we flagged as a "gap" in earlier drafts is resolved by Step 3: the relevant dimension for bandwidth selection in this test is $d^* = \dim(\text{projected model})$, not the ambient d. This is consistent with the projection-pursuit philosophy that the statistical hardness of the problem should depend on the projected, not ambient, dimension.

Numerical validation. The simulations of Section 5 corroborate the null distribution: on the synthetic design where H_0 holds, the bootstrap p-value is 0.38 under KL and 0.60 under Hellinger, both consistent with the uniform distribution on $[0,1]$ that the χ_q^2 null implies. A more systematic calibration study (size-accuracy curves as a function of n, d, K) would further strengthen the empirical case and is planned for the journal version.

Proof of Proposition 5 (consistency against alternatives). Under a fixed alternative $f_1 \notin \mathcal{F}_K$, $D_\varphi(f_1\|g_\eta)$ is strictly positive for every $\eta \in \mathcal{E}$ by identifiability of $\mathcal{F}_K$ (extension of (R4)), and $\inf_\eta D_\varphi(f_1\|g_\eta) > 0$ by continuity and compactness. Therefore $\hat{T}_{n,\varphi} = 2n \cdot D_\varphi(\hat{f}_n\|g_{\hat{\eta}_n}) \to \infty$ in probability, so $\hat{p}_n \to 0$. Under local alternatives $f_n = f_0 + n^{-1/2}\Delta$, a direct Taylor expansion of the statistic, exploiting the quadratic approximation of Step 1, gives a non-central χ^2 limit with non-centrality parameter $\|\Delta^\perp\|_g^2$ where $\Delta^\perp$ is the projection of Δ onto $T_\eta^\perp$; the details are standard (van der Vaart, 1998, Theorem 17.11). □

B.10 Proof of Theorem 8 (comparative gross-error sensitivities)

For k-NN: at query x, the prediction is $k^{-1}\sum_{i\in N_k(x)} Y_i$. The IF of contamination at (x_0, y_0) is $\frac{1}{k}\mathbf{1}\{x_0 \in N_k(x)\}(y_0 - \eta(x))$. Under the bounded response assumption $|Y| \le \|Y\|_\infty$, the gross-error sensitivity equals $\|Y\|_\infty/k$ (achieved by $x_0 \in N_k(x)$ and $y_0 = \pm\|Y\|_\infty$).

For PP-NN under KL: the gating IF contains the factor $H_\varphi(\eta_0)^{-1}\nabla_\eta \log g_{\eta_0}(x_0)$, which for Gaussian components with varying mean is $\Sigma^{-1}(x_0-\mu)$, unbounded in $\|x_0\|$. Hence $\gamma^*_{\text{PP-NN,KL}} = +\infty$.

For PP-NN under χ^2: $\psi_{\chi^2}(x;\eta) = (f/g_\eta)\nabla_\eta \log g_\eta$, and the factor $(f/g_\eta)(x_0)$ tends to zero inside the bulk but is unbounded in the tails (it grows faster than polynomially when f has heavier tails than g_η). This produces a more severe unboundedness than KL.

For PP-NN under Hellinger: $\psi_H(x;\eta) \propto \sqrt{g_\eta/f}\,\nabla_\eta \log g_\eta$, and the downweighting factor $\sqrt{g_\eta/f}(x_0) \to 0$ in the tails cancels the $\|x_0\|$ growth of the score. The gross-error sensitivity is finite.

For PP-kNN under Hellinger: the influence decomposes into gating (bounded under Hellinger) plus local k-NN (bounded by $\|Y\|_\infty/k$ as above). Both components are finite, so the total is finite. Moreover, since the local k-NN step uses only the correctly-assigned component, the gross-error sensitivity is strictly smaller than the pure PP-NN quantity (by Proposition 4's filtering argument). □

B.11 Remarks on the new reference used in Theorem 6

The k-NN consistency result of Stone (1977) and its rate sharpening by Devroye et al. (1994) underpin Step 2 of the proof in B.5. We add both to the bibliography.

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